AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

EDITED BY

William Gray-Roncal, PhD
James Diamond, PhD

Learning Engineering for Next-Generation Systems · v2.1

EDITION · 15 JUNE 2026

LDT · SCHOOL OF EDUCATION · JOHNS HOPKINS UNIVERSITY

SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the LEOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.

II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.

III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

> **On verification.** The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganizes the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset's standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE	Case 128	**Operation Eagle Claw (1980).** Cross-organizational capability that existed on no service's deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.
STANDOUT SUCCESS	Case 137	**U.S. Nuclear Navy / Rickover Training Model (1954–).** The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

124	USS Fitzgerald & USS John S. McCain	2017	T·K·N
126	F-35 Sustainment & the Maintainer Capability Gap	ongoing	T·K·D
127	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T·H·K
128	Operation Eagle Claw	1980	T·K
129	Patriot Missile / Dhahran	1991	D·H·K
130	USS Vincennes & Iran Air Flight 655	1988	H·T
131	Stanislav Petrov / 1983 False Alert	1983	H·T
133	Mark 14 Torpedo Failures	1941 – 1943	T·K·G
134	V-22 Osprey	1991 – present	T·H·N
135	9/11 Intelligence Sharing Failures	1996 – 2001	G·K
137	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T·K·N
138	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard	2020 revision (series since 1968)	G·K

AVIATION

STANDOUT FAILURE **Case 97**

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS **Case 117**

Crew Resource Management & CAST (1981–). Authority gradients named and engineered into the cockpit, then a closed-loop industry data architecture sixteen years later. The 83% fatality-risk reduction CAST reports is a portfolio result; CRM's separate contribution is not isolated in any published estimate.

HEALTHCARE

STANDOUT FAILURE — **Case 2** — **EHR / CPOE Implementation (2005–).** Roughly $30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS — **Case 23** — **WHO Surgical Safety Checklist (2008–).** One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

36	AlphaFold — Protein Structure Prediction 2020 – present	T
37	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback 2020	T·K·D
39	TeamSTEPPS 2006 – present	T·N
176	Tylenol Recall 1982	G·N

EDUCATION AT SCALE

STANDOUT FAILURE

Case 54

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS

Case 80

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

45	Tennessee Voluntary Pre-K Study 2009–2018	G·D
46	Algorithmic Bias in Educational Predictive Analytics ongoing	G·H·D
51	Atlanta Public Schools Cheating Scandal 2009 – 2015	G·N
53	inBloom 2014	G
54	Summit Learning / Personalized Learning Rollout 2014–2019	G·T·K
67	Cognitive Tutor / Carnegie Learning 1990s – present	T
80	Georgia State University Predictive Analytics 2012 – present	T·K
142	xAPI / Total Learning Architecture — Interoperability Gap ongoing	K·G
205	The Discipline We Build Next ongoing	T·K·N·G·H·D

ENERGY

STANDOUT FAILURE

Case 166

Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT SUCCESS

Case 175

INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES Level 4-or-above accident at a U.S. commercial reactor since.

146	Northeast Blackout 2003	H·K

160 Davis-Besse Reactor Head Corrosion 2002 N·K·G

161 Texas City BP Refinery Explosion 2005 N·T·K·G

162 Upper Big Branch Mine Explosion 2010 N·G·K

163 Deepwater Horizon 2010 T·N·K

165 Camp Fire / PG&E 2018 G·N·K

166 Three Mile Island 1979 T·H

169 Fukushima Daiichi 2011 N·G·K

175 INPO and the Nuclear Academy 1979 – present T·K·G

GOVERNMENT

STANDOUT FAILURE — Case 191

Australia Robodebt (2016–2023). An income-averaging algorithm that reversed the burden of proof onto welfare recipients, raising some 470,000 wrongful debts. The Royal Commission found the scheme unlawful and attributed deaths to its operation — circumstantially, not as a finding of individual causation.

NOTE — Case 203

The reforms that follow these failures — Royal Commissions, statutory restructures — are documented inside the narratives rather than as separate cases. The government dataset's paired-intervention success is NYC Local Law 144 (Case 203), filed under its own primary domain.

7 VA Wait-Time Scandal 2014 G·K·N

180 Healthcare.gov Launch 2013 K·T·G

193 Bernard Madoff / SEC Failures 1992 – 2008 G·K·N

197 Predictive Policing — PredPol 2011 – present G·H·D

INDUSTRIAL

STANDOUT FAILURE — Case 159

Bhopal (1984). Training, knowledge, normalization of deviance, and governance all collapsed in the same plant. Thousands died within hours; total-death estimates run to 15,000–20,000, with roughly half a million exposed. The cautionary case for outsourced operational responsibility.

STANDOUT SUCCESS — Case 155

Toyota Production System / Andon Cord (1950s–). Any operator can signal a defect at the source. Under the fixed-position stop system most pulls never halt the line — the smallest authority-architecture intervention in the dataset, and the most durably copied.

144 Kodak Digital Camera Stagnation 1975–2012 D·K·N

147	Takata Airbag Inflators	2008 – 2023	D·G
149	Volkswagen Dieselgate	2015	D·G
151	GM Ignition Switch	2002 – 2014	D·G
155	**Toyota Production System / Andon Cord**	1950s – present	N·G
159	Bhopal	1984	T·K·N·G
164	Grenfell Tower	2017	G·T·K·N

TECHNOLOGY

STANDOUT FAILURE — **Case 184**

UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. More than 900 wrongful convictions, 236 imprisoned, and 13 suicides the inquiry could not rule out as Horizon-caused — the longest-running operator-feedback failure in the dataset.

NOTE — **Case 36**

The technology dataset's success story is filed under Healthcare: AlphaFold (Case 36), the contemporary worked example of computational capability-engineering done as a discipline.

143	Knight Capital Trading Loss	2012	D·K
148	Wells Fargo Fake Accounts	2011 – 2016	G·N
152	TSB Bank IT Migration	2018	H·G
153	Equifax Data Breach	2017	G·K
157	AI-Augmented Coding Tools	2021 – present	T·H
172	CrowdStrike Falcon Outage	2024	D·K·G
184	UK Post Office Horizon Scandal	1999 – 2015	G·H·K

AI IN EDUCATION

88	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models	2025	T·K·N

K-12 EDUCATION

48	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism	2010s – 2022	G·K·N
60	Houston EVAAS — Value-Added Teacher Evaluation on Trial	2011–2017	G
61	Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap	1997–2023	K·T
62	Gates Intensive Partnerships — Measurement Without a Coupled Lever	2009–2018	G·K

K-12 MATHEMATICS

K-12 SCIENCE

AIR TRAFFIC MANAGEMENT

ANESTHESIOLOGY

AUTONOMOUS SYSTEMS

STANDOUT FAILURE — Case 183

Uber ATG / Tempe Fatality (2018). The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE — Case 205

This key's cases are all failures and frontiers; the paired-intervention successes — Waymo's safety case (199) and the CPUC permit regime (200) — are filed under autonomous vehicles. See Case 205 (**The Discipline We Build Next**) for the open question this volume closes on.

AUTONOMOUS VEHICLES

AVIATION INFRASTRUCTURE

AVIATION SAFETY

CLINICAL CARE

CLINICAL MEDICINE

CLINICAL/TRANSLATIONAL RESEARCH

CORPORATE L&D

CRIMINAL JUSTICE

DIGITAL IDENTITY

E-GOVERNMENT

ED-TECH

EDUCATION (NORWAY)

EDUCATION AT SCALE

EMERGENCY MANAGEMENT

FINANCE

FINANCIAL SERVICES

GENERAL AVIATION

GLOBAL HEALTH

GOVERNMENT

GOVERNMENT/WELFARE (NETHERLANDS)

HEALTH PROFESSIONS EDUCATION

HEALTHCARE/ONCOLOGY

HIGHER EDUCATION

HIGHER EDUCATION (AFRICA)

HIGHER EDUCATION (BRAZIL)

HIGHER EDUCATION (LATIN AMERICA)

HIGHER ED (UK)

HIGHER-ED ANALYTICS

HOSPITAL PHARMACY

NUCLEAR ENGINEERING

156 INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet 2010 – present **D·H·K**

NUCLEAR POWER

154 INL Turbine-Control Upgrade — Verification and Validation as the Cutover Deliverable 2014 **G·K·H**

NURSING EDUCATION

16 NCSBN National Simulation Study — Licensing the 50% Substitution Rule 2014 **G·K·D**

PHARMA SAFETY

38 iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms 2006 – 2011 **G·D·N**

PUBLIC HEALTH

179 The Johns Hopkins COVID-19 Dashboard — Evidence Infrastructure at Decision Speed 2020 – 2023 **G·K**

RAIL TRANSPORT

177 CIRAS — Confidential Incident Reporting for UK Rail 1996 – present **G·K·N**

SOFTWARE ENGINEERING EDUCATION

71 Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development 2025 (preprint) **K·N**

SOFTWARE SUSTAINMENT

174 Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed 1996 – 2000 **G·D·K**

SPACE / AEROSPACE

STANDOUT FAILURE — **Case 111** — **Challenger & Columbia (1986 / 2003).** Normalisation of deviance across seventeen years and two crews — the sociologist Diane Vaughan's term for a culture that accepted flying with flaws once problems were defined as normal and routine, and the culture the CAIB found still intact in 2003.

NOTE — **Case 137** — No paired-intervention success in the space dataset — Part III's aviation half carries seven. See Defense (Rickover, Case 137) for the safety-engineering counterpart human spaceflight did not build.

98 Mars Climate Orbiter — Unit Mismatch 1999 **D·K**

101 Ariane 5 Flight 501 1996 **D·K·H**

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the concentration, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

36 cases

LEN 2

53 cases

131 Stanislav Petrov / 1983 False Alert

141 DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks

144 Kodak Digital Camera Stagnation

146 Northeast Blackout

155 Toyota Production System / Andon Cord

157 AI-Augmented Coding Tools

172 CrowdStrike Falcon Outage

183 Uber ATG / Tempe Fatality

184 UK Post Office Horizon Scandal

195 Tesla Autopilot — Recurring Fatalities

LEN 3

47 cases

4 Mid Staffordshire NHS Foundation Trust

17 Spaced Education RCTs in Medical Training

24 Bristol Heart Babies Reform

32 Annual-Screening UI Redesign + CDS at University of Missouri Health Care

33 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

34 COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

37 DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback

39 TeamSTEPPS

49 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

55 Enrollment-Algorithm Yield Optimization Across U.S. Higher Education

57 GAO Online Program Manager Oversight Gap (GAO-22-104463)

58 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)

72 ASSISTments — National Replication and Long-Term Follow-Through

73 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension

74 Zhang/Scardamalia — Knowledge Building Across Three Cohorts

75 Chen et al. — Rural China AI Devices and the Equity-Direction Finding

84 Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive

85 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale

LEN 4

68 cases

LEN 5

84 cases

LEN 6

17 cases

LEN 7

102 cases

196 Fintech Lending Fairness Audit — When Including Race Reduces Disparity

197 Predictive Policing — PredPol

198 Launching the BRAIN Initiative — Governance of a Grand Challenge

LEN 8

90 cases

7 VA Wait-Time Scandal

11 California Nurse Staffing Ratios — Legislating a Capability Requirement

12 ACGME 80-Hour Resident Duty-Hour Reform

13 Implementation Science in Healthcare — The 17-Year Gap

28 Interprofessional Education and the Evidence Gap

32 Annual-Screening UI Redesign + CDS at University of Missouri Health Care

33 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

39 TeamSTEPPS

40 Team Science Training for Clinical and Translational Scientists

41 Translational-Workforce Training — Stated Goals Outrunning Named Practice

42 Australian Hospital-Pharmacy Technician Role Redesign

43 Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface

44 Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD

47 Algorithmic Bias in Automated Exam Proctoring

48 School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism

49 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

52 Purdue Course Signals — The Reverse-Causality Retention Claim

54 Summit Learning / Personalized Learning Rollout

55 Enrollment-Algorithm Yield Optimization Across U.S. Higher Education

57 GAO Online Program Manager Oversight Gap (GAO-22-104463)

58 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)

61 Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap

62 Gates Intensive Partnerships — Measurement Without a Coupled Lever

63 LAUSD iPad Program — The Common Core Technology Project

66 Growth-Mindset National Experiment — Heterogeneous Effects

LEN 9

30 cases

LEN 10

21 cases

APPENDIX · REFERENCES BY CASE

References, case by case.

Every reference cited in every case, organized by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds a **Retrieved from:** rule beneath it — a worksheet line for the reviewer to enter the canonical access path they used: a publisher URL, a DOI, an agency landing page, or a stable archival locator. The rules print blank by design, because the per-case verification is being carried out against this appendix rather than reported by it; the running status of that work is in `casebook/verification-log.md`. Future editions will print the locators as the review closes them out.

> **Style.** References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

CASE 1 Therac-25

1985 – 1987

1. N. G. Leveson & C. S. Turner, "An Investigation of the Therac-25 Accidents," IEEE Computer 26(7): 18–41 (1993). doi:10.1109/MC.1993.274940 — the removed hardware interlocks and the adapted software. — Retrieved from: ____________________

2. Leveson & Turner (1993) — the two software defects (data-entry race condition and counter overflow), the uninformative "Malfunction 54", overdoses up to 100×, six accidents, and at least three deaths. — Retrieved from: ____________________

3. Leveson & Turner (1993); N. G. Leveson, Safeware: System Safety and Computers (Addison-Wesley, 1995) — the manufacturer's denial and the systemic findings. — Retrieved from: ____________________

4. N. G. Leveson, Engineering a Safer World: Systems Thinking Applied to Safety (MIT Press, 2011) — why removing a safeguard requires explicitly reassigning its safety function. — Retrieved from: ____________________

5. The case's role in medical-device software regulation and software-safety practice (FDA software guidance; IEC 62304 lineage); see also Ethics Unwrapped, UT Austin. — Retrieved from: ____________________

CASE 2 EHR / CPOE Implementation

2005 – present

1. Health Information Technology for Economic and Clinical Health (HITECH) Act (2009) — the $30B EHR incentive program. — Retrieved from: ____________________

2. Y. Han et al., "Unexpected Increased Mortality After Implementation of a Commercially Sold CPOE System," Pediatrics 116(6): 1506–1512 (2005) — the disputed pediatric-ICU mortality result. — Retrieved from: ____________________

3. J. Howe, K. Adams, A. Z. Hettinger & R. Ratwani, "Electronic Health Record Usability Issues and Potential Contribution to Patient Harm," JAMA 319(12): 1276–1278 (2018) — 557 of 1.735M safety reports tied usability to possible harm. — Retrieved from: ____________________

4. D. Sittig & H. Singh, "Defining Health Information Technology-related Errors: New Developments Since To Err is Human," Archives of Internal Medicine 171(14): 1281–1284 (2011). — Retrieved from: ____________________

5. M. Grissinger, "The Absence of a Drug–Disease Interaction Alert Leads to a Child's Death," P&T 43(2): 71–72 (2018); F. Schulte & E. Fry, "Death by 1,000 Clicks," Fortune/KHN (2019). — Retrieved from: ____________________

6. Pew / MedStar / AMA, Ways to Improve Electronic Health Record Safety (2018); R. Wachter, The Digital Doctor (2015). — Retrieved from: ____________________

CASE 3 Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018

1. Ross & Swetlitz, "IBM's Watson supercomputer recommended 'unsafe and incorrect' cancer treatments, internal documents show," STAT (25 July 2018), and "IBM pitched its Watson supercomputer as a revolution in cancer care. It's nowhere close," STAT (5 Sept 2017) — the primary sources; investigative journalism drawing on leaked IBM internal documents. — Retrieved from: ____________________

2. Strickland (2019), "How IBM Watson Overpromised and Underdelivered on AI Health Care," IEEE Spectrum — independent retrospective synthesis of the public record. — Retrieved from: ____________________

3. Topol (2019), Deep Medicine, Basic Books — secondary academic situating of Watson within the broader clinical-AI delegation discourse. — Retrieved from: ____________________

4. Somashekhar et al. and the WFO lung/colorectal early-experience abstracts, J Clin Oncol 35 (2017), 15_suppl, presented at the ASCO Annual Meeting — the marketed concordance figures: 96% lung, 93% rectal, 81% colon. — Retrieved from: ____________________

5. v2 paired cases: TREWS (Case 20), Epic Sepsis Model (Case 5), SyRI (Case 189) — the AI-delegation typology. — Retrieved from: ____________________

CASE 4 Mid Staffordshire NHS Foundation Trust 2005 – 2009

1. R. Francis QC, Report of the Mid Staffordshire NHS Foundation Trust Public Inquiry (2013) — the staffing cuts and Foundation Trust targets. — Retrieved from: ____________________

2. Healthcare Commission, Investigation into Mid Staffordshire NHS Foundation Trust (2009) — ward conditions and excess mortality. — Retrieved from: ____________________

3. Francis QC (2013) — the three-volume, 1,782-page report and 290 recommendations. — Retrieved from: ____________________

4. Francis QC (2013) — "not just the Trust's Board but the system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm and from unacceptable, and in some cases inhumane, treatment" (quoted). — Retrieved from: ____________________

5. D. Berwick, A Promise to Learn — A Commitment to Act (National Advisory Group on the Safety of Patients in England, 2013). — Retrieved from: ____________________

6. K. Walshe & J. Higgins (2002) on NHS safety governance; cf. the VA wait-time scandal (Case 7). — Retrieved from: ____________________

CASE 5 Epic Sepsis Model 2017 – 2021

1. Wong et al. (2021), "External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients," JAMA Internal Medicine 181(8):1065–1070, doi:10.1001/jamainternmed.2021.2626. — Retrieved from: ____________________

2. Habib et al. (2021), commentary on Wong et al., JAMA Internal Medicine — on the implications for proprietary clinical AI. — Retrieved from: ____________________

3. FDA, Clinical Decision Support Software: Final Guidance (2022) — the post-Wong reframing of the EHR-embedded oversight question. — Retrieved from: ____________________

4. Ross, C. (2022), "Epic overhauls its sepsis algorithm," STAT News (Oct 2022) — the ESM v2 redesign and the shift to locally-trained models. — Retrieved from: ____________________

5. Adams et al. (2022), Nature Medicine — the paired positive case (20). — Retrieved from: ____________________

CASE 6 Racial Bias in Pain Assessment — The False-Belief Mechanism

2016

1. Hoffman, Trawalter, Axt, & Oliver (2016), "Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites," PNAS 113(16):4296–4301, doi:10.1073/pnas.1516047113. — Retrieved from: ____________________

2. Anderson, Green, & Payne (2009), "Racial and ethnic disparities in pain: causes and consequences of unequal care," Journal of Pain 10(12):1187–1204 — the population-level disparity. — Retrieved from: ____________________

3. Sabin & Greenwald (2012), "The influence of implicit bias on treatment recommendations for 4 common pediatric conditions," American Journal of Public Health — the diffuse-mechanism backdrop the Hoffman precision improves on. — Retrieved from: ____________________

4. Vyas, Eisenstein, & Jones (2020), NEJM — connecting race-in-clinical-algorithms to race-in-clinical-judgment. — Retrieved from: ____________________

5. Omiye, Lester, Spichak, Rotemberg, & Daneshjou (2023), "Large language models propagate race-based medicine," npj Digital Medicine 6:195 — commercial LLMs reproduce the Hoffman false beliefs. — Retrieved from: ____________________

CASE 7 VA Wait-Time Scandal

2014

1. VA Office of Inspector General, Report 14-02603-267 (2014) — secret waiting lists and falsified appointment data. — Retrieved from: ____________________

2. GAO Veterans Health Administration reports (2000–2019) — fifteen years of data-reliability warnings. — Retrieved from: ____________________

3. VA OIG reports (2005, 2007, 2008) — prior, unactioned findings. — Retrieved from: ____________________

4. D. Draper (GAO), House VA Committee testimony, GAO-19-687T (2019) — "given the high turnover among schedulers, it is important that VA remain vigilant about scheduler training" (quoted). — Retrieved from: ____________________

5. Veterans Access, Choice, and Accountability Act (2014) — the legislative response. — Retrieved from: ____________________

6. C. Argyris & D. Schön, Organizational Learning: A Theory of Action Perspective (1978); G. Bevan & C. Hood, "What's Measured Is What Matters," Public Administration 84(3): 517–538 (2006). — Retrieved from: ____________________

CASE 8 Medical Errors as Systemic Failure

1999 – present

1. Institute of Medicine, To Err Is Human: Building a Safer Health System (1999); Crossing the Quality Chasm (2001); Improving Diagnosis in Health Care (2015) — the field-defining trilogy and the 44,000–98,000 estimate; the systems framing. — Retrieved from: ____________________

2. Makary, M. & Daniel, M. (2016), "Medical error — the third leading cause of death in the U.S.," BMJ 353:i2139 — the 250,000 estimate, the quoted framing, the extrapolation from four studies published since the IOM report (Health-Grades 2004; OIG 2010; Landrigan 2010; Classen 2011). — Retrieved from: ____________________

3. Shojania, K. & Dixon-Woods, M. (2017), "Estimating deaths due to medical error: the ongoing controversy and why it matters," BMJ Quality & Safety 26(5):423–428; with Bates, D. W. & Singh, H. (2018), Health Affairs 37(11):1736–1743 — methodological contestation of the Makary extrapolation and CDC-ranking comparison. — Retrieved from: ____________________

4. Makary & Daniel (2016), BMJ — death certificates do not capture medical error as a cause; ICD billing taxonomy as the structural reason. — Retrieved from: ____________________

5. Bates, D. W., Levine, D. M., Salmasian, H., et al. (2023), "The Safety of Inpatient Health Care," NEJM 388(2):142–153 — eleven-hospital Massachusetts cohort of 2,809 admissions; at least one adverse event in 23.6% of admissions, 25% of those preventable; persistence of harm at scale. — Retrieved from: ____________________

6. Agency for Healthcare Research and Quality, National Healthcare Quality and Disparities Reports (annual); CDC WONDER ICD-coded mortality data — institutional context for the missing national active-surveillance instrument. — Retrieved from: ____________________

CASE 9 Vioxx Withdrawal 1999 – 2004

1. Bombardier, C. et al. (2000), "Comparison of upper gastrointestinal toxicity of rofecoxib and naproxen in patients with rheumatoid arthritis," VIGOR trial, NEJM 343(21):1520–1528 — the five-fold myocardial-infarction signal and the naproxen-protective hypothesis. — Retrieved from: ____________________

2. Bresalier, R. et al. (2005), "Cardiovascular events associated with rofecoxib in a colorectal adenoma chemoprevention trial," APPROVe, NEJM 352(11):1092–1102 — placebo-controlled confirmation of cardiovascular risk; early trial termination. — Retrieved from: ____________________

3. Graham, D. J. et al. (2005), "Risk of acute myocardial infarction and sudden cardiac death in patients treated with COX-2 selective and non-selective NSAIDs: nested case-control study," Lancet 365(9458):475–481 — Kaiser Permanente population-level cardiovascular risk analysis. — Retrieved from: ____________________

4. U.S. Senate Committee on Finance, hearings on Vioxx and FDA's drug-safety system (November 18, 2004) — Graham testimony; "88,000 to 139,000 Americans" estimate of excess cardiovascular events; described FDA management pressure. — Retrieved from: ____________________

5. Ross, J. S., Hill, K. P., Egilman, D. S., Krumholz, H. M. (2008), "Guest authorship and ghostwriting in publications related to rofecoxib," JAMA 299(15):1800–1812 — guest-authorship and ghostwriting documentation from Merck litigation discovery. — Retrieved from: ____________________

6. FDA Amendments Act of 2007 (P.L. 110-85) and FDA Sentinel Initiative documentation (2008–present) — REMS authority, post-market study requirements, and active distributed-data post-market surveillance. — Retrieved from: ____________________

CASE 11 California Nurse Staffing Ratios — Legislating a Capability Requirement 2004 – 2010

1. Aiken, Sloane, Cimiotti, Clarke, Flynn, Seago, Spetz, & Smith (2010), "Implications of the California Nurse Staffing Mandate for Other States," Health Services Research 45(4):904–921, doi:10.1111/j.1475-6773.2010.01114.x. — Retrieved from: ____________________

2. Aiken, Clarke, Sloane, Sochalski, & Silber (2002), "Hospital nurse staffing and patient mortality, nurse burnout, and job dissatisfaction," JAMA 288(16):1987–1993 — the workload-mortality relationship the 2010 modeling rests on. — Retrieved from: ____________________

3. California Department of Health Services (1999–2004), AB 394 regulatory implementation documentation. — Retrieved from: ____________________

4. Spetz, Chapman, Herrera, Kaiser, Seago, & Dower (2009), "Assessing the impact of California's nurse staffing ratios on hospitals and patient care," California HealthCare Foundation — implementation-period analysis. — Retrieved from: ____________________

CASE 12 ACGME 80-Hour Resident Duty-Hour Reform 2003–2017

1. B. H. Lerner, The Libby Zion Case and the Reform of Medical Education (2006); and the 1989 New York State (Bell Commission) duty-hour regulations — the origin of the reform. — Retrieved from: ____________________

2. Accreditation Council for Graduate Medical Education, Common Program Requirements (2003 and 2011 revisions) — the 80-hour weekly cap and the 16-hour first-year shift limit. — Retrieved from: ____________________

3. D. A. Asch, K. Y. Bilimoria & S. V. Desai, "Resident Duty Hours and Medical Education Policy — Raising the Evidence Bar," NEJM 376: 1704–1706 (2017); Institute of Medicine (Ulmer, Wolman & Johns, eds.), Resident Duty Hours: Enhancing Sleep, Supervision, and Safety (2008) — hand-offs and continuity trade-offs. — Retrieved from: ____________________

4. Bilimoria, Chung, Hedges et al., "National Cluster-Randomized Trial of Duty-Hour Flexibility in Surgical Training" (FIRST), NEJM 374: 713–727 (2016). doi:10.1056/NEJMoa1515724. — Retrieved from: ____________________

5. Silber, Bellini, Shea et al., "Patient Safety Outcomes under Flexible and Standard Resident Duty-Hour Rules" (iCOMPARE), NEJM 380: 905–914 (2019). doi:10.1056/NEJMoa1810642. — Retrieved from: ____________________

6. ACGME, Common Program Requirements (2017 revision) — relaxation of the 16-hour first-year shift limit. — Retrieved from: ____________________

7. P. Pronovost et al., Keystone ICU intervention, NEJM 355: 2725–2732 (2006), doi:10.1056/NEJMoa061115 A. B. Haynes et al., surgical safety checklist, NEJM 360: 491–499 (2009), doi:10.1056/NEJMsa0810119 — integrated interventions that engineered the surrounding architecture. — Retrieved from: ____________________

CASE 13 Implementation Science in Healthcare — The 17-Year Gap

ongoing

1. E. A. Balas & S. A. Boren (2000), "Managing clinical knowledge for health care improvement," Yearbook of Medical Informatics — the nine-service adoption analysis behind the seventeen-year figure, and the pipeline-attrition derivation behind the 14 percent. — Retrieved from: ____________________

2. Z. S. Morris, S. Wooding & J. Grant, "The answer is 17 years, what is the question: understanding time lags in translational research," J. Royal Society of Medicine 104(12):510–520 (2011) (quoted). — Retrieved from: ____________________

3. D. Fixsen et al., Implementation Research: A Synthesis of the Literature (2005) — the Active Implementation Frameworks. — Retrieved from: ____________________

4. G. Aarons et al. (2011), the EPIS framework; L. Damschroder et al. (2009), CFIR. — Retrieved from: ____________________

5. Cf. medical error as systemic failure (Case 8); Goodell & Kolodner, Learning Engineering Toolkit (2022). — Retrieved from: ____________________

CASE 14 Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA

2009 – 2020s

1. Barsuk, Cohen, Feinglass, McGaghie, & Wayne (2009), "Use of simulation-based education to reduce catheter-related bloodstream infections," Archives of Internal Medicine 169(15):1420–1423, doi:10.1001/archinternmed.2009.215. — Retrieved from: ____________________

2. Cohen, Feinglass, Barsuk, Barnard, O'Donnell, McGaghie, & Wayne (2010), "Cost savings from reduced catheter-related bloodstream infection after simulation-based education for residents in a medical intensive care unit," Simulation in Healthcare 5(2):98–102, doi:10.1097/SIH.0b013e3181bc8304. — Retrieved from: ____________________

3. Barsuk, McGaghie, Cohen, Balachandran, & Wayne (2009), "Use of simulation-based mastery learning to improve the quality of central venous catheter placement in a medical intensive care unit," Journal of Hospital Medicine 4(7):397–403. — Retrieved from: ____________________

4. McGaghie, Issenberg, Cohen, Barsuk, & Wayne (2011), "Does simulation-based medical education with deliberate practice yield better results than traditional clinical medical education? A meta-analytic comparative review of the evidence," Academic Medicine 86(6):706–711. — Retrieved from: ____________________

5. Barsuk, Cohen, Nguyen, Mitra, O'Hara, Okuda, Feinglass, Cameron, McGaghie, & Wayne (2016), "Attending physician adherence to a 29-component central venous catheter bundle checklist during simulated procedures," Critical Care Medicine 44(10):1871–1881 — the published record of the train-the-trainer dissemination across 58 Veterans Affairs Medical Centers. — Retrieved from: ____________________

CASE 15 I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer 2014

1. Starmer, A. J., Spector, N. D., Srivastava, R., West, D. C., Rosenbluth, G., Allen, A. D., Noble, E. L., Tse, L. L., Dalal, A. K., Keohane, C. A., Lipsitz, S. R., Rothschild, J. M., Wien, M. F., Yoon, C. S., Zigmont, K. R., Wilson, K. M., O'Toole, J. K., Solan, L. G., Aylor, M., Bismilla, Z., Coffey, M., Mahant, S., Blankenburg, R. L., Destino, L. A., Everhart, J. L., Patel, S. J., Bale, J. F., Spackman, J. B., Stevenson, A. T., Calaman, S., Cole, F. S., Balmer, D. F., Hepps, J. H., Lopreiato, J. O., Yu, C. E., Sectish, T. C., & Landrigan, C. P. (2014). Changes in medical errors after implementation of a handoff program. New England Journal of Medicine, 371(19):1803–1812. doi:10.1056/NEJMsa1405556 — the case's primary evaluation. — Retrieved from: ____________________

2. Colligan, L., Brick, D., & Patterson, E. S. (2015). Changes in medical errors with a handoff program. New England Journal of Medicine, 372(5):490 — the correspondence cautioning against implementing the mnemonic alone; authors' reply at 372(5):490–491. — Retrieved from: ____________________

3. Sectish, T. C., Starmer, A. J., Landrigan, C. P., & Spector, N. D. (2010). Establishing a multisite education and research project requires leadership, expertise, collaboration, and an important aim. Pediatrics, 126(4):619–622 — the I-PASS Study Group methodology and design rationale. — Retrieved from: ____________________

4. Cohen, M. D., & Hilligoss, P. B. (2010). The published literature on handoffs in hospitals: deficiencies identified in an extensive review. Quality and Safety in Health Care — the broader handoff-failure-mode literature. — Retrieved from: ____________________

CASE 16 NCSBN National Simulation Study — Licensing the 50% Substitution Rule 2014

1. Hayden, J. K., Smiley, R. A., Alexander, M., Kardong-Edgren, S., & Jeffries, P. R. (2014). The NCSBN National Simulation Study: A longitudinal, randomized, controlled study replacing clinical hours with simulation in prelicensure nursing education. Journal of Nursing Regulation, 5(2 Suppl):S1–S64. https://www.journalofnursingregulation.com/article/s2155-8256(15)30062-4/fulltext — the case's primary study. — Retrieved from: ____________________

2. Smiley, R. A., & Martin, B. (2023). Simulation in Nursing Education: Advancements in Regulation, 2014–2022. Journal of Nursing Regulation. doi:10.1016/S2155-8256(23)00086-8 — the 2023 follow-up documenting the regulatory propagation. — Retrieved from: ____________________

3. Jeffries, P. R. (2012). Simulation in Nursing Education: From Conceptualization to Evaluation (2nd ed.). National League for Nursing — the simulation-quality framework the NCSBN study's quality conditions rest on. — Retrieved from: ____________________

4. INACSL Standards Committee (2016). INACSL Standards of Best Practice: Simulation. Clinical Simulation in Nursing, 12(S) — the simulation-practice standards downstream programs adopt as the analog of the NCSBN quality conditions. — Retrieved from: ____________________

CASE 17 Spaced Education RCTs in Medical Training 2007 – 2009

1. Kerfoot, B. P., Baker, H. E., Koch, M. O., Connelly, D., Joseph, D. B., & Ritchey, M. L. (2007). Randomized, controlled trial of spaced education to urology residents in the United States and Canada. Journal of Urology, 177(4):1481–1487. doi:10.1016/j.juro.2006.11.074 — the case's primary RCT. — Retrieved from: ____________________

2. Kerfoot, B. P. (2009). Learning benefits of on-line spaced education persist for 2 years. Journal of Urology, 181(6):2671 — the two-year persistence follow-up. — Retrieved from: ____________________

3. Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. Psychological Bulletin, 132(3):354–380 — the basic-literature spacing-effect review the workplace-L&D RCT sits against. — Retrieved from: ____________________

4. Settles, B., & Meeder, B. (2016). A trainable spaced repetition model for language learning. Proceedings of ACL 2016, 1848–1858 — Duolingo half-life regression (Case 69), the operational deployment companion. — Retrieved from: ____________________

CASE 18 PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption 2023

1. Kiguli-Malwadde, E., Forster, M., Eliaz, A., et al. (2023). "Comparing in-person, blended and virtual training interventions; a real-world evaluation of HIV capacity building programs in 16 countries in sub-Saharan Africa," PLOS Global Public Health 3(7):e0001654, doi:10.1371/journal.pgph.0001654 — the published version of medRxiv 2023.02.08.23285641. — Retrieved from: ____________________

2. The evaluated program is STRIPE HIV, led by the African Forum for Research and Education in Health (AFREhealth) and the University of California, San Francisco, and funded by HRSA through PEPFAR — named in the study's program description and funding statement. — Retrieved from: ____________________

3. Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the chain-of-evidence framework the L1–L2 limitation references (paired Case 79). — Retrieved from: ____________________

4. Blume et al. (2010), Journal of Management 36(4):1065–1105 — the meta-analytic transfer finding the L3 question references (paired Case 65). — Retrieved from: ____________________

CASE 19 Keystone ICU / Pronovost Checklist 2004 – present

1. Pronovost, P. et al. (2006), "An Intervention to Decrease Catheter-Related Bloodstream Infections in the ICU," NEJM 355 — the trial and the near-zero result. — Retrieved from: ____________________

2. Pronovost & Vohr (2010), Safe Patients, Smart Hospitals — the checklist-plus-nurse-authority pairing (paraphrased). — Retrieved from: ____________________

3. Pronovost, P. et al. (2016), "Sustaining Reductions in CLABSI in Michigan ICUs: A 10-Year Analysis," Am J Med Qual 31(3):197–202 — the effect sustained through 2013. — Retrieved from: ____________________

4. Agency for Healthcare Research and Quality, CUSP toolkit — dissemination across states. — Retrieved from: ____________________

5. Bosk, C. et al. (2009), "Reality check for checklists," The Lancet — the critique: the Keystone result cannot be attributed cleanly to the checklist, and the bundle's active ingredient is not established. — Retrieved from: ____________________

CASE 20 TREWS / Sepsis Watch 2018 – 2022

1. Adams et al. (2022), "Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis," Nature Medicine 28(7):1455–1460, doi:10.1038/s41591-022-01894-0. — Retrieved from: ____________________

2. Henry et al. (2022), "Factors driving provider adoption of the TREWS machine learning-based early warning system and its effects on sepsis treatment timing," Nature Medicine 28(7):1447–1454, doi:10.1038/s41591-022-01895-z. — Retrieved from: ____________________

3. Sendak et al. (2020), "Real-world integration of a sepsis deep learning technology into routine clinical care: implementation study," JMIR Medical Informatics 8(7):e15182 (Sepsis Watch implementation). — Retrieved from: ____________________

4. Wong et al. (2021), "External Validation of a Widely Implemented Proprietary Sepsis Prediction Model," JAMA Internal Medicine 181(8):1065–1070 — the foil case (102). — Retrieved from: ____________________

CASE 21 Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff 2024 – 2025

1. Treloar, E., Herath, M., Altree, M., Potter, S., Penhall, M., Walsh, D., Kennedy, L., Bruening, M., Edwards, S., Ey, J. D., Bradshaw, E. L., & Maddern, G. J. (2025), "A Simple Solution for a Complex Problem: The 'Sterile Cockpit' to Improve Ward Rounds," World Journal of Surgery 49(10):2769–2776, doi:10.1002/wjs.70074, PMID:40930848, PMCID:PMC12515027 — the cited adaptation study. — Retrieved from: ____________________

2. Federal Aviation Administration, 14 CFR § 121.542 (codified 1981) — origin of the aviation sterile-cockpit rule. — Retrieved from: ____________________

3. Starmer, A. J. et al. (2014), "Changes in medical errors after implementation of a handoff program," New England Journal of Medicine 371(19):1803–1812 — I-PASS handoff intervention; structural cousin in the structured-information half of handoff capability. — Retrieved from: ____________________

4. Broom, M. A., Capek, A. L., Carachi, P., Akeroyd, M. A., & Hilditch, G. (2011), "Critical phase distractions in anaesthesia and the sterile cockpit concept," Anaesthesia 66(3):175–179 — prior anesthesia-domain sterile-cockpit adaptation establishing the cross-domain transfer pattern. — Retrieved from: ____________________

CASE 22 ChatGPT in Healthcare — Hallucination Cases

2023 – present

1. Morath, B., Chiriac, U., Jaszkowski, E., et al. (2024). "Performance and risks of ChatGPT used in drug information: an exploratory real-world analysis," European Journal of Hospital Pharmacy 31(6):491–497 — only 13 of 50 answers correct and complete enough to act on without risk of harm. — Retrieved from: ____________________

2. Sallam, M. (2023). "ChatGPT Utility in Healthcare Education, Research, and Practice: Systematic Review on the Promising Perspectives and Valid Concerns," Healthcare 11(6):887 — 60 records reviewed; benefits cited in 85%, with the accuracy and citation concerns alongside. — Retrieved from: ____________________

3. Bhattacharyya, M., Miller, V. M., Bhattacharyya, D., & Miller, L. E. (2023). "High Rates of Fabricated and Inaccurate References in ChatGPT-Generated Medical Content," Cureus 15(5):e39238 — 47% of 115 generated references fabricated. — Retrieved from: ____________________

4. World Health Organization (2024). Ethics and Governance of Artificial Intelligence for Health: Guidance on Large Multi-Modal Models, 18 January 2024 — names automation bias by health care professionals and the risk of false or inaccurate output. — Retrieved from: ____________________

5. Triplett, S., Ness-Engle, G. L., & Behnen, E. M. (2025). "A comparison of drug information question responses by a drug information center and by ChatGPT," American Journal of Health-System Pharmacy 82(8):448–460, doi:10.1093/ajhp/zxae316 — pharmacist-rated overall accuracy of 50% across the ChatGPT responses. — Retrieved from: ____________________

6. Wachter, R. M., & Brynjolfsson, E. (2024). "Will Generative Artificial Intelligence Deliver on Its Promise in Health Care?" JAMA 331(1):65–69, doi:10.1001/jama.2023.25054 — genAI adoption and the productivity paradox in health care. — Retrieved from: ____________________

CASE 23 WHO Surgical Safety Checklist

2008 – present

1. Haynes, A. et al. (2009), "A Surgical Safety Checklist to Reduce Morbidity and Mortality," NEJM — the 1.5%→0.8% result and major-complication reduction. — Retrieved from: ____________________

2. WHO Safe Surgery Saves Lives campaign documentation — the nineteen-item checklist and the three junctures. — Retrieved from: ____________________

3. Gawande, A. (2009), The Checklist Manifesto — the pause as the active mechanism. — Retrieved from: ____________________

4. Urbach, D. et al. (2014), "Introduction of Surgical Safety Checklists in Ontario, Canada," NEJM — null mortality result after mandate. — Retrieved from: ____________________

5. Bosk, C. et al. (2009), "Reality check for checklists," The Lancet — implementation fidelity. — Retrieved from: ____________________

6. Buissonniere, M. (2020), Checking In on the Checklist, Ariadne Labs and Lifebox — the decade review: facilities in 70% of the world's countries attest to using the checklist, with adoption far lower in low-income settings. — Retrieved from: ____________________

CASE 24 Bristol Heart Babies Reform

1984 – present

1. Kennedy, I. (2001), Learning from Bristol: The Report of the Public Inquiry into Children's Heart Surgery at the Bristol Royal Infirmary 1984–1995 — the inquiry and its recommendations, paraphrased; the pull quote is verbatim from paragraph 18 of the Final Report Summary, and differs in wording from Recommendation 155, which carries the same requirement. — Retrieved from: ____________________

2. Society for Cardiothoracic Surgery in GB & Ireland, national outcomes reports — the registry and surgeon-level publication. — Retrieved from: ____________________

3. Berwick, D. (2013), A Promise to Learn — A Commitment to Act — the broader NHS-safety reform. — Retrieved from: ____________________

4. Sherlaw-Johnson, C., Gallivan, S., Treasure, T. & Nashef, S. A. M. (2004), "Computer tools to assist the monitoring of outcomes in surgery," European Journal of Cardio-Thoracic Surgery 26(5):1032–1036 — risk-adjusted outcome monitoring (variable life-adjusted display) as it developed in UK cardiac surgery. — Retrieved from: ____________________

5. NHS national clinical audit and registry documentation — extension across specialties. — Retrieved from: ____________________

CASE 25 Removing the Race Coefficient from eGFR

2021

1. Delgado et al. (2021), "A Unifying Approach for GFR Estimation: Recommendations of the NKF-ASN Task Force on Reassessing the Inclusion of Race in Diagnosing Kidney Disease," American Journal of Kidney Diseases 79(2):268–288 (published online 2021; in print vol. 79, 2022), doi:10.1053/j.ajkd.2021.08.003. Cited by online-first year, the year of the Task Force recommendation. — Retrieved from: ____________________

2. Inker et al. (2021), "New Creatinine- and Cystatin C-Based Equations to Estimate GFR without Race," New England Journal of Medicine 385(19):1737–1749, doi:10.1056/NEJMoa2102953. — Retrieved from: ____________________

3. OPTN/UNOS (2022–2023), "Kidney waiting-time modifications for candidates affected by race-inclusive eGFR calculations" — the mandated backdating of Black candidates' accrued waiting time (effective January 2023). — Retrieved from: ____________________

4. Genzen, J. R., Souers, R. J., Pearson, L. N., et al. (2022), "Reported Awareness and Adoption of 2021 Estimated Glomerular Filtration Rate Equations Among US Clinical Laboratories, March 2022," JAMA 328(20):2060–2062, doi:10.1001/jama.2022.15404 — the College of American Pathologists survey: 1,124 of 3,708 responding U.S. laboratories (30.3%) had adopted the 2021 CKD-EPI creatinine equation. — Retrieved from: ____________________

5. Eneanya, Yang, & Reese (2019), "Reconsidering the Consequences of Using Race to Estimate Kidney Function," JAMA 322(2):113–114 — the equity argument that motivated the revision. — Retrieved from: ____________________

6. Vyas, Eisenstein, & Jones (2020), "Hidden in Plain Sight — Reconsidering the Use of Race Correction in Clinical Algorithms," NEJM 383(9):874–882 — broader race-in-algorithms survey. — Retrieved from: ____________________

CASE 26 Pulse Oximetry Across Skin Tones

1990 – 2020 – 2025

1. Sjoding, Dickson, Iwashyna, Gay, & Valley (2020), "Racial Bias in Pulse Oximetry Measurement," New England Journal of Medicine 383(25):2477–2478, doi:10.1056/NEJMc2029240. — Retrieved from: ____________________

2. Jubran & Tobin (1990), "Reliability of Pulse Oximetry in Titrating Supplemental Oxygen Therapy in Ventilator-Dependent Patients," Chest 97(6):1420–1425 — original finding, published thirty years earlier. — Retrieved from: ____________________

3. FDA (2025), "Pulse Oximeters for Medical Purposes — Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations: Draft Guidance for Industry and Food and Drug Administration Staff," issued January 7 2025, Docket No. FDA-2023-N-4976; Federal Register notice 2024-31540, published January 7 2025 — the regulatory corrective-action artifact, language may evolve in final. — Retrieved from: ____________________

4. Fawzy et al. (2022), "Racial and Ethnic Discrepancy in Pulse Oximetry and Delayed Identification of Treatment Eligibility Among Patients With COVID-19," JAMA Internal Medicine — downstream effect during the pandemic. — Retrieved from: ____________________

CASE 27 Anesthesia Monitoring Standards and the APSF — The Mortality Decline 1986 – present

1. Eichhorn, Cooper, Cullen, Maier, Philip, & Seeman (1986), "Standards for Patient Monitoring During Anesthesia at Harvard Medical School," JAMA 256(8):1017–1020. — Retrieved from: ____________________

2. Eichhorn (1989), "Prevention of intraoperative anesthesia accidents and related severe injury through safety monitoring," Anesthesiology 70(4):572–577. — Retrieved from: ____________________

3. Anesthesia Patient Safety Foundation (1985 – present), founding documents and the APSF Newsletter — institutional-history record of the broader change effort. — Retrieved from: ____________________

4. American Society of Anesthesiologists, "Standards for Basic Anesthetic Monitoring," approved 21 October 1986 and amended since — the original standard and the amendment sequence that made pulse oximetry required (1989), expired-CO2 verification of tube placement required (early 1990s), and continual capnography through a general anesthetic required (effective July 1999). — Retrieved from: ____________________

5. Braz, L. G., Módolo, N. S. P., do Nascimento, P. Jr., et al. (2009), "Mortality in anesthesia: a systematic review," Clinics (São Paulo) 64(10):999–1006 — 33 studies, 1954–2007; anaesthesia-related mortality below one death per 10,000 anaesthetics in the final two decades. — Retrieved from: ____________________

6. Sjoding et al. (2020), "Racial Bias in Pulse Oximetry Measurement," NEJM 383(25):2477–2478 — the racial-bias edge case the standard still carries (cross-reference Case 26). — Retrieved from: ____________________

CASE 28 Interprofessional Education and the Evidence Gap 2013 – 2015

1. Reeves, Perrier, Goldman, Freeth, & Zwarenstein (2013), "Interprofessional education: effects on professional practice and healthcare outcomes (update)," Cochrane Database of Systematic Reviews, doi:10.1002/14651858.CD002213.pub3. — Retrieved from: ____________________

2. Institute of Medicine (2015), Measuring the Impact of Interprofessional Education on Collaborative Practice and Patient Outcomes, National Academies Press, doi:10.17226/21726, NCBI Bookshelf NBK338360. — Retrieved from: ____________________

3. WHO (2010), Framework for Action on Interprofessional Education and Collaborative Practice — the international policy backdrop. — Retrieved from: ____________________

4. v2 paired cases: Team-science training (Case 40), Implementation-science training (Case 41). — Retrieved from: ____________________

CASE 29 Bar-Code Medication Administration — Cue at the Point of Care 2010

1. Poon, E. G., Keohane, C. A., Yoon, C. S., Ditmore, M., Bane, A., Levtzion-Korach, O., Moniz, T., Rothschild, J. M., Kachalia, A. B., Hayes, J., Churchill, W. W., Lipsitz, S., Whittemore, A. D., Bates, D. W., & Gandhi, T. K. (2010). Effect of bar-code technology on the safety of medication administration. New England Journal of Medicine, 362(18):1698–1707. doi:10.1056/NEJMsa0907115 — the case's primary evaluation. — Retrieved from: ____________________

2. Bonkowski, J., Carnes, C., Melucci, J., Mirtallo, J., Prier, B., Reichert, E., Moffatt-Bruce, S., & Weber, R. J. (2013). Effect of barcode-assisted medication administration on emergency department medication errors. Academic Emergency Medicine, 20(8):801–806 — adjacent transfer evidence. — Retrieved from: ____________________

3. Thompson, K. M., Swanson, K. M., Cox, D. L., Kirchner, R. B., Russell, J. J., Wermers, R. A., Storlie, C. B., Johnson, M. G., & Naessens, J. M. (2018), "Implementation of Bar-Code Medication Administration to Reduce Patient Harm," Mayo Clinic Proceedings: Innovations, Quality & Outcomes 2(4):342–351, doi:10.1016/j.mayocpiqo.2018.09.001, PMID:30560236, PMCID:PMC6257885 — later single-site rollout reporting a 55.4% reduction in harmful medication

errors (0.65 to 0.29 per 100,000 medications), by interrupted time series across fifty inpatient units. — Retrieved from: ____________________

4. Institute for Safe Medication Practices (2019), Guidelines for Safe Electronic Communication of Medication Information — the display and communication standards that bar-code scanning systems and eMARs are built against. — Retrieved from: ____________________

CASE 30 Surgical Skill Video Peer-Rating Predicts Complications 2013

1. Birkmeyer, J. D., Finks, J. F., O'Reilly, A., Oerline, M., Carlin, A. M., Nunn, A. R., Dimick, J., Banerjee, M., & Birkmeyer, N. J. O., for the Michigan Bariatric Surgery Collaborative. (2013). Surgical skill and complication rates after bariatric surgery. New England Journal of Medicine, 369(15):1434–1442. doi:10.1056/NEJMsa1300625 — the case's primary evaluation. — Retrieved from: ____________________

2. Birkmeyer, J. D., Stukel, T. A., Siewers, A. E., Goodney, P. P., Wennberg, D. E., & Lucas, F. L. (2003). Surgeon volume and operative mortality in the United States. New England Journal of Medicine, 349(22):2117–2127 — the volume-outcome literature the confound rests against. — Retrieved from: ____________________

3. Vassiliou, M. C., Feldman, L. S., Andrew, C. G., et al. (2005). A global assessment tool for evaluation of intraoperative laparoscopic skills. American Journal of Surgery, 190(1):107–113 — the laparoscopic skill-rating literature this video assessment sits in. — Retrieved from: ____________________

4. Jacobs, D. O. (2013). Cut well, sew well, do well? New England Journal of Medicine, 369(15):1466–1467. doi:10.1056/NEJMe1309785 — the accompanying editorial. — Retrieved from: ____________________

CASE 31 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units 2006 – 2016

1. Vedula, Malpani, Tao, Chen, Gao, Poddar, Ahmidi, Paxton, Vidal, Khudanpur, Hager, & Chen (2016), "Analysis of the Structure of Surgical Activity for a Suturing and Knot-Tying Task," PLOS One 11(3):e0149174, doi:10.1371/journal.pone.0149174. — Retrieved from: ____________________

2. Gao, Vedula, Reiley, Ahmidi, Varadarajan, Lin, Tao, Zappella, Bejar, Yuh, Chen, Vidal, Khudanpur, & Hager (2014), "JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS): A Surgical Activity Dataset for Human Motion Modeling," MICCAI Workshop — JIGSAWS dataset release paper. — Retrieved from: ____________________

3. Reiley, Lin, Yuh, & Hager (2011), "Review of methods for objective surgical skill evaluation," Surgical Endoscopy 25(2):356–366 — situates the decomposition within the broader skill-assessment literature. — Retrieved from: ____________________

4. Birkmeyer, Finks, O'Reilly, et al. (2013), NEJM — the paired skill-matters study (Case 30). — Retrieved from: ____________________

CASE 32 Annual-Screening UI Redesign + CDS at University of Missouri Health Care 2020

1. Pierce RP, Eskridge BR, Rehard L, Ross B, Day MA, Belden JL (2020), "The Effect of Electronic Health Record Usability Redesign on Annual Screening Rates in an Ambulatory Setting," Applied Clinical Informatics 11(4):580–588, doi:10.1055/s-0040-1715828. A HIMSS case study of the project is the secondary write-up. — Retrieved from: ____________________

2. Hussain MI, Reynolds TL, Zheng K (2019), "Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review," JAMIA 26(10):1141–1149, doi:10.1093/jamia/ocz095 — adjacent systematic-review evidence on interaction-design redesign. — Retrieved from: ____________________

3. Office of the National Coordinator for Health IT, SAFER Guides on CDS design — practitioner-tier guidance the redesign instantiates. — Retrieved from: ____________________

4. Middleton et al. (2013), "Enhancing patient safety and quality of care by improving the usability of electronic health record systems," JAMIA 20(e1):e2–e8 — the framing peer-reviewed paper on EHR-usability-as-safety. — Retrieved from: ____________________

CASE 33 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

2019 – 2024

1. Hussain, M. I., Reynolds, T. L., & Zheng, K. (2019), "Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review," JAMIA 26(10):1141–1149, doi:10.1093/jamia/ocz095 — thirty-nine studies; interruptive modals accepted least often; role tailoring the only alternative that appeared to increase prescriber acceptance. — Retrieved from: ____________________

2. Van Dort, B. A., Zheng, W. Y., Sundar, V., & Baysari, M. T. (2021), "Optimizing clinical decision support alerts in electronic medical records: a systematic review of reported strategies adopted by hospitals," JAMIA 28(1):177–183, doi:10.1093/jamia/ocaa279 — eight published optimization programs; six evaluated their changes and five of those reported a reduction in alert rate. — Retrieved from: ____________________

3. Patterson E (2024), "Navigating Alert Fatigue: A Case Study in Electronic Health Record Alert Design Optimization," Studies in Health Technology and Informatics 315:447–451, doi:10.3233/SHTI240188 — nursing-workflow QI redesign of four high-firing alerts; all four removed, 877 unactionable disruptive nursing hours released. — Retrieved from: ____________________

4. Fallon A, Haralambides K, Mazzillo J, Gleber C (2024), "Addressing Alert Fatigue by Replacing a Burdensome Interruptive Alert with Passive Clinical Decision Support," Applied Clinical Informatics 15(1):101–110, doi:10.1055/a-2226-8144 — 80% cut in alert firings; COVID precautions ordered rose from 23% to 61%. — Retrieved from: ____________________

5. Office of the National Coordinator for Health IT, SAFER Guides on CDS — practitioner-tier guidance the redesigns instantiate. — Retrieved from: ____________________

6. Ancker et al. (2017), "Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system," BMC Medical Informatics and Decision Making 17:36 — adjacent measurement evidence. — Retrieved from: ____________________

CASE 34 COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

2022 – 2024

1. Boussina, A., Shashikumar, S. P., Malhotra, A., Owens, R. L., El-Kareh, R., Longhurst, C. A., Quintero, K., et al. (2024), "Impact of a deep learning sepsis prediction model on quality of care and survival," NPJ Digital Medicine 7:14, doi:10.1038/s41746-023-00986-6. — Retrieved from: ____________________

2. Shashikumar, S. P., Wardi, G., Malhotra, A., & Nemati, S. (2021), "Artificial intelligence sepsis prediction algorithm learns to say 'I don't know,'" NPJ Digital Medicine 4:134 — the methodological-foundation paper for the conformal-prediction abstention structure. — Retrieved from: ____________________

3. Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., et al. (2021), "External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients," JAMA Internal Medicine 181(8):1065–1070 — the external-validation paper on Epic Sepsis (Case 5). — Retrieved from: ____________________

4. Adams, R., Henry, K. E., Sridharan, A., Soleimani, H., Zhan, A., Rawat, N., Johnson, L., et al. (2022), "Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis," Nature Medicine 28:1455–1460 — the prospective-positive TREWS deployment paper (Case 20). — Retrieved from: ____________________

CASE 35 Radiology AI Miscalibration

2018 – present

1. Larrazabal, A. J., Nieto, N., Peterson, V., Milone, D. H., & Ferrante, E. (2020), "Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis," PNAS 117(23):12592–12594 — AUC drops for under-represented genders on NIH ChestX-ray14 and CheXpert, across three network architectures. — Retrieved from: ____________________

2. Seyyed-Kalantari, L., Zhang, H., McDermott, M. B. A., Chen, I. Y., & Ghassemi, M. (2021), "Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations," Nature Medicine 27:2176–2182 — disparities across Black, Hispanic, female, and lower-socioeconomic subgroups; underdiagnosis highest for intersectional subpopulations such as Hispanic female patients. — Retrieved from: ____________________

3. Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019), "Dissecting racial bias in an algorithm used to manage the health of populations," Science 366(6464):447–453 — proxy-label bias in a care-management algorithm; 200 million Americans/year are subject to tools of this class. — Retrieved from: ____________________

4. Wachter, R. M. & Brynjolfsson, E. (2024), "Will Generative Artificial Intelligence Deliver on Its Promise in Health Care?" JAMA 331(1):65–69, doi:10.1001/jama.2023.25054 — generative-AI extension of the proxy-and-population problem. — Retrieved from: ____________________

5. FDA, "Proposed Regulatory Framework for Modifications to AI/ML-Based Software as a Medical Device" (2019); FDA draft guidance on AI/ML-Based SaMD (2025), with Predetermined Change Control Plan and Good Machine Learning Practice principles — the regulatory trajectory; absence of mandatory demographic stratification at clearance and post-market monitoring of in-use performance. — Retrieved from: ____________________

CASE 36 AlphaFold — Protein Structure Prediction 2020 – present

1. Jumper et al. (2021), "Highly accurate protein structure prediction with AlphaFold," Nature — the method and accuracy. — Retrieved from: ____________________

2. Varadi et al. (2024), "AlphaFold Protein Structure Database in 2024," Nucleic Acids Research — the 200M+ structures and open release. — Retrieved from: ____________________

3. CASP14 press release, Protein Structure Prediction Center, 30 November 2020 — the benchmark result and Moult's statement. — Retrieved from: ____________________

4. CASP benchmark documentation — the decades-long agreed evaluation. — Retrieved from: ____________________

5. AlphaFold 3 repository terms of use and prohibited-use policy (Google DeepMind, November 2024) — the open-release governance decision as actually written. — Retrieved from: ____________________

6. Abramson et al. (2024), "Accurate structure prediction of biomolecular interactions with AlphaFold 3," Nature; and the 2024 Nobel Prize in Chemistry (Hassabis & Jumper) — the extension to complexes and the later code release. — Retrieved from: ____________________

CASE 37 DeepMind Mammography — High-Profile Nature Paper, Replicability Push-back 2020

1. McKinney, S. M., Sieniek, M., Godbole, V., Godwin, J., Antropova, N., Ashrafian, H., Back, T., et al. (2020), "International evaluation of an AI system for breast cancer screening," Nature 577:89–94, doi:10.1038/s41586-019-1799-6. — Retrieved from: ____________________

2. Haibe-Kains, B., Adam, G. A., Hosny, A., Khodakarami, F., MAQC Society Board of Directors, Waldron, L., Wang, B., et al. (2020), "Transparency and reproducibility in artificial intelligence," Nature 586:E14–E16, doi:10.1038/s41586-020-2766-y. — Retrieved from: ____________________

3. McKinney et al. (2020), reply to Haibe-Kains et al., Nature 586:E17–E18 — the developers' response on the reproducibility-infrastructure question. — Retrieved from: ____________________

4. Freeman, K., Geppert, J., Stinton, C., Todkill, D., Johnson, S., Clarke, A., & Taylor-Phillips, S. (2021), "Use of artificial intelligence for image analysis in breast cancer screening programmes: systematic review of test accuracy," BMJ 374:n1872 — independent systematic review of subsequent AI-screening-deployment evidence. — Retrieved from: ____________________

CASE 38 iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms 2006 – 2011

1. Shin, J., Cheetham, T. C., Wong, L., Niu, F., Kass, E., Yoshinaga, M. A., Sorel, M., McCombs, J. S., & Sidney, S. (2011), "The impact of the iPLEDGE program on isotretinoin fetal exposure in an integrated health care system," Journal of the American Academy of Dermatology, PMID:21565419. — Retrieved from: ____________________

2. FDA, iPLEDGE program documentation (2006 – present) — REMS architecture and enrollment requirements. — Retrieved from: ____________________

3. Collins, M. K., Moreau, J. F., Opel, D., Swan, J., Prevost, N., Hastings, M., Schwarz, E. B., & Ferris, L. K. (2014), "Compliance with pregnancy prevention measures during isotretinoin therapy," Journal of the American Academy of Dermatology, 70(1):55–59, PMID:24157382 — source of the 150 annual exposures figure. — Retrieved from: ____________________

4. Sullivan et al. (2003), House Science Committee statement on SUBSAFE — the structural counterpoint (Case 173). — Retrieved from: ____________________

CASE 39 TeamSTEPPS 2006 – present

1. AHRQ, TeamSTEPPS 3.0 Curriculum (2023) — the framework and four competencies. — Retrieved from: ____________________

2. DoD / AHRQ partnership documentation — the joint development and implementation infrastructure. — Retrieved from: ____________________

3. Salas, E., DiazGranados, D., Klein, C., Burke, C. S., Stagl, K. C., Goodwin, G. F. & Halpin, S. M. (2008), "Does Team Training Improve Team Performance? A Meta-Analysis," Human Factors 50(6):903–933 — the cross-domain team-training evidence base. — Retrieved from: ____________________

4. Weaver, S., Dy, S. & Rosen, M. (2014) — patient-safety team-training implementation and outcomes. — Retrieved from: ____________________

5. Weld, L. R. et al. (2016), "TeamSTEPPS Improves Operating Room Efficiency and Patient Safety," American Journal of Medical Quality 31(5):408–414 — the on-time first-start figure. — Retrieved from: ____________________

CASE 40 Team Science Training for Clinical and Translational Scientists 2020 – 2022

1. Mook, A., Knerich, V., Komaie, G., Cicutto, L., & Cross, J. (2025), "Team science training for clinical and translational scientists: An assessment of effectiveness," Journal of Clinical and Translational Science 9(1):e158, doi:10.1017/cts.2025.10088. — Retrieved from: ____________________

2. Falk-Krzesinski et al. (2011), "Mapping a research agenda for the science of team science," Research Evaluation 20(2):143–156 — broader team-science literature backdrop. — Retrieved from: ____________________

3. National Research Council (2015), Enhancing the Effectiveness of Team Science (National Academies Press) — the National Academies team-science synthesis. — Retrieved from: ____________________

4. v2 paired cases: IPE evidence gap (122), Implementation Science Training (123) — the frontier/measurement companions. — Retrieved from: ____________________

CASE 41 Translational-Workforce Training — Stated Goals Outrunning Named Practice 2020s

1. Sancheznieto, F., Sorkness, C. A., Attia, J., et al. (2021), "Clinical and translational science award T32/TL1 training programs: program goals and mentorship practices," Journal of Clinical and Translational Science 6(1):e13, doi:10.1017/cts.2021.884. — Retrieved from: ____________________

2. Morris, Wooding, & Grant (2011), "The answer is 17 years, what is the question: understanding time lags in translational research," Journal of the Royal Society of Medicine — the original 17-year-gap source for Case 13. — Retrieved from: ____________________

3. Brownson, Colditz, & Proctor (2018), Dissemination and Implementation Research in Health (2nd ed.) — the broader implementation-science synthesis. — Retrieved from: ____________________

4. v2 paired cases: Team-science training (121), IPE evidence gap (122). — Retrieved from: ____________________

CASE 42 Australian Hospital-Pharmacy Technician Role Redesign 2016

1. Society of Hospital Pharmacists of Australia (November 2016), "Exploring the role of hospital pharmacy technicians and assistants to enhance the delivery of patient centred care: A White Paper on the findings and outcomes of the Pharmacy Technician and Assistant Role Redesign within Australian Hospitals (Redesign) Project." — Retrieved from: ____________________
2. Anderson et al. (2021), "Perceptions of hospital pharmacists and pharmacy technicians towards expanding roles for hospital pharmacy technicians: a cross-sectional survey," Journal of Pharmacy Practice and Research, doi:10.1002/jppr.1697. — Retrieved from: ____________________
3. Boughen, Sutton, Fenn, & Wright (2017), "Defining the Role of the Pharmacy Technician and Identifying Their Future Role in Medicines Optimisation," Pharmacy 5(3):40 — UK companion analysis. — Retrieved from: ____________________
4. Pharmacy Board of Australia and SHPA practice statements limiting technician work to activities not requiring professional judgement — the regulatory framing the project sits inside. — Retrieved from: ____________________

CASE 43 Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface 2013 – 2018

1. Hategeka, C., Ruton, H., & Law, M. R. (2019), "Effect of a community health worker mHealth monitoring system on uptake of maternal and newborn health services in Rwanda," Global Health Research and Policy 4:8, doi:10.1186/s41256-019-0098-y. — Retrieved from: ____________________
2. Rwanda Ministry of Health, community health program documentation and CHW scope-of-practice guidance, 2013–2018. — Retrieved from: ____________________
3. MIT News (2022), reporting on subsequent AI-augmented maternal-care work in Rwanda — journalism-tier companion to the peer-reviewed evaluation. — Retrieved from: ____________________
4. Cross-reference: Case 18 (PEPFAR HIV training-modality comparison) for the paired Sub-Saharan workforce-capability evidence. — Retrieved from: ____________________

CASE 44 Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD 2014 – present

1. Kikuchi, et al. (2025), "Scoping Review of Regulatory Transparency in AI-based Radiology Software: Analysis of PMDA-approved SaMD Products," medRxiv 2025.10.02.25336333. — Retrieved from: ____________________
2. Aisu, N., Miyake, M., Takeshita, K., et al. (2022), "Regulatory-approved deep learning/machine learning-based medical devices in Japan as of 2020: A systematic review," PLOS Digital Health 1(1):e0000001 (medRxiv preprint 2021). — Retrieved from: ____________________
3. Pharmaceuticals and Medical Devices Agency of Japan, PMD Act amendment (2019) and DASH for SaMD program documentation. — Retrieved from: ____________________
4. Cross-reference: "A decade of review in global regulation and research of artificial intelligence medical devices (2015–2025)," PMC12310608 — comparative regulatory context. — Retrieved from: ____________________

CASE 45 Tennessee Voluntary Pre-K Study 2009–2018

1. M. Lipsey, D. Farran & K. Durkin, "Effects of the Tennessee Pre-Kindergarten Program… Through Third Grade," Early Childhood Research Quarterly 45: 155–176 (2018) — the RCT and fade-out. — Retrieved from: ____________________
2. K. Durkin, M. Lipsey, D. Farran & S. Wiesen, "Effects of a Statewide Pre-Kindergarten Program… Through Sixth Grade," Developmental Psychology 58(3): 470–484 (2022) — the sixth-grade reversal (quoted). — Retrieved from: ____________________
3. Responses and counter-analyses to the TN-VPK findings (2018–2022) — the contested reception. — Retrieved from: ____________________

4. National Institute for Early Education Research, State of Preschool yearbooks (2010–2022) — program scale and context. — Retrieved from: ____________________

5. D. Phillips et al., Puzzling It Out: The Current State of Scientific Knowledge on Pre-Kindergarten Effects (2017); J. Heckman on durable early-childhood programs. — Retrieved from: ____________________

CASE 46 Algorithmic Bias in Educational Predictive Analytics

ongoing

1. The Evolving State of Predictive Analytics, The Chronicle of Higher Education (2020), survey conducted by Maguire Associates, 589 institutions — 63 percent use predictive analytics for retention and graduation; only 20 percent institution-wide. — Retrieved from: ____________________

2. K. Bird et al., “Are Algorithms Biased in Education? Exploring Racial Bias in Predicting Community College Student Success,” Journal of Policy Analysis and Management 44 (2025), 379–402 — racial calibration bias, 5× higher at the bottom decile depending on the “at-risk” construct. — Retrieved from: ____________________

3. D. Gándara, H. Anahideh, M. Ison & L. Picchiarini, “Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction,” AERA Open (2024) — bias traced to training data encoding historical discrimination. — Retrieved from: ____________________

4. R. Baker & A. Hawn, “Algorithmic Bias in Education,” International Journal of Artificial Intelligence in Education 32 (2022), 1052–1092 — foundational review of algorithmic bias in education. — Retrieved from: ____________________

5. M. Ekowo & I. Palmer, Predictive Analytics in Higher Education: Five Guiding Practices for Ethical Use, New America (2017) — the deficit-versus-asset mindset and the handling of “at-risk,” “low-risk,” and “high-risk” labels from early-alert systems. — Retrieved from: ____________________

6. Cf. UK A-Level / Ofqual (Case 49); V. Eubanks, Automating Inequality (2018). — Retrieved from: ____________________

CASE 47 Algorithmic Bias in Automated Exam Proctoring

2022

1. Yoder-Himes, D. R., Asif, A., Kinney, K., Brandt, T. J., Cecil, R. E., Himes, P. R., Cashon, C., Hopp, R. M. P., & Ross, E. (2022). Racial, skin tone, and sex disparities in automated proctoring software. Frontiers in Education, 7:881449. doi:10.3389/feduc.2022.881449 — the case’s primary study. — Retrieved from: ____________________

2. Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. Proceedings of Machine Learning Research, 81:77–91 — the foundational intersectional-bias finding in face recognition. — Retrieved from: ____________________

3. Raji, I. D., & Buolamwini, J. (2019). Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial AI products. AAAI/ACM Conference on AI, Ethics, and Society — the audit-and-disclosure mechanism the case calls for. — Retrieved from: ____________________

4. Sjoding, M. W., Dickson, R. P., Iwashyna, T. J., Gay, S. E., & Valley, T. S. (2020). Racial bias in pulse oximetry measurement. New England Journal of Medicine, 383(25):2477–2478 — the structural analog in the medical-device context (Case 26). — Retrieved from: ____________________

CASE 48 School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism

2010s – 2022

1. O. Johnson & J. Jabbari (2022), “Infrastructure of social control: A multi-level counterfactual analysis of surveillance and Black education,” Journal of Criminal Justice 83:101983 — the peer-reviewed source for the case. — Retrieved from: ____________________

2. Hoffman, Trawalter, Axt, & Oliver (2016), “Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites,” PNAS 113(16):4296–4301 — race-construct trio at the cognitive-baseline layer (Case 6). — Retrieved from: ____________________

3. Inker, Eneanya, Coresh, et al. (2021), “New Creatinine- and Cystatin C–Based Equations to Estimate GFR without Race,” NEJM — race-construct trio at the formula layer (Case 25). — Retrieved from: ____________________

4. Sjoding, Dickson, Iwashyna, Gay, & Valley (2020), "Racial Bias in Pulse Oximetry Measurement," NEJM 383:2477–2478 — race-construct trio at the device layer (Case 26). — Retrieved from: ____________________

CASE 49 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days 2020

1. Ofqual, Awarding GCSE, AS, A level, advanced extension awards and extended project qualifications in summer 2020: interim report, August 2020. — Retrieved from: ____________________

2. UK House of Commons Education Committee, Getting the grades they've earned: Covid-19: the cancellation of exams and 'calculated' grades, First Report of Session 2019–21, HC 617, 11 July 2020 — the pre-deployment warning; Office for Statistics Regulation, Ensuring statistical models command public confidence, 2 March 2021 — the post-withdrawal review. — Retrieved from: ____________________

3. Royal Statistical Society, Submission to the Office for Statistics Regulation on the summer 2020 grading process, 2020 — independent statistical review of the standardisation methodology. — Retrieved from: ____________________

4. Smith, H. (2020), "Algorithmic bias: should students pay the price?" AI & Society 35(4):1077–1078 — early academic commentary on the equity dimensions of the withdrawal. — Retrieved from: ____________________

CASE 50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction 2012 – 2023

1. Perdomo, J. C., Britton, T., Hardt, M., & Abebe, R. (2025), "Difficult Lessons on Social Prediction from Wisconsin Public Schools," Proceedings of FAccT 2025, doi:10.1145/3715275.3732175 (also arXiv:2304.06205). — Retrieved from: ____________________

2. Wisconsin Department of Public Instruction (2021), internal equity analysis of the Dropout Early Warning System — unpublished; the presentation summarizing it, whose slide is headed "Is DEWS Fair?", was obtained by The Markup under public records. — Retrieved from: ____________________

3. Feathers, T. (2023), "False Alarm: How Wisconsin Uses Race and Income to Label Students 'High Risk,'" The Markup, April 27, 2023 — investigation documenting the disparate-impact finding and the agency response. — Retrieved from: ____________________

4. Knowles, J. E. (2015), "Of Needles and Haystacks: Building an Accurate Statewide Dropout Early Warning System in Wisconsin," Journal of Educational Data Mining 7(3):18–67 — the original DEWS technical-methodology paper. — Retrieved from: ____________________

CASE 51 Atlanta Public Schools Cheating Scandal 2009 – 2015

1. Office of the Governor, Special Investigators (Bowers, Wilson, Hyde), Special Investigation into Test Tampering in Atlanta's School System, Vol. 1 (June 30, 2011) — the organized cheating and the quoted finding. — Retrieved from: ____________________

2. Atlanta Journal-Constitution investigative series (2009–2011) — the erasure-rate analysis. — Retrieved from: ____________________

3. State of Georgia v. Hall et al. (2013–2015) — indictments and RICO convictions. — Retrieved from: ____________________

4. Koretz, D. (2017), The Testing Charade — high-stakes-testing distortion. — Retrieved from: ____________________

5. Campbell, D. (1976), "Assessing the Impact of Planned Social Change" — Campbell's Law. — Retrieved from: ____________________

CASE 52 Purdue Course Signals — The Reverse-Causality Retention Claim 2012 – 2013

1. Arnold, K. E., & Pistilli, M. D. (2012). Course Signals at Purdue: Using learning analytics to increase student success. Proceedings of LAK 2012, 267–270. doi:10.1145/2330601.2330666 — the original study at the center of the critique. — Retrieved from: ____________________

2. Caulfield, M. (2013). “Why the Course Signals Math Does Not Add Up,” Hapgood, September 26, 2013, and “Purdue Course Signals Data Issue Explainer,” e-Literate, November 12, 2013 — the reverse-causality critique. — Retrieved from: ____________________

3. Essa, A. (2013). “Can We Improve Retention Rates by Giving Students Chocolates?” alfredessa.com — the placebo simulation reproducing the Course Signals retention curve. — Retrieved from: ____________________

4. Feldstein, M. (2013). “Purdue’s Non-Answer on Course Signals,” e-Literate, November 6, 2013 — the field-level critique of promoting research and then declining scrutiny. — Retrieved from: ____________________

CASE 53 inBloom 2014

1. M. Bulger, P. McCormick & M. Pitcan, The Legacy of inBloom, Data & Society Research Institute (2017) — inBloom as a failure of technocratic reform. — Retrieved from: ____________________

2. Education Week and Hechinger Report coverage of the state withdrawals (2013–2014) — nine states exiting. — Retrieved from: ____________________

3. Bulger et al. (2017) — the diagnosis: low public tolerance for risk and uncertainty meeting a failure to communicate benefits and win stakeholder buy-in. — Retrieved from: ____________________

4. N. Selwyn, Distrusting Educational Technology (2014); d. boyd & K. Crawford, “Critical Questions for Big Data” (2012). — Retrieved from: ____________________

5. Parent Coalition for Student Privacy archives and the wave of state student-data-privacy legislation that followed inBloom. — Retrieved from: ____________________

CASE 54 Summit Learning / Personalized Learning Rollout 2014–2019

1. N. Bowles, “Silicon Valley Came to Kansas Schools. That Started a Rebellion,” New York Times (2019) — the parent revolt. — Retrieved from: ____________________

2. N. Singer, “The Silicon Valley Billionaires Remaking America’s Schools,” New York Times (2017) — the CZI/Summit rollout. — Retrieved from: ____________________

3. S. Schwartz, “Two Districts Roll Back Summit Personalized Learning Program,” Education Week (December 22, 2017) — the Cheshire suspension and the Indiana Area rollback. — Retrieved from: ____________________

4. M. Barnum, “Summit Learning declined to be studied, then cited collaboration with Harvard researchers anyway,” Chalkbeat (January 18, 2019) — the missing evaluation framework. — Retrieved from: ____________________

5. Chan Zuckerberg Initiative & Summit Learning public program documentation (2015–2019); cf. inBloom (Case 53). — Retrieved from: ____________________

CASE 55 Enrollment-Algorithm Yield Optimization Across U.S. Higher Education 2010s – present (Brookings synthesis 2021)

1. Engler, A. (2021), “Enrollment algorithms are contributing to the crises of higher education,” Brookings Institution, 14 Sept 2021. — Retrieved from: ____________________

2. Franke, R. (2012), Towards the Education Nation: Revisiting the Impact of Financial Aid, College Experience, and Institutional Context on Baccalaureate Degree Attainment, UCLA doctoral dissertation — the aid-and-graduation effect sizes Engler cites. — Retrieved from: ____________________

3. Goldrick-Rab, S. (2016), Paying the Price — broader synthesis on net-price, unsubsidized loans, and low-income completion. — Retrieved from: ____________________

4. Vendor case studies cited in Engler (Othot, EAB, Ruffalo Noel Levitz, University of Washington) — vendor-reported and not externally audited; flagged at evidence-tier under the title. — Retrieved from: ____________________

CASE 57 GAO Online Program Manager Oversight Gap (GAO-22-104463) 2022

1. U.S. Government Accountability Office (2022), "Higher Education: Education Needs to Strengthen Its Approach to Monitoring Colleges' Arrangements with Online Program Managers," GAO-22-104463. — Retrieved from: ____________________

2. U.S. Department of Education (2011), "Dear Colleague" guidance on incentive-compensation bundled-services exemption — the regulatory artifact the GAO audits. — Retrieved from: ____________________

3. Higher Education Amendments of 1992, Pub. L. No. 102-325, incentive-compensation prohibition, 20 U.S.C. § 1094(a)(20) — the statutory basis the 2011 guidance interpreted. — Retrieved from: ____________________

4. 2U Inc. (2024), Chapter 11 bankruptcy filing (July 25, 2024); U.S. Department of Education, "Dear Colleague" Letter (Jan. 14, 2025) reaffirming the 2011 bundled-services guidance (GEN-11-05) — the commercial-boundary closure alongside the federal guidance's continued validity. — Retrieved from: ____________________

CASE 58 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC) 2010s – 2024

1. Stephanie Luna v. University of Southern California, class action complaint, Los Angeles County Superior Court, May 2023. — Retrieved from: ____________________

2. Higher Ed Dive reporting on the Luna complaint, May 2023; classaction.org and topclassactions.com summaries. — Retrieved from: ____________________

3. Project on Predatory Student Lending statement on USC-2U partnership termination, November 2023. — Retrieved from: ____________________

4. 2U Inc. and USC public statements on partnership termination, November 2023; broader 2U commercial-collapse reporting referenced through Case 57. — Retrieved from: ____________________

CASE 60 Houston EVAAS — Value-Added Teacher Evaluation on Trial 2011–2017

1. Hous. Fed'n of Teachers v. Hous. Indep. Sch. Dist., 251 F. Supp. 3d 1168 (S.D. Tex. 2017) — memorandum opinion denying summary judgment on the procedural due process claim. — Retrieved from: ____________________

2. Amrein-Beardsley, A., & Collins, C. (2012), "The SAS Education Value-Added Assessment System (SAS® EVAAS®) in the Houston Independent School District (HISD): Intended and Unintended Consequences," Education Policy Analysis Archives 20(12), doi:10.14507/epaa.v20n12.2012. — Retrieved from: ____________________

3. American Statistical Association (2014), ASA Statement on Using Value-Added Models for Educational Assessment, April 8, 2014. — Retrieved from: ____________________

4. American Educational Research Association Council (2015), "AERA Statement on Use of Value-Added Models (VAM) for the Evaluation of Educators and Educator Preparation Programs," Educational Researcher 44(8):448–452, doi:10.3102/0013189X15618385. — Retrieved from: ____________________

5. Amrein-Beardsley, A., & Geiger, T. (2020), "Methodological Concerns About the Education Value-Added Assessment System (EVAAS): Validity, Reliability, and Bias," SAGE Open 10(2), doi:10.1177/2158244020922224. — Retrieved from: ____________________

6. Education Week (October 2017), "Houston District Settles Lawsuit With Teachers' Union Over Value-Added Scores"; Houston Public Media (October 10, 2017), "Federal Lawsuit Settled Between Houston's Teacher Union and HISD" — settlement terms and the $237,000 fee figure. — Retrieved from: ____________________

CASE 61 Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap 1997–2023

1. Hanford, E. (2022 – 2024), "Sold a Story: How Teaching Kids to Read Went So Wrong," APM Reports, podcast series launched October 20, 2022. — Retrieved from: ____________________

2. Hanford, E. (2018), "Hard Words: Why Aren't Our Kids Being Taught to Read?" APM Reports, September 10, 2018. — Retrieved from: ____________________

3. National Reading Panel (2000), Teaching Children to Read: An Evidence-Based Assessment of the Scientific Research Literature on Reading and Its Implications for Reading Instruction, National Institute of Child Health and Human Development, NIH Pub. No. 00-4769. — Retrieved from: ____________________

4. Castles, A., Rastle, K., & Nation, K. (2018), "Ending the Reading Wars: Reading Acquisition From Novice to Expert," Psychological Science in the Public Interest 19(1):5 – 51. — Retrieved from: ____________________

5. Schwartz, S. (updated through 2024), "Which States Have Passed 'Science of Reading' Laws? What's in Them?" Education Week — legislation tracker; 40 states plus DC by late 2024. — Retrieved from: ____________________

6. Seidenberg, M. (2017), Language at the Speed of Sight: How We Read, Why So Many Can't, and What Can Be Done About It, Basic Books. — Retrieved from: ____________________

CASE 62 Gates Intensive Partnerships — Measurement Without a Coupled Lever 2009–2018

1. Stecher, B. M., Holtzman, D. J., Garet, M. S., Hamilton, L. S., et al. (2018), Improving Teaching Effectiveness: Final Report: The Intensive Partnerships for Effective Teaching Through 2015–2016, RAND Corporation, RR-2242-BMGF. — Retrieved from: ____________________

2. Kane, T. J., McCaffrey, D. F., Miller, T., & Staiger, D. O. (2013), Have We Identified Effective Teachers? Validating Measures of Effective Teaching Using Random Assignment, MET Project research paper, Bill & Melinda Gates Foundation, January 2013. — Retrieved from: ____________________

3. Sokol, M. (2015), "Sticker shock: How Hillsborough County's Gates grant became a budget buster," Tampa Bay Times, October 2015 — with companion reporting, "Hillsborough schools to dismantle Gates-funded system that cost millions to develop." — Retrieved from: ____________________

4. Stecher, B. M., et al. (2016), Improving Teaching Effectiveness: Implementation: The Intensive Partnerships for Effective Teaching Through 2013–2014, RAND Corporation, RR-1295. — Retrieved from: ____________________

5. Will, M. (2018), "'An Expensive Experiment': Gates Teacher-Effectiveness Program Shows No Gains for Students," Education Week, June 21, 2018. — Retrieved from: ____________________

CASE 63 LAUSD iPad Program — The Common Core Technology Project 2013–2015

1. Gilbertson, A. (2014), "LA schools cancel iPad contracts after KPCC publishes internal emails," KPCC 89.3 (Southern California Public Radio), August 25, 2014 — the public-records email investigation and the contract halt; see also KPCC's "LAUSD iPads: Timeline of a troubled program." — Retrieved from: ____________________

2. American Institutes for Research (2014), interim evaluation of the LAUSD Common Core Technology Project, Phase 1 — 245 classroom observations across CCTP schools; Pearson curriculum observed in use in one. — Retrieved from: ____________________

3. Herold, B. (2014), "Hard Lessons Learned in Ambitious L.A. iPad Initiative," Education Week, September 2014 — retrospective on the implementation failures; see also Herold (2014), "FBI Investigation Leaves L.A. iPad Initiative in Further Disarray," Education Week, December 2014. — Retrieved from: ____________________

4. Blume, H., Los Angeles Times coverage of the CCTP (2013–2015) — the rollout, the security bypass, the contract suspension, the Deasy resignation, and the FBI document seizure. — Retrieved from: ____________________

5. Education Week Market Brief (2017), "Feds Drop Investigation Into Los Angeles District Over $1 Billion iPad Purchase," February 2017 — the reported closure of the federal inquiry without charges. — Retrieved from: ____________________

CASE 65 Training Transfer — The Gap Between Learning and Doing 2010

1. Blume, Ford, Baldwin, & Huang (2010), "Transfer of Training: A Meta-Analytic Review," Journal of Management 36(4):1065–1105, doi:10.1177/0149206309352880. — Retrieved from: ____________________

2. Baldwin & Ford (1988), "Transfer of Training: A Review and Directions for Future Research," Personnel Psychology 41(1):63–105 — the foundational synthesis the 2010 meta-analysis updates. — Retrieved from: ____________________

3. Burke & Hutchins (2007), "Training Transfer: An Integrative Literature Review," Human Resource Development Review 6(3):263–296 — the integrative-review companion synthesis. — Retrieved from: ____________________

4. Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the framework the meta-analysis informs (paired Case 79). — Retrieved from: ____________________

CASE 66 Growth-Mindset National Experiment — Heterogeneous Effects 2019

1. Yeager, D. S., Hanselman, P., Walton, G. M., Murray, J. S., Crosnoe, R., Muller, C., Tipton, E., Schneider, B., Hulleman, C. S., Hinojosa, C. P., Paunesku, D., Romero, C., Flint, K., Roberts, A., Trott, J., Iachan, R., Buontempo, J., Yang, S. M., Carvalho, C. M., Hahn, P. R., Gopalan, M., Mhatre, P., Ferguson, R., Duckworth, A. L., & Dweck, C. S. (2019). A national experiment reveals where a growth mindset improves achievement. Nature, 573(7774):364–369. doi:10.1038/s41586-019-1466-y — the case's primary trial. — Retrieved from: ____________________

2. National Study of Learning Mindsets, ICPSR 37353 — the trial dataset. — Retrieved from: ____________________

3. Dweck, C. S. (2006). Mindset: The New Psychology of Success. Random House — the broader theoretical framework the intervention rests on. — Retrieved from: ____________________

4. Sisk, V. F., Burgoyne, A. P., Sun, J., Butler, J. L., & Macnamara, B. N. (2018). To what extent and under which circumstances are growth mindsets important to academic achievement? Two meta-analyses. Psychological Science, 29(4):549–571 — the prior moderator-analysis literature the Yeager trial extends. — Retrieved from: ____________________

5. Macnamara, B. N., & Burgoyne, A. P. (2023). Do growth mindset interventions impact students' academic achievement? A systematic review and meta-analysis. Psychological Bulletin 149(3–4):133–173 — near-null effects among the best-designed studies. — Retrieved from: ____________________

6. Education Endowment Foundation (Foliano et al., 2019), "Changing Mindsets" evaluation — a 101-school RCT finding no significant effect on attainment. — Retrieved from: ____________________

CASE 67 Cognitive Tutor / Carnegie Learning 1990s – present

1. Anderson, J., Corbett, A., Koedinger, K. & Pelletier, R. (1995), "Cognitive Tutors: Lessons Learned," Journal of the Learning Sciences — the ACT-R basis and design. — Retrieved from: ____________________

2. Koedinger, K. & Corbett, A. (2006), Cambridge Handbook of the Learning Sciences — knowledge tracing and adaptive instruction. — Retrieved from: ____________________

3. Pane, J. F., Griffin, B. A., McCaffrey, D. F. & Karam, R. (2014), "Effectiveness of Cognitive Tutor Algebra I at Scale," Educational Evaluation and Policy Analysis 36(2):127–144, doi:10.3102/0162373713507480 — no effect in year one; a statistically significant year-two effect for high schools, not for middle schools. — Retrieved from: ____________________

4. What Works Clearinghouse (2009), Cognitive Tutor® Algebra I intervention report, U.S. Department of Education, July 2009 — "more than 500,000 students in approximately 2,600 urban, rural, and suburban school districts" as of August 2008. — Retrieved from: ____________________

5. Lynch, C., Ashley, K., Pinkwart, N. & Aleven, V. (2009), "Concepts, Structures, and Goals: Redefining Ill-Definedness," International Journal of Artificial Intelligence in Education 19(3):253–266 — what makes a domain ill-defined and why the tutor pipeline has less to attach to there. — Retrieved from: ____________________

6. Ritter, S. & Fancsali, S. (2016), "MATHia X: The Next Generation Cognitive Tutor," EDM 2016 — the web re-platforming of the Cognitive Tutor software into MATHia; and Carnegie Learning's 2023 announcement of LiveHint AI (cf. Case 88). — Retrieved from: ____________________

CASE 68 CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

2017 – 2023 (six cycles)

1. Cervantes, Floryanzia, Sharp, Gray-Roncal, & Johnson (2023), "Empowering Trailblazers toward Scalable, Systematized, Research-Based Workforce Development," ASEE Annual Conference, doi:10.18260/1-2--43271. — Retrieved from: ____________________
2. CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional program description. — Retrieved from: ____________________
3. National Academies / NSF research on undergraduate research experience effects on STEM persistence (broader literature against which the case sits). — Retrieved from: ____________________
4. Paired case (78) — CIRCUIT proofreading + MICrONS — the deployed-capability companion. — Retrieved from: ____________________

CASE 69 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

2016

1. Settles, B., & Meeder, B. (2016), "A Trainable Spaced Repetition Model for Language Learning," Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics, pp. 1848–1858, doi:10.18653/v1/P16-1174. — Retrieved from: ____________________
2. Ebbinghaus, H. (1885), Über das Gedächtnis — the empirical forgetting-curve foundation HLR formalizes. — Retrieved from: ____________________
3. Pimsleur, P. (1967), "A memory schedule," Modern Language Journal 51(2):73–75 — graduated-interval recall heuristic. — Retrieved from: ____________________
4. Leitner, S. (1972), So lernt man lernen — Leitner-box spacing heuristic. — Retrieved from: ____________________

CASE 70 High-Impact Learning System — Engineering the Environment, Not Just the Event

2001 – present

1. Brinkerhoff, R. O., & Apking, A. M. (2001), High Impact Learning: Strategies for Leveraging Performance and Business Results from Training Investments, Basic Books. — Retrieved from: ____________________
2. Brinkerhoff, R. O. (2006), "Increasing impact of training investments: an evaluation strategy for building organizational learning capability," Industrial and Commercial Training 38(6):302–307 — the author's own statement that evaluation should address the whole performance-improvement process, not the training event. — Retrieved from: ____________________
3. Blume et al. (2010), Journal of Management 36(4):1065–1105 — the meta-analytic finding HILS operationalizes (paired Case 65). — Retrieved from: ____________________
4. Brinkerhoff (2005), Advances in Developing Human Resources 7(1):86–101 — SCM as the paired evaluation instrument (Case 83). — Retrieved from: ____________________

CASE 71 Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development

2025 (preprint)

1. Barr, M., Nabi, S. W., & Andrei, O. (2025), "The Development of Reflective Practice on a Work-Based Software Engineering Program: A Longitudinal Study," Proc. 37th IEEE Conference on Software Engineering Education and Training (CSEE&T), pp. 53–61, doi:10.1109/CSEET66350.2025.00012 (preprint: arXiv:2504.20956). — Retrieved from: ____________________

2. D. Schön, The Reflective Practitioner (1983) — the foundational account of reflection-in-action the genre rests on. — Retrieved from: ____________________

3. Boud, Keogh & Walker (eds.), Reflection: Turning Experience into Learning (1985) — reflection as a learning process, and the measurement problem it raises. — Retrieved from: ____________________

4. the proposed revisions — the amended LEO-2 (first-person narration of design iteration) the case evaluates. — Retrieved from: ____________________

CASE 72 ASSISTments — National Replication and Long-Term Follow-Through 2014 – present

1. Roschelle, J., Feng, M., Murphy, R. F., & Mason, C. A. (2016), "Online Mathematics Homework Increases Student Achievement," AERA Open 2(4):1–12, doi:10.1177/2332858416673968. — Retrieved from: ____________________

2. Feng, M., Huang, C., & Collins, K. (2023), Technology-Based Support Shows Promising Long-Term Impact on Math Learning, WestEd — the North Carolina replication, the equity heterogeneity finding, and the 8th-grade persistence result. — Retrieved from: ____________________

3. Heffernan, N. T., & Heffernan, C. L. (2014), "The ASSISTments ecosystem," International Journal of AI in Education 24:470–497 — platform design and research-loop description. — Retrieved from: ____________________

4. Arnold Ventures grant documentation of the virtual-training-adaptation arm — including the 8th-grade extension that WestEd and Arnold Ventures ended when the design departed from the planned RCT. — Retrieved from: ____________________

CASE 73 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension 2016 – 2025

1. Van Campenhout, R., Jerome, B., Dittel, J. S., & Johnson, B. G. (2023), "The Doer Effect at Scale: Investigating Correlation and Causation Across Seven Courses," LAK23, doi:10.1145/3576050.3576103. — Retrieved from: ____________________

2. Van Campenhout, R., Autry, K. S., Clark, M. W., & Johnson, B. G. (2025), "Scaling the Doer Effect: A Replication Analysis Using AI-Generated Questions," L@S '25, pp. 24–34, doi:10.1145/3698205.3729545. — Retrieved from: ____________________

3. Butler, D. et al. (2025), "Does the Doer Effect Generalize To Non-WEIRD Populations? Toward Analytics in Radio and Phone-Based Learning," LAK '25, doi:10.1145/3706468.3706505 (also arXiv 2412.20923). — Retrieved from: ____________________

4. Koedinger, K. R., McLaughlin, E. A., Jia, J. Z., & Bier, N. L. (2016), "Is the doer effect a causal relationship? How can we tell and why it's important," LAK '16, pp. 388–397, doi:10.1145/2883851.2883957 — the replication target the present case builds on. — Retrieved from: ____________________

CASE 74 Zhang/Scardamalia — Knowledge Building Across Three Cohorts 2009

1. Zhang, J., Scardamalia, M., Reeve, R., & Messina, R. (2009), "Designs for Collective Cognitive Responsibility in Knowledge-Building Communities," Journal of the Learning Sciences 18(1):7–44, doi:10.1080/10508400802581676. — Retrieved from: ____________________

2. Scardamalia, M., & Bereiter, C. (2006), "Knowledge Building: Theory, pedagogy, and technology," in K. Sawyer (ed.), Cambridge Handbook of the Learning Sciences — programmatic backdrop. — Retrieved from: ____________________

3. Bereiter, C., & Scardamalia, M. (2014), "Knowledge Building and Knowledge Creation: One Concept, Two Hills to Climb," in S. C. Tan, H. J. So, & J. Yeo (eds.), Knowledge Creation in Education — extension into the post-2009 program. — Retrieved from: ____________________

4. Design-based research methodology references — Cobb, Confrey, diSessa, Lehrer, & Schauble (2003), "Design Experiments in Educational Research," Educational Researcher 32(1):9–13 — the methodological frame the 2009 study operates inside. — Retrieved from: ____________________

CASE 75 Chen et al. — Rural China AI Devices and the Equity-Direction Finding 2025

1. Chen, R., Wu, Y., Chen, Z., & Zhou, P. (2025), "Advancing educational equity in rural China: the impact of AI devices on teaching quality and learning outcomes for sustainable development," Frontiers in Psychology 16:1588047, doi:10.3389/fpsyg.2025.1588047. — Retrieved from: ____________________

2. Paired Case 86 (Gándara et al., AERA Open) — community-college predictive equity at the higher-education scale. — Retrieved from: ____________________

3. Paired Case 88 (LiveHint AI bias across foundation models, AIED 2025) — bias-surfacing in AI-supported instruction. — Retrieved from: ____________________

4. Paired Case 73 (Doer Effect non-WEIRD LAK 2025 radio-and-phone extension) — non-WEIRD methodological discipline at the heterogeneity-finding axis. — Retrieved from: ____________________

CASE 76 Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account 2023

1. Martinez-Maldonado, R., Echeverria, V., Fernandez-Nieto, G., et al. (2023), "Lessons Learnt from a Multimodal Learning Analytics Deployment In-the-wild," ACM Transactions on Computer-Human Interaction 31(1), Article 8, doi:10.1145/3622784; author preprint at arXiv:2303.09099. — Retrieved from: ____________________

2. Blikstein, P., & Worsley, M. (2016), "Multimodal Learning Analytics and Education Data Mining: Using computational technologies to measure complex learning tasks," Journal of Learning Analytics 3(2):220–238 — broader MMLA literature backdrop. — Retrieved from: ____________________

3. Schon, D. (1983), The Reflective Practitioner — the genre's theoretical underpinning, referenced across the reflective-practice case tier. — Retrieved from: ____________________

4. Editors' memo (B1) — Practice Flywheel commissioning structure that the case is offered as a published-first-person exemplar within. — Retrieved from: ____________________

CASE 77 Hybrid Human-AI Tutoring — Augmentation, Not Delegation 2024

1. Thomas, D. R. et al. (2024), "Improving Student Learning with Hybrid Human-AI Tutoring: A Three-Study Quasi-Experimental Investigation," LAK '24, doi:10.1145/3636555.3636896. — Retrieved from: ____________________

2. Case 20 (TREWS) reference set — Henry et al. (2022), Nature Medicine — clinician-AI teaming analog. — Retrieved from: ____________________

3. Case 5 (Epic Sepsis) reference set — Wong et al. (2021), JAMA Internal Medicine — delegation-collapse analog. — Retrieved from: ____________________

4. Koedinger, K. R. et al. — Cognitive Tutor literature as the fully-automated track the augmentation track contrasts with. — Retrieved from: ____________________

CASE 78 CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale 2017 – present

1. MICrONS program literature (IARPA) — connectomics method and automated segmentation evidence base. — Retrieved from: ____________________

2. APL BossDB documentation — petabyte-scale connectomics data infrastructure. — Retrieved from: ____________________

3. CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional/program description; the training-outcome evidence is at this tier and the evidence-tier flag is binding. — Retrieved from: ____________________

4. Cervantes et al. (2023), ASEE Annual Conference — the paired peer-reviewed CIRCUIT case (Case 68). — Retrieved from: ____________________

5. Wachter, R. M., & Brynjolfsson, E. (2024), "Will Generative Artificial Intelligence Deliver on Its Promise in Health Care?" JAMA 331(1):65–69 — the productivity-paradox framing: general purpose technologies pay off only after the complementary organizational work is done. — Retrieved from: ____________________

CASE 79 The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops

1959 – present

1. Kirkpatrick (1959–1960), original four-level evaluation series in the Journal of the American Society of Training Directors (Nov 1959 – Feb 1960); updated as Kirkpatrick & Kirkpatrick, Evaluating Training Programs (3rd ed., 2006). — Retrieved from: ____________________
2. Devlin Peck, "The Kirkpatrick Model: 4 Levels, Examples, and Free Template" — synthesis of the stop-at-L2 pattern in corporate practice. — Retrieved from: ____________________
3. D2L, "Evaluating Training Programs: Models, Metrics and Business Impact," practitioner guide documenting the same pattern. — Retrieved from: ____________________
4. Valamis, "Kirkpatrick model: how to evaluate whether training worked" — practitioner guide on evaluation-practice gaps. — Retrieved from: ____________________
5. Blume, Ford, Baldwin, & Huang (2010), "Transfer of Training: A Meta-Analytic Review," Journal of Management 36(4):1065–1105 — the paired peer-reviewed case (65). — Retrieved from: ____________________

CASE 80 Georgia State University Predictive Analytics

2012 – present

1. Renick, T. & Strom, A. (2020) on GSU's advising transformation — the system design and outcomes. — Retrieved from: ____________________
2. Georgia State University institutional research and Strategic Plan reports — graduation-rate and equity data. — Retrieved from: ____________________
3. Fausset, R., "Georgia State, Leading U.S. in Black Graduates, Is Engine of Social Mobility," New York Times, 15 May 2018 — the 800-factor advising model. — Retrieved from: ____________________
4. EDUCAUSE Review on GSU predictive advising — the human-loop architecture. — Retrieved from: ____________________
5. Complete College America, Game Changers documentation — dissemination of the model. — Retrieved from: ____________________
6. Georgia State University National Institute for Student Success (founded 2021) — reported engagement with 130-plus campuses and 1.5 million students by 2024. — Retrieved from: ____________________

CASE 81 Open University 'Ethical Use of Student Data' and OU Analyse

2014 – 2025

1. Slade & Prinsloo (2013), "Learning Analytics: Ethical Issues and Dilemmas," American Behavioral Scientist 57(10):1510–1529, doi:10.1177/0002764213479366. — Retrieved from: ____________________
2. Open University (2014), "Ethical Use of Student Data for Learning Analytics" — first institutional policy of its kind in higher education. — Retrieved from: ____________________
3. Rienties, Boroowa, Cross, Kubiak, Mayles, & Murphy (2016), "Analytics4Action Evaluation Framework: A Review of Evidence-Based Learning Analytics Interventions at the Open University UK," Journal of Interactive Media in Education 2016(1):2, doi:10.5334/jime.394. — Retrieved from: ____________________
4. Herodotou, Hlosta, Boroowa, Rienties, Zdrahal, & Mangafa (2019), "Empowering online teachers through predictive learning analytics," British Journal of Educational Technology 50(6):3064–3079, doi:10.1111/bjet.12853 — OU Analyse evaluation across 559 teachers (189 with OUA access) and 14,000+ students in 15 undergraduate courses; average-use teachers benefited students most. — Retrieved from: ____________________

CASE 82 MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil) 2024

1. Freitas, Fonseca, Garcia, Pontual Falcão, Marques, Gasevic, & Mello (2024), "MMALA: Developing and Evaluating a Maturity Model for Adopting Learning Analytics," Journal of Learning Analytics 11(1):67–86. — Retrieved from: ___________________
2. Open University Ethical Use of Student Data policy (2014) — institutional-policy companion (Case 81). — Retrieved from: ___________________
3. Hilliger et al. (2020), Internet and Higher Education — LALA participatory adoption companion (Case 91). — Retrieved from: ___________________
4. Norwegian Expert Commission on Learning Analytics, final NOU (2023) — national-scale companion (Case 92). — Retrieved from: ___________________

CASE 83 Brinkerhoff Success Case Method — Tails as the Evaluation Instrument 2005 – present

1. Brinkerhoff, R. O. (2005), "The Success Case Method: A Strategic Evaluation Approach to Increasing the Value and Effect of Training," Advances in Developing Human Resources 7(1):86–101, doi:10.1177/1523422304272172. — Retrieved from: ___________________
2. Brinkerhoff Evaluation Institute deployment list — International Red Cross and Red Crescent, World Bank, Cargill, Merck, Ford Motor — practitioner channel. — Retrieved from: ___________________
3. Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the chain-of-evidence framework SCM operationalizes (paired Case 79). — Retrieved from: ___________________
4. Blume, Ford, Baldwin, & Huang (2010), Journal of Management 36(4):1065–1105 — the meta-analytic environment-as-finding SCM samples around (paired Case 65). — Retrieved from: ___________________

CASE 84 Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive 2007 – 2014

1. Evidence for ESSA, program record for Carnegie Learning High School Math Solution Algebra I — ESSA rating "Strong," weighted mean effect size +0.04 across three qualifying studies (Cabalo & Vu 2007; Campuzano et al. 2009; Pane et al. 2014). — Retrieved from: ___________________
2. Pane, J. F., Griffin, B. A., McCaffrey, D. F., & Karam, R. (2014), "Effectiveness of Cognitive Tutor Algebra I at Scale," Educational Evaluation and Policy Analysis 36(2):127–144, doi:10.3102/0162373713507480. — Retrieved from: ___________________
3. RAND Working Paper WR-1050 — addendum to the Pane et al. evaluation. — Retrieved from: ___________________
4. Koedinger, K. R., Anderson, J. R., Hadley, W. H., & Mark, M. A. (1997), "Intelligent tutoring goes to school in the big city," IJAIED — the v1 Case 67 system description Cognitive Tutor builds from. — Retrieved from: ___________________
5. What Works Clearinghouse (2016), Cognitive Tutor® intervention report, Secondary Mathematics, June 2016 — rates Cognitive Tutor Algebra I "mixed effects" on algebra; records the Pane et al. year-two difference as no longer significant after adjustment for multiple comparisons. — Retrieved from: ___________________
6. Escueta, Quan, Nickow, & Oreopoulos (2017), "Education Technology: An Evidence-Based Review," NBER Working Paper 23744 — reports that the Pane year-two gain tracked reduced use of the prescribed non-computer activities, not higher fidelity; published as Escueta, Nickow, Oreopoulos, & Quan (2020), "Upgrading Education with Technology: Insights from Experimental Research," Journal of Economic Literature 58(4):897–996. — Retrieved from: ___________________
7. Kraft, M. A. (2020), "Interpreting Effect Sizes of Education Interventions," Educational Researcher 49(4):241–253 — 1,942 effect sizes from 747 RCTs, median 0.10 SD, 0.03 SD for samples above 2,000 students, and the benchmark that treats 0.20 SD as large. — Retrieved from: ___________________

CASE 85 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale 2019 – 2023

1. Herodotou, C., Hlosta, M., Boroowa, A., Rienties, B., Zdrahal, Z., & Mangafa, C. (2019), "Empowering online teachers through predictive learning analytics," British Journal of Educational Technology 50(6):3064–3079, doi:10.1111/bjet.12853. — Retrieved from: ____________________

2. Herodotou, C., Maguire, C., Hlosta, M., & Mulholland, P. (2023), "Predictive Learning Analytics and University Teachers: Usage and perceptions three years post implementation," LAK '23, pp. 68–78, doi:10.1145/3576050.3576061 — survey of 366 teachers, UTAUT model of adoption. — Retrieved from: ____________________

3. Herodotou et al. (2019), "A large-scale implementation of predictive learning analytics in higher education: the teachers' role and perspective," Educational Technology Research and Development, ERIC EJ1227972 — complementary teacher-perspective paper. — Retrieved from: ____________________

4. Arnold, K. E., & Pistilli, M. D. (2012), "Course Signals at Purdue," LAK '12 — the structural precursor v1 carries as a cautionary case. — Retrieved from: ____________________

CASE 86 Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction 2024

1. Gándara, Anahideh, Ison, & Picchiarini (2024), "Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction," AERA Open 10, doi:10.1177/23328584241258741 — primary case source on cross-group accuracy disparity, stratified evaluation, and bias mitigation in student-success prediction. — Retrieved from: ____________________

2. Barocas, Hardt, & Narayanan, Fairness and Machine Learning (fairmlbook.org) — methodological backdrop on construct definition and stratified evaluation. — Retrieved from: ____________________

3. Friedler, Scheidegger, & Venkatasubramanian (2021), "The (im)possibility of fairness: different value systems require different mechanisms for fair decision making," Communications of the ACM — the construct-definition argument at field level. — Retrieved from: ____________________

4. v2 cross-referenced cases: 25 (eGFR race correction), 26 (pulse oximetry across skin tones), 6 (Hoffman pain bias) — the race-construct trio at the construct-definition layer. — Retrieved from: ____________________

CASE 87 Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models 2021 – 2024

1. Yu, R., Lee, H., & Kizilcec, R. F. (2021), "Should College Dropout Prediction Models Include Protected Attributes?" in Proceedings of the Eighth ACM Conference on Learning @ Scale (L@S '21), doi:10.1145/3430895.3460139 — primary paper on the include-or-exclude empirical analysis. — Retrieved from: ____________________

2. Kizilcec, R. F., & Lee, H. (2022), "Algorithmic Fairness in Education," in Holmes & Porayska-Pomsta (eds.), The Ethics of Artificial Intelligence in Education, Routledge, doi:10.4324/9780429329067-10 — broader synthesis of the fairness-in-learning-analytics frontier. — Retrieved from: ____________________

3. Dwork, Hardt, Pitassi, Reingold, & Zemel (2012), "Fairness through awareness," Proceedings of ITCS — foundational paper on the inadequacy of fairness-through-unawareness. — Retrieved from: ____________________

4. Barocas, Hardt, & Narayanan, Fairness and Machine Learning (fairmlbook.org) — technical-fairness backdrop. — Retrieved from: ____________________

5. v2 cross-referenced cases: 86 (Gándara), 25 (eGFR), 26 (pulse oximetry), 6 (Hoffman) — equity-construct five-case set. — Retrieved from: ____________________

CASE 88 LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models 2025

1. Vinodh, A., Harvey, E., Almoubayyed, H., Yu, R., Brooks, C., Koenecke, A., & Kizilcec, R. F. (2025), "Evaluating an AI Tutor for Bias Across Different Foundation Models," AIED 2025, LNAI 15882, pp. 341–348, doi:10.1007/978-3-031-98465-5_43; author copy at renzheyu.com/papers/AIED2025_Tutor.pdf. — Retrieved from: ____________________

2. Bommasani, R. et al. (2021), "On the Opportunities and Risks of Foundation Models," Stanford CRFM — the foundation-model framing the case builds on. — Retrieved from: ____________________

3. Race-construct trio reference set: Hoffman et al. (2016), Sjoding et al. (2020) pulse oximetry, Inker et al. (2021) eGFR-without-race — paired with Cases 25, 26, 6. — Retrieved from: ____________________

4. Carnegie Learning LiveHint product documentation — the subject system; case framing is binding on LiveHint being in development, not deployment. — Retrieved from: ____________________

CASE 89 Data Privacy and Learning Analytics on the African Continent 2022

1. Prinsloo, P., & Kaliisa, R. (2022), "Data privacy on the African continent: Opportunities, challenges and implications for learning analytics," British Journal of Educational Technology 53(4):894–913, doi:10.1111/bjet.13226. — Retrieved from: ____________________

2. African Union Convention on Cyber Security and Personal Data Protection (Malabo Convention, adopted 2014; in force 8 June 2023) — continental policy backdrop, ratified by a minority of AU member states. — Retrieved from: ____________________

3. Slade & Prinsloo (2013), American Behavioral Scientist — earlier framing on which the 2022 paper builds. — Retrieved from: ____________________

4. Open University Ethical Use of Student Data policy (2014) — single-regime governance-by-design companion. — Retrieved from: ____________________

CASE 90 Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions 2025

1. Pyle, C., & Andalibi, N. (2025), "Algorithmic College Admissions in the U.S.: Distances Between Vendors' Claims and Applicants' Perceptions," Proceedings of the ACM on Human-Computer Interaction 9(7), CSCW369, doi:10.1145/3757550. — Retrieved from: ____________________

2. Engler (2021), Brookings — paired deployment-side mapping (Case 55). — Retrieved from: ____________________

3. GAO-22-104463 (2022) — paired regulator-side audit (Case 57). — Retrieved from: ____________________

4. Dunne, A., & Raby, F. (2013), Speculative Everything — methodological backdrop for speculative-design probes. — Retrieved from: ____________________

CASE 91 LALA — Building Learning-Analytics Governance Capacity Across Latin America 2017 – 2020

1. Hilliger, Ortiz-Rojas, Pesántez-Cabrera, Scheihing, Tsai, Muñoz-Merino, Broos, Whitelock-Wainwright, & Pérez-Sanagustín (2020), "Identifying needs for learning analytics adoption in Latin American universities: A mixed-methods approach," The Internet and Higher Education 45:100726, doi:10.1016/j.iheduc.2020.100726. — Retrieved from: ____________________

2. LALA project — EU Erasmus+ grant 586120-EPP-1-2017-1-ES (2017–2020) — program documentation and the LALA CANVAS artifact. — Retrieved from: ____________________

3. Open University Ethical Use of Student Data policy (2014) and OU Analyse — UK companion governance-by-design case (Case 81). — Retrieved from: ____________________

4. Slade & Prinsloo (2013), “Learning Analytics: Ethical Issues and Dilemmas,” American Behavioral Scientist 57(10):1510–1529 — the broader field-scale ethics framing. — Retrieved from: ____________________

CASE 92 Norway's National Expert Commission on Learning Analytics 2021 – 2023

1. Ekspertgruppen for digital læringsanalyse (2022), Læringsanalyse — noen sentrale dilemmaer, interim report delivered to the Minister of Education and Research, June 2022. — Retrieved from: ____________________

2. NOU 2023: 19, Læring, hvor ble det av deg i alt mylderet? Bruk av elev- og studentdata for å fremme læring (June 2023) — the commission’s final report. — Retrieved from: ____________________

3. Wasson, Giannakos, Blikstad-Balas, Uppstad, Langford & Bøhn (2024), “Implementing Learning Analytics in Norway: Four Central Dilemmas,” Journal of Learning Analytics 11(2), 268–280 — written by commission members; the interim report and its four dilemmas. — Retrieved from: ____________________

4. Hilliger et al. (2020), Internet and Higher Education — the LALA companion at multi-country participatory scale (Case 91). — Retrieved from: ____________________

CASE 93 Singapore SkillsFuture — National Workforce Capability at Scale 2015 – present

1. SkillsFuture Singapore (SSG), Year-in-Review 2024 — program metrics and outcome reporting. — Retrieved from: ____________________

2. Ministry of Education (MOE), Singapore, parliamentary replies on TRAQOM, 2020. — Retrieved from: ____________________

3. International Labour Organization (ILO), “Investigating an Upskilling Programme in Singapore” — international comparative analysis. — Retrieved from: ____________________

4. “Future-Skilling the Workforce: SkillsFuture Movement in Singapore,” Springer, 2024 — peer-reviewed program analysis. — Retrieved from: ____________________

5. Ministry of Trade and Industry (MTI), Singapore, 2018 — WSQ wage-premium study. — Retrieved from: ____________________

CASE 94 Learning Analytics on the African Continent — A Scoping Review as the Present-State Map 2022

1. Prinsloo, P., & Kaliisa, R. (2022), “Learning Analytics on the African Continent: An Emerging Research Focus and Practice,” Journal of Learning Analytics 9(2), 218–235. — Retrieved from: ____________________

2. Lemmens, J. C., & Henn, M. (2016), “Learning analytics: A South African higher education perspective,” in Botha & Muller (eds.), Institutional Research in South African Higher Education, 231–253, SUN PReSS — one of the fifteen included studies. — Retrieved from: ____________________

3. Janse van Vuuren, E. C. (2020), “Development of a contextualised data analytics framework in South African higher education: evolvement of teacher (teaching) analytics as an indispensable component,” South African Journal of Higher Education 34(1) — one of the fifteen included studies. — Retrieved from: ____________________

4. Cross-reference: the African data-privacy governance case earlier in the corpus, for the construct-travel problem stated in governance terms. — Retrieved from: ____________________

CASE 95 PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop 1997–2024

1. Bradshaw, Mitchell, & Leaf (2010), “Examining the Effects of Schoolwide Positive Behavioral Interventions and Supports on Student Outcomes: Results From a Randomized Controlled Effectiveness Trial in Elementary Schools,” Journal of Positive Behavior Interventions, 12(3), 133 – 148. — Retrieved from: ____________________

2. Bradshaw, Waasdorp, & Leaf (2012), “Effects of School-Wide Positive Behavioral Interventions and Supports on Child Behavior Problems,” Pediatrics, 130(5), e1136 – e1145. — Retrieved from: ____________________

3. Horner, Todd, Lewis-Palmer, Irvin, Sugai, & Boland (2004), “The School-Wide Evaluation Tool (SET): A Research Instrument for Assessing School-Wide Positive Behavior Support,” Journal of Positive Behavior Interventions, 6(1), 3 – 12. — Retrieved from: ____________________

4. McIntosh, Massar, Algozzine, George, Horner, Lewis, & Swain-Bradway (2017), “Technical Adequacy of the SWPBIS Tiered Fidelity Inventory,” Journal of Positive Behavior Interventions, 19(1), 3 – 13. — Retrieved from: ____________________

5. McIntosh, Mercer, Nese, Strickland-Cohen, & Hoselton (2016), “Predictors of Sustained Implementation of School-Wide Positive Behavioral Interventions and Supports,” Journal of Positive Behavior Interventions, 18(4), 209 – 218. — Retrieved from: ____________________

6. Bradshaw, Pas, et al. (2012), “A State-Wide Partnership to Promote Safe and Supportive Schools: The PBIS Maryland Initiative,” Administration and Policy in Mental Health, 39(4), 225 – 237. — Retrieved from: ____________________

CASE 96 **AeroPerú Flight 603** 1996

1. Peru Dirección General de Transporte Aéreo, Accident Investigation Board, final report on AeroPerú 603 (2 October 1997; with NTSB/FAA/Boeing participation) — primary cause and the instrument cascade (paraphrased). — Retrieved from: ____________________

2. Peru DGTA Accident Investigation Board report (2 October 1997) — the static-port tape and the contradictory warnings. — Retrieved from: ____________________

3. Peru DGTA Accident Investigation Board report (2 October 1997) — the radio altimeter as “the only reliable element remaining to them,” and the unheeded GPWS terrain warnings. — Retrieved from: ____________________

4. Leveson, N. (2011), Engineering a Safer World (STAMP) — common-cause failure. — Retrieved from: ____________________

5. Dismukes, Berman & Loukopoulos (2007), The Limits of Expertise — crew performance under instrument failure. — Retrieved from: ____________________

CASE 97 **Boeing 737 MAX / MCAS** 2018 – 2019

1. U.S. House Committee on Transportation and Infrastructure, The Boeing 737 MAX Aircraft: Costs, Consequences, and Lessons from Its Design, Development, and Certification (Final Committee Report, Sept. 2020) — the MCAS omission, the 2013 minutes, the 2016 survey, and the “diminished safety” conclusion. — Retrieved from: ____________________

2. U.S. House report (2020) and MCAS design summary — MCAS firing on a single angle-of-attack sensor; the Lion Air 610 (189 killed) and Ethiopian 302 (157 killed) accident sequences. — Retrieved from: ____________________

3. U.S. House Transportation Committee report (2020) — Boeing internal communications and the certification-and-training-impact reasoning (quoted). — Retrieved from: ____________________

4. U.S. Department of Transportation, Office of Inspector General, correspondence and review of FAA oversight of MCAS and the angle-of-attack systems (2019–2020) — the delegated-certification process. — Retrieved from: ____________________

5. J. Herkert, J. Borenstein & K. Miller, “The Boeing 737 MAX: Lessons for Engineering Ethics,” Science and Engineering Ethics 26 (2020) — certification-by-omission as an engineering-ethics failure. — Retrieved from: ____________________

6. Federal Aviation Administration, Summary of the FAA’s Review of the Boeing 737 MAX (Return-to-Service, 2020) — the MCAS redesign and the simulator-training requirement; and the Aircraft Certification, Safety, and Accountability Act (2020). — Retrieved from: ____________________

CASE 98 Mars Climate Orbiter — Unit Mismatch 1999

1. NASA, Mars Climate Orbiter Mishap Investigation Board: Phase I Report (Nov. 1999) — mission, the Lockheed/JPL split, and the program context. — Retrieved from: ____________________

2. NASA MCO MIB Report (1999) — the pound-force-second vs newton-second mismatch (factor 4.45), the accumulated navigation error, and atmospheric destruction on 23 Sept. 1999 (root-cause statement quoted). — Retrieved from: ____________________

3. NASA MCO MIB Report (1999) — the missing verified interface specification, the absent end-to-end validation, and the unresolved cruise-trajectory anomalies. — Retrieved from: ____________________

4. N. G. Leveson, Engineering a Safer World (MIT Press, 2011) — interfaces as engineering deliverables requiring an owner and a verification step. — Retrieved from: ____________________

5. Sauser, B. J., Reilly, R. R., & Shenhar, A. J. (2009), "Why projects fail? How contingency theory can provide new insights — A comparative analysis of NASA's Mars Climate Orbiter loss," International Journal of Project Management 27(7), 665–679. — Retrieved from: ____________________

CASE 100 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010

1. National Transportation Safety Board (2010). Introduction of Glass Cockpit Avionics into Light Aircraft, Safety Study NTSB/SS-10/01. https://www.ntsb.gov/safety/safety-studies/Documents/SS1001.pdf — the case's primary investigation document. — Retrieved from: ____________________

2. NTSB Safety Recommendations A-10-36 through A-10-41 (2010), issued to the FAA — the engineering response the open verdict points to. — Retrieved from: ____________________

3. Wiener, E. L., & Curry, R. E. (1980). Flight-deck automation: Promises and problems. Ergonomics, 23(10):995–1011 — the foundational literature on automation-induced failure modes that the glass-cockpit transition re-introduced at the GA scale. — Retrieved from: ____________________

4. Sarter, N. B., Woods, D. D., & Billings, C. E. (1997). Automation surprises. In Handbook of Human Factors and Ergonomics (2nd ed.) — the mode-confusion / automation-surprise literature the NTSB findings cross-reference. — Retrieved from: ____________________

CASE 101 Ariane 5 Flight 501 1996

1. J. L. Lions (chair), Ariane 5 Flight 501 Failure Inquiry Board Report (1996) — the data-conversion overflow (quoted). — Retrieved from: ____________________

2. Lions Report (1996) — the 39-second disintegration and the automatic self-destruct. — Retrieved from: ____________________

3. ESA Science & Technology, "Cluster — Summary" — the loss of the four Cluster spacecraft and the 214 MECU recovery mission approved by the SPC. — Retrieved from: ____________________

4. Lions Report (1996) — the 64-bit-to-16-bit conversion overflow and the simultaneous shutdown of both inertial reference systems. — Retrieved from: ____________________

5. Lions Report (1996) — the reused code path neither removed nor re-verified for Ariane 5's envelope. — Retrieved from: ____________________

6. N. Leveson, Safeware (1995) — software-reuse hazards; G. Le Lann, "An Analysis of the Ariane 5 Flight 501 Failure" (1997). — Retrieved from: ____________________

7. Cf. Patriot (Case 129) and Mars Climate Orbiter (Case 98). — Retrieved from: ____________________

CASE 102 Air France Flight 447 2009

1. Bureau d'Enquêtes et d'Analyses (BEA), Final Report on the accident on 1 June 2009 to the Airbus A330-203 registered F-GZCP operated by Air France flight AF447 (July 2012) — full report: flight, aircraft, and the known pitot-icing vulnerability with retrofit pending. — Retrieved from: ____________________

2. BEA, Final Report AF447 (2012) — accident sequence: autopilot/autothrust disconnection, degraded control law, sustained nose-up input, stall, and the stall-warning logic dropping out at high angle of attack. — Retrieved from: ____________________

3. BEA, Final Report AF447 (2012) — probable cause: airspeed inconsistency from pitot icing, inappropriate control inputs, and failure to recognize and recover from the stall. — Retrieved from: ____________________

4. BEA, Final Report AF447 (2012) — finding on warning-system ergonomics and stall-training conditions (quoted). — Retrieved from: ____________________

5. G. Palmer / E. Strickland, "Air France Flight 447 Crash Caused by a Combination of Factors," IEEE Spectrum (2014) — analysis of the divergence between the trained and operational envelopes. — Retrieved from: ____________________

6. BEA, Final Report AF447 (2012), safety recommendations — pitot-probe replacement, manual high-altitude flying, approach-to-stall and stall recovery, unreliable-airspeed procedures, and angle-of-attack indication. — Retrieved from: ____________________

7. European Union Aviation Safety Agency, Upset Prevention and Recovery Training (UPRT) requirements for airline pilots (phased in by 2019); ICAO, Manual on Aeroplane Upset Prevention and Recovery Training (Doc 10011). — Retrieved from: ____________________

8. Federal Aviation Administration, Qualification, Service, and Use of Crewmembers and Aircraft Dispatchers, final rule, 78 FR 67800 (2013, effective 2014) — Part 121 stall and upset recovery training, directed by the Airline Safety and FAA Extension Act of 2010; FAA Advisory Circular 120-109A, Stall Prevention and Recovery Training. — Retrieved from: ____________________

CASE 103 Kegworth / British Midland 92 1989

1. Air Accidents Investigation Branch, Report on the accident to Boeing 737-400 G-OBME near Kegworth, Leicestershire, on 8 January 1989, AAIB Aircraft Accident Report 4/90 (1990) — aircraft, route, and the limited conversion training; see also U.S. FAA, Lessons Learned: G-OBME. — Retrieved from: ____________________

2. AAIB Report 4/90 (1990) — accident sequence: left-engine fan-blade failure, shutdown of the serviceable right engine, and total power loss on approach; 47 killed, 74 seriously injured. — Retrieved from: ____________________

3. AAIB Report 4/90 (1990) — the older-737 bleed-air mental model misapplied to the 737-400's changed configuration. — Retrieved from: ____________________

4. AAIB Report 4/90 (1990) — the new electronic engine instrumentation and the readability of the vibration indicator. — Retrieved from: ____________________

5. AAIB Report 4/90 (1990) — the cabin-to-flight-deck communication gap (crew resource management). — Retrieved from: ____________________

6. AAIB Report 4/90 (1990) — the 31 safety recommendations on difference training, instrument design, CRM, and crashworthiness. — Retrieved from: ____________________

7. J. Reason, Human Error (Cambridge Univ. Press, 1990) — Kegworth as a case in the misapplication of a correct mental model to a changed system. — Retrieved from: ____________________

CASE 104 Asiana Airlines Flight 214 2013

1. NTSB, Aircraft Accident Report: Asiana Airlines Flight 214, NTSB/AAR-14/01 (2014) — probable cause and contributing factors (quoted). — Retrieved from: ____________________

2. NTSB/AAR-14/01 (2014) — the autothrottle HOLD reversion and airspeed decay to 103 kt. — Retrieved from: ____________________

3. NTSB/AAR-14/01 (2014) — the captain's mental model of the autothrottle. — Retrieved from: ____________________

4. Sarter, N. & Woods, D. (1995), "How in the world did we ever get into that mode?," Human Factors — automation surprise. — Retrieved from: ____________________

5. NTSB/AAR-14/01 (2014), safety recommendations A-14-37 to A-14-43 and A-14-55 on autoflight training, low-energy alerting, interface intuitiveness, and manual flight. — Retrieved from: ____________________

CASE 105 Helios Airways Flight 522 2005

1. FAA Airworthiness Directive 2011-03-14, Amendment 39-16598, 76 FR 6529 (7 February 2011) — the mandated installation of separate CABIN ALTITUDE and TAKEOFF CONFIG warning lights on the 737 Classic fleet. — Retrieved from: ____________________

2. Hellenic Air Accident Investigation & Aviation Safety Board, Aircraft Accident Report 11/2006 — probable cause and the warning misinterpretation (paraphrased). — Retrieved from: ____________________

3. AAIASB Report 11/2006 — the manual pressurization selector and failure to pressurize. — Retrieved from: ____________________

4. AAIASB Report 11/2006 — the dual-meaning warning horn and the training emphasis. — Retrieved from: ____________________

5. Boeing 737 Flight Crew Operations Manual — configuration and cabin-altitude warnings. — Retrieved from: ____________________

6. Reason, J. (1990), Human Error — the cue-ambiguity / human-error framing the case draws on. — Retrieved from: ____________________

CASE 106 TransAsia Airways Flight 235 2015

1. Aviation Safety Council (Taiwan), Aviation Occurrence Report: TransAsia Airways Flight GE235, final report (2016) — findings and the wrong-engine shutdown (paraphrased). — Retrieved from: ____________________

2. ASC final report (2016) — the autofeather event and the crash sequence. — Retrieved from: ____________________

3. ASC final report (2016) — non-use of the checklist and the captain's proficiency record. — Retrieved from: ____________________

4. AAIB (UK), British Midland Flight 92 (Kegworth) report (1990) — the antecedent wrong-engine case. — Retrieved from: ____________________

5. Dismukes, Berman & Loukopoulos (2007), The Limits of Expertise — memory-driven response under time pressure. — Retrieved from: ____________________

CASE 107 Eastern Air Lines Flight 401 1972

1. NTSB, Aircraft Accident Report: Eastern Air Lines Flight 401, NTSB-AAR-73-14 (1973) — the night approach and full flight deck. — Retrieved from: ____________________

2. NTSB-AAR-73-14 (1973) — the landing-gear-bulb fixation, the inadvertent autopilot disengagement, and 101 deaths of 176 aboard. — Retrieved from: ____________________

3. NTSB-AAR-73-14 (1973) — the crew's preoccupation with the landing-gear indication that distracted them from maintaining altitude (paraphrased). — Retrieved from: ____________________

4. C. D. Wickens et al., Engineering Psychology and Human Performance — attention, monitoring, and alert design. — Retrieved from: ____________________

5. R. Helmreich & H. Foushee (1993), aircrew-coordination research — the Crew Resource Management origins. — Retrieved from: ____________________

6. Cooper, G. E., White, M. D. & Lauber, J. K. (1980), Resource Management on the Flightdeck, NASA CP-2120 — proceedings of the 1979 NASA/industry workshop that named Crew Resource Management. — Retrieved from: ____________________

CASE 108 NASA EVA-23 — Water Intrusion Inside a Spacesuit 2013

1. NASA International Space Station Program (2013), "International Space Station EVA Suit Water Intrusion High Visibility Close Call" — Mishap Investigation Board final report, December 2013. — Retrieved from: ____________________

2. NASA Aerospace Safety Advisory Panel (2014), Annual Report — EVA-23 follow-up actions including helmet absorption pad and snorkel. — Retrieved from: ____________________

3. Parmitano, L. (2013), "EVA 23: exploring the frontier," ESA astronaut blog, 20 August 2013 — the crewmember's account of the water event and the return traverse. — Retrieved from: ____________________

4. Chappell, Norcross, Abercromby et al. (2017), "Risk of Injury and Compromised Performance Due to EVA Operations," NASA Human Research Program Evidence Report — broader EVA risk context. — Retrieved from: ____________________

CASE 109 Colgan Air Flight 3407 2009

1. NTSB, Aircraft Accident Report: Colgan Air Flight 3407, NTSB/AAR-10/01 (2010) — probable cause (quoted). — Retrieved from: ____________________

2. NTSB/AAR-10/01 (2010) — the captain's training history, and the first officer's fatigue and pay. — Retrieved from: ____________________

3. NTSB/AAR-10/01 (2010), §2.7.3 — PRIA and FAA guidance did not require operators to obtain notices of disapproval; recommendation A-05-1 reiterated as "Open—Unacceptable Response". — Retrieved from: ____________________

4. Airline Safety and Federal Aviation Administration Extension Act of 2010, Pub. L. 111-216 — the 1,500-hour rule and the Pilot Records Database. — Retrieved from: ____________________

5. DOT Office of Inspector General, FAA Delays in Establishing a Pilot Records Database Limit Air Carriers' Access to Background Information, AV-2015-079 (2015); the Families of Continental Flight 3407 campaign. — Retrieved from: ____________________

CASE 110 Atlas Air Flight 3591 2019

1. NTSB, Aircraft Accident Report: Atlas Air Flight 3591, NTSB/AAR-20/02 (2020) — probable cause and contributing factors (paraphrased). — Retrieved from: ____________________

2. NTSB/AAR-20/02 (2020) — the inadvertent go-around activation and the dive into Trinity Bay. — Retrieved from: ____________________

3. NTSB/AAR-20/02 (2020) — the first officer's training history and Atlas's record-access limits. — Retrieved from: ____________________

4. FAA Pilot Records Database final rule (Federal Register, 10 June 2021; subpart B/C compliance from June 2022, full historical coverage by September 2024); Pilot Records Improvement Act of 1996. — Retrieved from: ____________________

5. DOT Office of Inspector General, FAA Delays in Establishing a Pilot Records Database Limit Air Carriers' Access to Background Information, AV-2015-079 (2015) — the decade of implementation delay. — Retrieved from: ____________________

CASE 111 Challenger & Columbia

1986 / 2003

1. Rogers Commission, Report of the Presidential Commission on the Space Shuttle Challenger Accident (1986) — the O-ring failure, the Thiokol teleconference, and the cold-weather launch decision. — Retrieved from: ____________________

2. D. Vaughan, The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA (Univ. of Chicago Press, 1996) — normalization of deviance; the working group's drift of decision rules. — Retrieved from: ____________________

3. Columbia Accident Investigation Board, Report Vol. I (2003) — the foam strike, findings F3.2-5 and F3.2-7 on foam loss across the imaged missions and its filing as "in-family," the recurrence of the cultural pattern, and the call for an independent technical authority. — Retrieved from: ____________________

4. CAIB (2003) — "the NASA organizational culture had as much to do with this accident as the foam" (quoted); Rogers Commission (1986) and CAIB (2003) jointly on the suppressed upward flow of safety information across both accidents. — Retrieved from: ____________________

5. W. Starbuck & M. Farjoun (eds.), Organization at the Limit: Lessons from the Columbia Disaster (2005) — independent academic re-analyses of the institutional-learning gap. — Retrieved from: ____________________

6. NASA Engineering and Safety Center founding documents (2003 – present) — institutional response to the CAIB's call for independent technical authority; Shuttle retirement (STS-135, July 2011). — Retrieved from: ____________________

CASE 112 NASA Saturn V Documentation

1972 – present

1. NASA Constellation Program documentation and reviews (2005–2010) — the difficulty of reproducing Saturn V capability. — Retrieved from: ____________________

2. R. E. Bilstein, Stages to Saturn: A Technological History of the Apollo/Saturn Launch Vehicles (NASA SP-4206, 1980) — the program and its workforce. — Retrieved from: ____________________

3. The documentation-vs-capability distinction (editors' synthesis of the Saturn V record). — Retrieved from: ____________________

4. M. Polanyi, The Tacit Dimension (1966); I. Nonaka & H. Takeuchi, The Knowledge-Creating Company (1995). — Retrieved from: ____________________

5. J. S. Brown & P. Duguid, The Social Life of Information (2000) — tacit and institutional knowledge. — Retrieved from: ____________________

CASE 113 Boeing Starliner

2019 – 2025

1. U.S. GAO, NASA Commercial Crew Program: Significant Work Remains to Begin Operational Missions to the Space Station, GAO-20-121 (Jan. 2020); Schedule Uncertainty Persists for Start of Operational Missions, GAO-19-504 (June 2019). — Retrieved from: ____________________

2. NASA and Boeing joint Independent Review Team, Orbital Flight Test findings (7 July 2020) — 80 recommendations across software requirements, testing and simulation, process, and hardware. — Retrieved from: ____________________

3. NASA OIG, NASA's Management of Its Commercial Crew Program, IG-26-011 (30 June 2026); NASA, Starliner Propulsion System Anomalies during the Crewed Flight Test (redacted, February 2026). — Retrieved from: ____________________

4. NASA Aerospace Safety Advisory Panel, annual reports (2020–2025) — Commercial Crew oversight and the Panel's criticism of NASA's mishap-classification requirements. — Retrieved from: ____________________

5. NASA, Starliner's uncrewed landing at White Sands Space Harbor (6 September 2024) and the Crew-9 return of Wilmore and Williams aboard a SpaceX Dragon (18 March 2025). — Retrieved from: ____________________

6. Cf. Saturn V (Case 112); N. Augustine, Augustine's Laws (1986). — Retrieved from: ____________________

CASE 114 FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule 1988 – 2010s

1. NTSB (1989), Aircraft Accident Report AAR-89/03, Aloha Airlines Flight 243, Boeing 737-200, N73711. — Retrieved from: ____________________

2. FAA, 14 CFR Part 26, "Continued Airworthiness and Safety Improvements for Transport Category Airplanes," Final Rule, 72 FR 63409 (November 8, 2007) — the rule that created Part 26. — Retrieved from: ____________________

3. FAA, "Aging Airplane Program: Widespread Fatigue Damage," Final Rule, 75 FR 69746 (November 15, 2010, effective January 14, 2011) — the subpart C limit-of-validity provisions. — Retrieved from: ____________________

4. Airworthiness Assurance Working Group (1999), Recommendations for Regulatory Action to Prevent Widespread Fatigue Damage in the Commercial Airplane Fleet, Final Report, March 11, 1999 (Rev. A, June 29, 1999) — the ARAC record behind the rule. — Retrieved from: ____________________

5. Swift, T. (1993), "Widespread Fatigue Damage Monitoring — Issues and Concerns," International Conference on Structural Airworthiness of New and Aging Aircraft, Hamburg, June 16–18, 1993 — technical synthesis of the WFD inspection regime. — Retrieved from: ____________________

CASE 115 FAA NextGen and the ADS-B Out Transition 2003 – 2020

1. FAA, "Automatic Dependent Surveillance-Broadcast (ADS-B) Out Performance Requirements To Support Air Traffic Control (ATC) Service," Final Rule, 75 FR 30160 (May 28, 2010) — 14 CFR §§91.225 and 91.227. — Retrieved from: ____________________

2. Vision 100 — Century of Aviation Reauthorization Act, Public Law 108-176 (2003) — NextGen program statutory basis. — Retrieved from: ____________________

3. Government Accountability Office, "NextGen Air Transportation System" report series (2010s) — sustained external audit record on schedule slippage and benefit-realization gaps. — Retrieved from: ____________________

4. Department of Transportation Office of Inspector General (2021), NextGen Benefits Have Not Kept Pace With Initial Projections, Report No. AV2021023, March 30, 2021 — FAA's 2017 benefits projection $113 billion below the JPDO estimate; $6 billion in benefits realized 2010–2018. — Retrieved from: ____________________

5. FAA, NextGen Annual Reports (2010 – present) — program-self-report tier; read against the GAO and DOT IG reviews. — Retrieved from: ____________________

CASE 116 Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change 2005

1. DelaPeyronnie, Newcomb, Morillo, Trimech, Nguyen, & Purtill (2010), "Modernization of the Eurocat Air Traffic Management System (EATMS)," in Information Systems Transformation: Architecture-Driven Modernization Case Studies (Elsevier / Morgan Kaufmann), Chapter 5. — Retrieved from: ____________________

2. Ulrich & Newcomb (eds., 2010), Information Systems Transformation — the host volume on architecture-driven modernization patterns. — Retrieved from: ____________________

3. Brodie & Stonebraker (1995), Migrating Legacy Systems — the framing reference on small-step legacy modernization. — Retrieved from: ____________________

4. Bisbal et al. (1999), "Legacy Information Systems: Issues and Directions," IEEE Software 16(5):103–111 — peer-reviewed framing companion. — Retrieved from: ____________________

CASE 117 Crew Resource Management & CAST 1981 – present

1. FAA Advisory Circular 120-51E, Crew Resource Management Training — CRM protocols and authority-gradient training. — Retrieved from: ____________________

2. Helmreich, Merritt & Wilhelm (1999), "The Evolution of Crew Resource Management Training in Commercial Aviation," International Journal of Aviation Psychology. — Retrieved from: ____________________

3. Spanish CIAIAC / ALPA reports on the 1977 Tenerife collision — the overridden crew challenge. — Retrieved from: ____________________

4. CAST/FAA Safety Enhancement reports (2016, 2018) — the closed-loop data process and the 83% reduction. — Retrieved from: ____________________

5. Collier Trophy citation (2008); Kanki, Helmreich & Anca (2010), Crew Resource Management. — Retrieved from: ____________________

CASE 118 Korean Air Safety Transformation 2000 – present

1. NTSB (2000), AAR-00/01, Controlled Flight Into Terrain, Korean Air Flight 801, Boeing 747-300, HL7468, Nimitz Hill, Guam, August 6, 1997 — the captain's approach failure and the crew's failure to monitor and challenge; contributing, the captain's fatigue and Korean Air's inadequate flight crew training. — Retrieved from: ____________________

2. Air Transport World Phoenix Award documentation (2006) — recognition of the turnaround. — Retrieved from: ____________________

3. Gladwell, M. (2008), Outliers — the Korean Air chapter on power distance and the English-language change. — Retrieved from: ____________________

4. Helmreich, Wilhelm, Klinect & Merritt (2001), "Culture, Error, and Crew Resource Management," in Salas, Bowers & Edens (eds.), Improving Teamwork in Organizations (Erlbaum) — national culture and CRM adaptation. — Retrieved from: ____________________

5. Korean Air corporate safety reports — the post-2000 accident-free record. — Retrieved from: ____________________

CASE 119 Aviation Safety Reporting System (ASRS) 1976 – present

1. NASA ASRS Program documentation and annual reports — system design, immunity provision, and report volume. — Retrieved from: ____________________

2. FAA Advisory Circular 00-46F, "Aviation Safety Reporting Program" — the program's establishment in April 1975, the FAA–NASA Memorandum of Agreement of 15 August 1975, and the immunity provisions. — Retrieved from: ____________________

3. Reason, J. (1997), Managing the Risks of Organizational Accidents — non-punitive reporting as a model (paraphrased). — Retrieved from: ____________________

4. NASA ASRS technical reports (Connell et al.) — patterns first surfaced via ASRS data. — Retrieved from: ____________________

5. Dekker, S. (2012), Just Culture — the cultural commitment to non-punitive use. — Retrieved from: ____________________

6. CHIRP and patient-safety reporting-system documentation — cross-domain emulation. — Retrieved from: ____________________

CASE 120 EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation 1996 – 2002

1. Bateman, C. D. (1999), "The Introduction of Enhanced Ground Proximity Warning Systems (EGPWS) into Civil Aviation Operations Around the World," in Proceedings of the 11th Annual European Aviation Safety Seminar, Flight Safety Foundation, pp. 259–274 — developer history. — Retrieved from: ____________________

2. Federal Aviation Administration (2000), 14 CFR §§ 91.223, 121.354, 135.154 — Terrain Awareness and Warning System (TAWS) equipage requirement. — Retrieved from: ____________________

3. Flight Safety Foundation (1998 – 2000), CFIT / ALAR Task Force reports — operational and outcome analyses motivating mandate. — Retrieved from: ____________________

4. Aeronáutica Civil of the Republic of Colombia (1996), Aircraft Accident Report: Controlled Flight Into Terrain, American Airlines Flight 965, Boeing 757-223, N651AA, near Cali, Colombia, December 20, 1995 (NTSB participating, DCA96RA020). — Retrieved from: ____________________

5. Royal Commission to Inquire into the Crash on Mount Erebus, Antarctica of a DC10 Aircraft Operated by Air New Zealand Limited (1981), final report (Mahon report). — Retrieved from: ____________________

CASE 121 TCAS — Coordinated Collision Avoidance and the Überlingen Lesson

1981 – 2008 (TCAS II Version 7.1)

1. RTCA (2008), DO-185B "Minimum Operational Performance Standards for Traffic Alert and Collision Avoidance System II (TCAS II)" — Version 7.1 with reversal logic. — Retrieved from: ____________________

2. Bundesstelle für Flugunfalluntersuchung (BFU) (2004), AX001-1-2/02 — Investigation Report on the mid-air collision on 1 July 2002 near Überlingen. — Retrieved from: ____________________

3. Commission Regulation (EU) No 1332/2011 — ACAS II version 7.1 required in EU airspace from 1 March 2012 for new aeroplanes and from 1 December 2015 for the existing fleet. — Retrieved from: ____________________

4. Federal Aviation Administration, 14 CFR § 121.356 — TCAS II equipage requirement. — Retrieved from: ____________________

5. MIT Lincoln Laboratory / FAA, "Safety analysis of upgrading to TCAS Version 7.1 using the 2008 U.S. Correlated Encounter Model" (2009) — the quantitative basis for the V7.1 change. — Retrieved from: ____________________

CASE 122 Singapore Airlines Safety Transformation

1980s – present

1. Aviation Safety Council (Taiwan), Crashed on a Partially Closed Runway during Takeoff, Singapore Airlines Flight 006, Boeing 747-400, 9V-SPK, CKS Airport, Taoyuan, October 31, 2000 (2002) — the accident, the eight probable causes, and Singapore's dissenting comments at §7. — Retrieved from: ____________________

2. Henderson, J. C. (2003), "Communicating in a crisis: flight SQ 006," Tourism Management 24(3):279–287 — the analysis of Singapore Airlines' early crisis response. — Retrieved from: ____________________

3. Singapore Airlines, "Singapore Airlines keeps fleet young" (SilverKris/company account) and SIA annual reports — the young-fleet policy in the airline's own words; in-service average fleet age about 8.3 years as of 2026. — Retrieved from: ____________________

4. Helmreich, Wilhelm, Klinect & Merritt (2001), "Culture, error, and crew resource management," in Salas, Bowers & Edens (eds.), Improving Teamwork in Organizations — national culture and CRM adaptation. — Retrieved from: ____________________

5. Weick & Sutcliffe (2007), Managing the Unexpected — sustained high-reliability investment. — Retrieved from: ____________________

CASE 123 FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike

1992 – 2000s

1. Flight Safety Foundation, "Killers in Aviation: FSF Task Force Presents Facts About Approach-and-landing and Controlled-flight-into-terrain Accidents," Flight Safety Digest (1998–1999). — Retrieved from: ____________________

2. Flight Safety Foundation, ALAR Tool Kit (2000; updated 2010) — distributed multi-element package built on the 34 ALAR Briefing Notes published in Flight Safety Digest, August–November 2000. — Retrieved from: ____________________

3. Khatwa & Helmreich, "Analysis of critical factors during approach and landing in accidents and normal flight," Flight Safety Digest (1998) — the analytical basis of the ALAR Task Force scope. — Retrieved from: ____________________

4. FAA, Terrain Awareness and Warning System (TAWS) Final Rule, 65 FR 16736 (29 March 2000), 14 CFR §§ 91.223, 121.354, 135.154 — new-build compliance 29 March 2002, retrofit by 29 March 2005. — Retrieved from: ____________________

5. ICAO Annex 6, TAWS requirement (effective 2007) — the international consolidation. — Retrieved from: ____________________

CASE 124 USS Fitzgerald & USS John S. McCain 2017

1. T. C. Miller, M. Faturechi & R. Rotella, “Years of Warnings, Then Death and Disaster,” ProPublica (2019) — the 2003 SWOS-in-a-Box shift. — Retrieved from: ____________________

2. GAO, Navy Readiness, GAO-17-809T (Sept. 2017) — 37% of Japan-based warfare certifications expired by June 2017. — Retrieved from: ____________________

3. NTSB, Collision between USS Fitzgerald and MV ACX Crystal, NTSB/MAR-20/02 (2020), DCA17PM018. — Retrieved from: ____________________

4. NTSB, Collision between USS John S. McCain and Tanker Alnic MC, NTSB/MAR-19/01 (2019), DCA17PM029. — Retrieved from: ____________________

5. U.S. Navy, Comprehensive Review of Recent Surface Force Incidents (Nov. 2017) — both collisions judged avoidable. — Retrieved from: ____________________

6. NTSB/MAR-19/01 (2019) — probable cause: insufficient training and inadequate bridge procedures from a lack of Navy oversight. — Retrieved from: ____________________

7. R. Rotella et al., “The Navy Installed Touch-Screen Steering Systems to Save Money,” ProPublica (2019). — Retrieved from: ____________________

8. U.S. Navy, Strategic Readiness Review (Dec. 2017) — risks that “accumulated over time and did so insidiously.” — Retrieved from: ____________________

9. U.S. Navy corrective actions (2017); NTSB/MAR-19/01 recommendations; USNI News (2017–2020). — Retrieved from: ____________________

10. Surface Warfare Officers School Command, Basic Division Officer Course curriculum and the post-2017 return to in-person instruction; Naval Surface Group Western Pacific stand-up (2018) as the forward-readiness certification authority. — Retrieved from: ____________________

11. NTSB/MAR-19/01 (2019) — watchstander fatigue findings, including average sleep hours and the touch-screen helm misdiagnosis sequence. — Retrieved from: ____________________

CASE 126 F-35 Sustainment & the Maintainer Capability Gap ongoing

1. U.S. Government Accountability Office, F-35 Aircraft: DOD and the Military Services Need to Reassess the Future Sustainment Strategy, GAO-23-105341 (Sept. 2023) — 2,500 planned aircraft and a lifecycle cost exceeding $1.7 trillion, $1.3 trillion of it in operating and support. — Retrieved from: ____________________

2. GAO-23-105341 (2023) — 55% fleet mission-capable rate (March 2023); over 10,000 components awaiting repair; 141-day average depot repair cycle time against a 60-to-90-day goal; depot stand-up behind schedule. — Retrieved from: ____________________

3. GAO-23-105341 (2023) — sustainment shortfall traced to insufficient technical data, training and support equipment for maintainers, depot stand-up delays, and contractor dependency. — Retrieved from: ____________________

4. GAO, F-35 Sustainment: DOD Faces Several Uncertainties..., GAO-22-105995 (2022), and the broader GAO F-35 series — the recurring, year-over-year diagnosis. — Retrieved from: ____________________

5. GAO-23-105341 (2023) — recommendation that DOD reassess the future sustainment strategy. — Retrieved from: ____________________

6. GAO, F-35 Sustainment: Costs Continue to Rise While Planned Use and Availability Have Decreased, GAO-24-106703 (2024) — costs rising while readiness stays below goal. — Retrieved from: ____________________

CASE 127 Military Fratricide — Desert Storm to Afghanistan 1991 – 2014

1. C. R. Shrader, Amicicide: The Problem of Friendly Fire in Modern War (U.S. Army Combat Studies Institute, 1982) — the under-2% estimate from 269 incidents, a baseline later challenged as too low (OTA 1993: 15 to 20% may be the norm). — Retrieved from: ____________________

2. “Friendly Fire: Facts, Myths and Misperceptions,” U.S. Naval Institute Proceedings (June 1994) — Desert Storm: 35 of 146 KIA (24%) and 72 of 467 wounded (15%) by friendly fire; critique of the 2% “historical norm.” — Retrieved from: ____________________

3. Khafji and Warrior fratricide incidents (Feb. 1991) — see USNI Proceedings (1994) and Time, “They Didn’t Have to Die”. (Per-incident casualty figures vary across sources; see AUDIT.) — Retrieved from: ____________________

4. U.S. GAO, Operation Desert Storm fratricide investigations — Apache incident (OSI-93-4) and Army fratricide investigation (OSI-95-10) — causes and the Army’s reviews. — Retrieved from: ____________________

5. Post-war combat-identification reviews — leading causes (situational awareness, fire-control measures, combat ID) and the absence of a comprehensive incident database. (Synthesized from secondary analyses; see AUDIT.) — Retrieved from: ____________________

6. Later-conflict recurrences (2001 coordinate-reset strike; 2014 B-1 infrared-strobe strike) and S. Snook, Friendly Fire (Princeton Univ. Press, 2000) on the 1994 Black Hawk shootdown — the systems-integration archetype. — Retrieved from: ____________________

7. “IFF Update: Stalled Again,” U.S. Naval Institute Proceedings (June 1994) — the combat-identification and identification-friend-or-foe programs pursued after Desert Storm, and how slowly they matured. — Retrieved from: ____________________

CASE 128 Operation Eagle Claw 1980

1. Holloway Special Operations Review Group, Rescue Mission Report (1980) — the ad-hoc assembly and equipment choices (paraphrased). — Retrieved from: ____________________

2. Holloway Commission (1980) — the Desert One sequence and the helicopter–C-130 collision. — Retrieved from: ____________________

3. Goldwater-Nichols Department of Defense Reorganization Act of 1986, Pub. L. 99-433. — Retrieved from: ____________________

4. Nunn-Cohen Amendment (1986), establishing U.S. Special Operations Command (1987). — Retrieved from: ____________________

5. Locher, J. (2002), Victory on the Potomac — the reform arc from Desert One to Goldwater-Nichols. — Retrieved from: ____________________

CASE 129 Patriot Missile / Dhahran 1991

1. U.S. General Accounting Office, Patriot Missile Defense: Software Problem Led to System Failure at Dhahran, Saudi Arabia, GAO/IMTEC-92-26 (1992) — the design-context mismatch and continuous-operation use. — Retrieved from: ____________________

2. R. Skeel, “Roundoff Error and the Patriot Missile,” SIAM News 25(4): 11 (1992) — the 24-bit fixed-point time truncation and the 0.34-second range-gate drift after 100 hours. — Retrieved from: ____________________

3. GAO/IMTEC-92-26 (1992) — the prior Israeli warning, the software patch arriving in Dhahran on 26 February (the day after the strike), and the 28 killed. — Retrieved from: ____________________

4. GAO/IMTEC-92-26 (1992) — the vague "very long run time" advisory and the Army's presumption that batteries would not be run continuously for such periods (quoted). — Retrieved from: ____________________

5. M. Barr / Barr Group, "An Update on the Patriot Missile Software Problem" — an engineering post-mortem of the accumulated-truncation defect. — Retrieved from: ____________________

6. GAO/IMTEC-92-26 (1992) and Skeel (1992) — the design assumption that failed to travel across the change in operational context, and the case's teaching legacy. — Retrieved from: ____________________

CASE 130 USS Vincennes & Iran Air Flight 655 1988

1. Rear Adm. W. Fogarty, Formal Investigation into the Circumstances Surrounding the Downing of Iran Air Flight 655 (U.S. Navy, August 1988) — the engagement, the IFF mode-confusion findings, and the 290 deaths. — Retrieved from: ____________________

2. Fogarty report (1988) — the Aegis SPY-1A radar functioned; the aircraft was ascending while the crew perceived a descent; "scenario fulfillment" as the psychological mechanism. — Retrieved from: ____________________

3. Fogarty report (1988) — "human error under extreme stress," confirmation bias, and "unconscious distortion of data" (quoted); shared-error finding across CIC operators. — Retrieved from: ____________________

4. J. Barry & R. Charles, "Sea of Lies," Newsweek (July 13, 1992) — contemporaneous reinvestigation of the operational record, including the disputed account of Vincennes' position relative to Iranian territorial waters. — Retrieved from: ____________________

5. LTC A. Tingle, USA, "The Human-Machine Team Failed Vincennes", U.S. Naval Institute Proceedings 144/7 (July 2018) — the human-AI-teaming reframing and the standing case-study role. — Retrieved from: ____________________

6. M. L. Cummings, "Human Supervisory Control of Weapon Systems" (MIT) — interface design and automation under time pressure as the engineering frame for the case. — Retrieved from: ____________________

7. G. Klein, Sources of Power (1998); M. Endsley, "Toward a Theory of Situation Awareness" (1995) — the naturalistic-decision-making and situation-awareness literatures that treat Vincennes as the canonical worked example. — Retrieved from: ____________________

CASE 131 Stanislav Petrov / 1983 False Alert 1983

1. D. Hoffman, The Dead Hand: The Cold War Arms Race and Its Dangerous Legacy (2009) — the Oko system and Soviet launch-on-warning posture. — Retrieved from: ____________________

2. Accounts of the 26 Sept. 1983 incident (Hoffman; Petrov interviews) — the five-missile alert and Petrov's assessment. — Retrieved from: ____________________

3. Investigations of the Oko incident (incl. V. Aksenov, 2004) — sunlight-on-clouds satellite geometry as the false-positive cause. — Retrieved from: ____________________

4. The subsequent Oko algorithm modification and the incident's use in early-warning training. — Retrieved from: ____________________

5. S. Petrov, interview, The Washington Post (1999) — "a funny feeling in my gut" (quoted). — Retrieved from: ____________________

6. S. Sagan, The Limits of Safety (1993); M. L. Cummings (2017) — automation in critical decisions and human-in-the-loop. — Retrieved from: ____________________

CASE 132 F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap 2008 – 2012

1. USAF Scientific Advisory Board, Report on Aircraft Oxygen Generation, SAB-TR-11-04 (1 February 2012) — the 2011 Quicklook Study on the F-22 hypoxia-like incidents. — Retrieved from: ____________________

2. NASA Engineering and Safety Center (2012), report contributing to the F-22 hypoxia review. — Retrieved from: ____________________

3. USAF Accident Investigation Board (2011), Capt. Jeffrey Haney F-22 accident report (Nov 2010). — Retrieved from: ____________________

4. U.S. House Armed Services Subcommittee on Tactical Air and Land Forces, F-22 Pilot Physiological Issues, H.A.S.C. No. 112-154 (13 September 2012) — congressional record. — Retrieved from: ____________________

CASE 133 Mark 14 Torpedo Failures

1941 – 1943

1. Blair, C. (1975), Silent Victory: The U.S. Submarine War Against Japan — operator reports and the Bureau's resistance (paraphrased). — Retrieved from: ____________________

2. Rowland, B. & Boyd, W. (1953), U.S. Navy Bureau of Ordnance in World War II — testing policy and the three defects. — Retrieved from: ____________________

3. Blair (1975) — the USS Tinosa's July 1943 attack (eleven consecutive duds on a stopped target) and Lockwood's fleet-level testing. — Retrieved from: ____________________

4. Naval History and Heritage Command, Mark 14 torpedo files — depth error, Mark 6 magnetic exploder, contact-pin failure. — Retrieved from: ____________________

5. Edmondson, A. (2018), The Fearless Organization — suppression of field reports as an organizational-learning failure. — Retrieved from: ____________________

CASE 134 V-22 Osprey

1991 – present

1. Compiled V-22 accident record — 16 hull losses and 62 fatalities since 1991, including the 2000 Marana test crash (19 Marines). — Retrieved from: ____________________

2. U.S. Air Force Accident Investigation Board findings, via USNI News (Aug. 2024) — the 29 Nov. 2023 Yakushima CV-22B crash: transmission-gear cracks (X-53 inclusions) and continued flight despite warnings. — Retrieved from: ____________________

3. U.S. GAO, Osprey Aircraft: Additional Oversight and Information Sharing Would Improve Safety Efforts, GAO-26-107285 (Dec. 2025) — the joint program office's failure to assess and address risks. — Retrieved from: ____________________

4. GAO-26-107285 (2025) — serious-mishap rates exceeding comparable Navy/Air Force fleets (FY2015–FY2024); 28 unresolved system safety risks with a median age of about nine years, over half carried six to fourteen years. — Retrieved from: ____________________

5. NAVAIR independent review of the V-22 (Dec. 2025) — materiel and cross-service-coordination factors and unresolved catastrophic parts issues. — Retrieved from: ____________________

6. USNI News, V-22 program coverage (2024–2025). — Retrieved from: ____________________

CASE 135 9/11 Intelligence Sharing Failures

1996 – 2001

1. National Commission on Terrorist Attacks Upon the United States, The 9/11 Commission Report (2004) — the quoted "failure of imagination" and the specific sharing failures. — Retrieved from: ____________________

2. 9/11 Commission Report (2004) — the Kuala Lumpur tracking and the Phoenix/Minneapolis flagging. — Retrieved from: ____________________

3. Joint Inquiry into Intelligence Community Activities Before and After the Terrorist Attacks of September 11, 2001, S. Rept. 107-351 (2002) — cross-agency information-sharing failures. — Retrieved from: ____________________

4. Zegart, A. (2007), Spying Blind — structural-organizational analysis of the failures. — Retrieved from: ____________________

5. Intelligence Reform and Terrorism Prevention Act of 2004 — creation of the ODNI and NCTC. — Retrieved from: ____________________

CASE 137 U.S. Nuclear Navy / Rickover Training Model

1954 – present

1. Polmar, N. & Allen, T. (2007), Rickover: Father of the Nuclear Navy — the program and Rickover's philosophy (paraphrased). — Retrieved from: ____________________
2. Naval Nuclear Propulsion Program documentation (NRC/DOE) — qualification standards and the accident record. — Retrieved from: ____________________
3. Admiral Hyman G. Rickover, "Doing a Job" (Columbia University, Egleston Medal address, 5 November 1982) — "people, not organizations… get things done" (quoted). — Retrieved from: ____________________
4. GAO-21-168 (2021), Navy Readiness: Actions Needed to Evaluate and Improve Surface Warfare Officer Career Path — the surface community's career path set against the submarine, aviation and EOD communities. — Retrieved from: ____________________
5. Duncan, F. (1990), Rickover and the Nuclear Navy — the qualification culture. — Retrieved from: ____________________

CASE 138 MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

2020 revision (series since 1968)

1. Department of Defense (2020), MIL-STD-1472H "Department of Defense Design Criteria Standard: Human Engineering," 15 September 2020 — replaces MIL-STD-1472G (2012). — Retrieved from: ____________________
2. Department of Defense (2012), MIL-STD-1472G — the prior revision; revision history documents the 1968 origin and intermediate letters. — Retrieved from: ____________________
3. Chapanis, A. (1965), "Man-Machine Engineering" — foundational text for the discipline the standard codifies. — Retrieved from: ____________________
4. Fitts, P. M., & Jones, R. E. (1947), "Analysis of factors contributing to 460 'pilot error' experiences in operating aircraft controls" — origin of designed-error analysis. — Retrieved from: ____________________

CASE 139 GIFT and the Adoption Gap

2012 – present

1. K. VanLehn, "The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems," Educational Psychologist 46(4) (2011) — tutoring effectiveness. — Retrieved from: ____________________
2. GIFT Project, gifttutoring.org — the framework, releases, and development under ARL / DEVCOM. — Retrieved from: ____________________
3. Editors' synthesis of the GIFT adoption record — active development but limited ubiquitous fielding (quoted). — Retrieved from: ____________________
4. R. Sottilare, A. Graesser, X. Hu & H. Holden (eds.), Design Recommendations for Intelligent Tutoring Systems (U.S. Army Research Laboratory, 2013–); IJAIED Special Issue on GIFT (2017). — Retrieved from: ____________________
5. J. Goodell & J. Kolodner, Learning Engineering Toolkit (2022) — adoption as an engineering problem. — Retrieved from: ____________________

CASE 140 Navy Surface Warfare Readiness Reform

2018 – present

1. GAO-20-154, Navy Readiness: Actions Needed to Evaluate the Effectiveness of Changes to Surface Warfare Officer Training (November 2019) — the absent evaluation processes and the planned threefold increase in ship-driving training hours. — Retrieved from: ____________________
2. Readiness Reform Oversight Council, One-Year Report (2019) — restored training, assessments, and gates. — Retrieved from: ____________________

3. Navy and NTSB reports on the Fitzgerald and McCain collisions (2017–2019) — the training-degradation antecedent. — Retrieved from: ____________________

4. SWOS Norfolk and San Diego Mariner Skills Training Center documentation — simulators and curriculum. — Retrieved from: ____________________

5. USNI News reform coverage (2020, 2022) — Ready-for-Sea Assessments and sidelined ships. — Retrieved from: ____________________

CASE 141 DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks 2009 – 2014

1. Morrison, J. E., & Fletcher, J. D. (September 2012). DARPA Digital Tutor: Assessment Data. IDA Document D-4686 (prepared for DARPA under contract DASW01-04-C-0003) — the sponsor-commissioned evaluation the case rests on. — Retrieved from: ____________________

2. Defense Advanced Research Projects Agency, Digital Tutor program documentation — program description and design rationale. — Retrieved from: ____________________

3. Fletcher, J. D. (2009). From behaviorism to constructivism: a philosophical journey from drill and practice to situated learning. — methodological grounding for the Digital Tutor's tutorial discipline. — Retrieved from: ____________________

4. Anderson, J. R., Corbett, A. T., Koedinger, K. R., & Pelletier, R. (1995). Cognitive tutors: Lessons learned. Journal of the Learning Sciences, 4(2):167–207. doi:10.1207/s15327809jls0402_2 — the broader intelligent-tutoring evidence base the Digital Tutor program sits within. — Retrieved from: ____________________

CASE 142 xAPI / Total Learning Architecture — Interoperability Gap ongoing

1. Advanced Distributed Learning Initiative, Total Learning Architecture documentation — the cross-boundary vision. — Retrieved from: ____________________

2. IEEE Std 9274.1.1-2023, Standard for Learning Technology — JSON Data Model Format and RESTful Web Service for Learner Experience Data Tracking and Access (xAPI 2.0, published October 2023) — the technical standard of record, transferred from ADL to the IEEE LTSC in 2019. — Retrieved from: ____________________

3. ADL Initiative, Total Learning Architecture Standards: Digital Learning Acquisition Techniques (December 2023) — xAPI data that does not follow xAPI Profiles "will have interoperability issues outside of the implementing organization." — Retrieved from: ____________________

4. B. Saxberg, learning-engineering infrastructure essays; IEEE ICICLE LEBoK chapters on data and analytics. — Retrieved from: ____________________

5. Cf. inBloom (Case 53) — technology in advance of governance. — Retrieved from: ____________________

CASE 143 Knight Capital Trading Loss 2012

1. U.S. Securities and Exchange Commission, Order Instituting Administrative and Cease-and-Desist Proceedings, In re Knight Capital Americas LLC (2013) — the firm, the Retail Liquidity Program launch, and the deployment. — Retrieved from: ____________________

2. SEC Order (2013) and Knight Capital 8-K filing (2012) — the missed eighth server, the reactivation of the dormant "Power Peg" code, and the 45-minute event; the SEC puts the loss on the positions at over $460 million, Knight's own filing at a realized pre-tax loss of $440 million. — Retrieved from: ____________________

3. SEC Order (2013) — the absence of a second-technician deployment verification, the lack of an automated code-consistency check, inadequate order controls, and the $12 million penalty (quoted). — Retrieved from: ____________________

4. B. Beyer, C. Jones, J. Petoff & N. R. Murphy (eds.), Site Reliability Engineering (O'Reilly, 2016) — deployment verification, dead-code removal, and automated safeguards as engineering deliverables. — Retrieved from: ____________________

5. SEC Market Access Rule (Rule 15c3-5) and subsequent automated-controls guidance — the regulatory response on pre-trade risk and market-access controls. — Retrieved from: ____________________

6. D. Seven, "Knightmare: A DevOps Cautionary Tale" — a widely cited engineering post-mortem on the deployment process, the orphaned eighth server, and the reused feature flag. — Retrieved from: ____________________

CASE 144 **Kodak Digital Camera Stagnation** 1975–2012

1. S. Sasson, "We Had No Idea," Kodak corporate blog, Kodak: Plugged In (2007) — first-hand account of the 1975 prototype, the demonstration, and the internal reception. — Retrieved from: ____________________

2. H. C. Lucas Jr. & J. M. Goh, "Disruptive technology: How Kodak missed the digital photography revolution," Journal of Strategic Information Systems 18(1):46–55 (March 2009). — Retrieved from: ____________________

3. Eastman Kodak Company, Voluntary Petition for Chapter 11 Reorganization, U.S. Bankruptcy Court, Southern District of New York, Case No. 12-10202 (19 January 2012). — Retrieved from: ____________________

4. A. R. Sorkin & M. J. de la Merced, "Eastman Kodak Files for Bankruptcy," The New York Times DealBook (19 January 2012). — Retrieved from: ____________________

5. M. Spector & D. Mattioli, "Kodak Teeters on the Brink," The Wall Street Journal (5 January 2012); follow-up coverage of the patent-sale process in WSJ and Reuters (December 2012). — Retrieved from: ____________________

CASE 146 **Northeast Blackout** 2003

1. U.S.-Canada Power System Outage Task Force, Final Report on the August 14, 2003 Blackout in the United States and Canada (2004) — the tree contact, the silent alarm, and the cascade. — Retrieved from: ____________________

2. Task Force (2004) — 50 million people affected across eight states and Ontario; the minute-by-minute sequence. — Retrieved from: ____________________

3. Task Force (2004), Causes Groups 1–4 — "inadequate situational awareness at FirstEnergy," FE's failure to manage tree growth, and the reliability organizations' failure to provide effective real-time diagnostic support (quoted). — Retrieved from: ____________________

4. FERC Order No. 693, Mandatory Reliability Standards for the Bulk-Power System (2007) — enforceable standards. — Retrieved from: ____________________

5. North American Electric Reliability Council reports (2004) and the creation of the Electric Reliability Organization. — Retrieved from: ____________________

6. M. R. Endsley (1995), situation-awareness theory — the human-factors frame for silent-automation failure. — Retrieved from: ____________________

CASE 147 **Takata Airbag Inflators** 2008 – 2023

1. U.S. National Highway Traffic Safety Administration, Takata air-bag inflator recall coordination materials — the ammonium-nitrate-without-desiccant design and the propellant-degradation rupture mechanism. — Retrieved from: ____________________

2. NHTSA recall record and investigative reporting (Reuters, New York Times) — 100M+ inflators across 19 automakers; the largest automotive recall in history; deaths and injuries from ruptures. — Retrieved from: ____________________

3. U.S. Department of Justice, settlement and guilty plea with Takata Corporation (2017) — wire fraud, $1B in fine and restitution, and Takata's subsequent bankruptcy. — Retrieved from: ____________________

4. U.S. DOJ (2017) and Takata internal documents released in litigation — the sustained misrepresentation of inflator test data to automakers and regulators. — Retrieved from: ____________________

5. NHTSA Takata recall status reporting — the years-long completion of replacements and the continuing risk in unrepaired vehicles. — Retrieved from: ____________________

6. NHTSA Coordinated Remedy Program for the Takata recalls — the regulator's move to actively prioritize and manage replacement across nineteen automakers rather than leave pace to each manufacturer. — Retrieved from: ____________________

CASE 148 Wells Fargo Fake Accounts 2011 – 2016

1. Consumer Financial Protection Bureau, Consent Order against Wells Fargo (2016) — the unauthorized accounts and penalties. — Retrieved from: ____________________
2. Office of the Comptroller of the Currency, Consent Order AA-EC-2016-66 (2016) — unsafe or unsound sales practices tied to the incentive-compensation structure (paraphrased). — Retrieved from: ____________________
3. Independent Directors of Wells Fargo, Sales Practices Investigation Report (2017) — how the sales-target architecture drove the conduct. — Retrieved from: ____________________
4. Enforcement record: 3.5 million accounts, $3 billion in penalties, and the CEO's resignation. — Retrieved from: ____________________
5. Federal Reserve asset cap on Wells Fargo (imposed February 2018; lifted 3 June 2025) — the structural growth constraint and its termination after remediation. — Retrieved from: ____________________
6. A. C. Edmondson, The Fearless Organization (2018); incentive-design and corporate-governance literature. — Retrieved from: ____________________

CASE 149 Volkswagen Dieselgate 2015

1. U.S. EPA, Notice of Violation to Volkswagen (2015) — the defeat device and emissions exceedances. — Retrieved from: ____________________
2. West Virginia University / ICCT real-world diesel-emissions study (2014) — the discovery comparing road to lab. — Retrieved from: ____________________
3. U.S. Department of Justice Plea Agreement with Volkswagen AG (January 2017) — the institutional decision, USD 4.3 billion criminal-and-civil settlement, and convictions (quoted). — Retrieved from: ____________________
4. Volkswagen internal documents released through litigation — authorization of the defeat device within the engineering hierarchy. — Retrieved from: ____________________
5. EU Real Driving Emissions (RDE) testing regulation — the post-Dieselgate measurement reform. — Retrieved from: ____________________
6. J. Ewing, Faster, Higher, Farther: The Volkswagen Scandal (2017); cf. Takata (Case 147). — Retrieved from: ____________________

CASE 151 GM Ignition Switch 2002 – 2014

1. A. R. Valukas, Report to the Board of Directors of General Motors Company Regarding Ignition Switch Recalls (Jenner & Block, 2014) — the 2002 identification of the defect and the switch-torque mechanism. — Retrieved from: ____________________
2. U.S. NHTSA investigation of the GM ignition switch (2014) and the GM recall record — 2.6 million vehicles; 124 deaths via the GM compensation fund. — Retrieved from: ____________________
3. Valukas Report (2014) — the unchanged part number, the "GM nod" and "GM salute," and the failure to use established escalation processes (quoted). — Retrieved from: ____________________
4. U.S. Department of Justice, deferred-prosecution agreement with General Motors (2015) — $900 million forfeiture for concealing the defect; total penalties/settlements exceeding $2.6 billion. — Retrieved from: ____________________
5. Valukas Report (2014) and A. C. Edmondson, The Fearless Organization (Wiley, 2018) — the part number as an organizational signal, and the suppression of upward safety information. — Retrieved from: ____________________

6. U.S. NHTSA consent order with General Motors (2014, $35M civil penalty) and the GM ignition-switch victims' compensation program (K. Feinberg, administrator) — the regulatory penalty and the basis for the 124-death figure. — Retrieved from: ____________________

CASE 152 TSB Bank IT Migration 2018

1. Slaughter and May, Independent Review of the TSB Migration (2019) — the single-weekend cutover and the testing failures. — Retrieved from: ____________________
2. Slaughter and May (2019) and FCA materials — 1.9 million customers locked out, £330M+ in costs, and the CEO's resignation. — Retrieved from: ____________________
3. Slaughter and May (2019) — inadequate load testing and an unchallenged readiness certification. — Retrieved from: ____________________
4. Financial Conduct Authority, Final Notice on TSB Bank plc (20 December 2022) — the £29.75m penalty (with the PRA's £18.9m) for planning, testing, risk-management and outsourcing failings. — Retrieved from: ____________________
5. House of Commons Treasury Committee, report on the TSB IT migration (2018). — Retrieved from: ____________________
6. Cf. Healthcare.gov (Case 180) and the migration-safety literature. — Retrieved from: ____________________

CASE 153 Equifax Data Breach 2017

1. U.S. Senate Permanent Subcommittee on Investigations, How Equifax Neglected Cybersecurity and Suffered a Devastating Data Breach (2019) — the unpatched Apache Struts vulnerability. — Retrieved from: ____________________
2. The breach record — 147 million affected, exploitation from May 2017, public disclosure in September 2017. — Retrieved from: ____________________
3. Senate PSI (2019) — "Equifax lacked a comprehensive IT asset inventory" (quoted). — Retrieved from: ____________________
4. U.S. FTC / CFPB / state settlement (at least $575 million, up to $700 million, 2019) and the CEO's resignation. — Retrieved from: ____________________
5. U.S. GAO, Actions Taken by Equifax and Federal Agencies in Response to the 2017 Breach, GAO-18-559 (2018). — Retrieved from: ____________________
6. Apache Struts CVE-2017-5638 advisory. — Retrieved from: ____________________

CASE 154 INL Turbine-Control Upgrade — Verification and Validation as the Cutover Deliverable 2014

1. Ronald L. Boring (2014), "Human Factors Design, Verification, and Validation for Two Types of Control Room Upgrades at a Nuclear Power Plant," Proceedings of the Human Factors and Ergonomics Society Annual Meeting 58(1), pp. 2295–2299, doi:10.1177/1541931214581478 — the NUREG-0711 process for the paired plant-process-computer and turbine-control-system upgrade. — Retrieved from: ____________________
2. Idaho National Laboratory, Light Water Reactor Sustainability Program reports on control-room modernization — series available via OSTI. — Retrieved from: ____________________
3. Nuclear Regulatory Commission (NUREG-0711), "Human Factors Engineering Program Review Model" — the regulatory framework the V&V deliverables are produced against. — Retrieved from: ____________________
4. O'Hara et al. (2008), "Human Factors Considerations with Respect to Emerging Technology in Nuclear Power Plants," NUREG/CR-6947 — peer-adjacent framing. — Retrieved from: ____________________

CASE 155 Toyota Production System / Andon Cord

1950s – present

1. Liker, J. (2020), The Toyota Way (2nd ed.) — the andon cord, the cultural pairing, and the American imitation (paraphrased). — Retrieved from: ____________________

2. Spear, S. & Bowen, H. (1999), "Decoding the DNA of the Toyota Production System," HBR — the response protocol and embedded learning. — Retrieved from: ____________________

3. Shingo, S. (1989), A Study of the Toyota Production System — the technical mechanism. — Retrieved from: ____________________

4. Rother, M. (2009), Toyota Kata — the routines that sustain the practice. — Retrieved from: ____________________

5. Womack & Jones (1996), Lean Thinking — diffusion and the limits of surface imitation. — Retrieved from: ____________________

6. Adler, P. (1993), "Time-and-Motion Regained," HBR 71(1) — NUMMI observed from the floor; the rehired workforce, the team structure, and the standardized-work practice that changed alongside the signalling system. — Retrieved from: ____________________

7. NASA Engineering and Safety Center / NHTSA (2011), Technical Assessment of Toyota Electronic Throttle Control Systems — the unintended-acceleration review underlying the scope limit. — Retrieved from: ____________________

CASE 156 INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet

2010 – present

1. Idaho National Laboratory, Light Water Reactor Sustainability (LWRS) program annual reports (2010 – present) — primary research-and-pilot record. — Retrieved from: ____________________

2. Nuclear Regulatory Commission, Regulatory Guide 1.180, "Guidelines for Evaluating Electromagnetic and Radio-Frequency Interference in Safety-Related Instrumentation and Control Systems." — Retrieved from: ____________________

3. Nuclear Regulatory Commission, Branch Technical Position 7-19, "Guidance for Evaluation of Diversity and Defense-in-Depth in Digital Computer-Based Instrumentation and Control Systems." — Retrieved from: ____________________

4. O'Hara, Higgins, Brown, Fink, Persensky, Lewis, Kramer, & Szabo (2008), "Human Factors Considerations with Respect to Emerging Technology in Nuclear Power Plants," NUREG/CR-6947 — foundational human-factors backdrop. — Retrieved from: ____________________

5. Electric Power Research Institute, control-room modernization technical reports — industry-side sustainment record. — Retrieved from: ____________________

CASE 157 AI-Augmented Coding Tools

2021 – present

1. Peng et al. (2023), "The Impact of AI on Developer Productivity" — short-term productivity gains. — Retrieved from: ____________________

2. Pearce et al. (2022), "Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions." — Retrieved from: ____________________

3. Sandoval et al. (2023), "Lost at C," USENIX Security — AI assistance did not significantly increase critical security-bug rates. — Retrieved from: ____________________

4. Dell'Acqua et al. (2023), "Navigating the Jagged Technological Frontier" (HBS / BCG) — professional LLM use. — Retrieved from: ____________________

5. L. Bainbridge, "Ironies of Automation," Automatica 19(6) (1983) — the classic deskilling and over-reliance problem, applied to AI-augmented work. — Retrieved from: ____________________

6. METR (2025), "Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity," arXiv:2507.09089 — the randomized trial finding a 19% slowdown against a self-reported speedup. — Retrieved from: ____________________

CASE 159 Bhopal 1984

1. Union Carbide and government investigation reports (1985) — MIC storage, the disabled safety systems, and plant understaffing. — Retrieved from: ____________________
2. Accounts of the 2–3 Dec. 1984 release — the contested toll (thousands of immediate deaths; 15,000–20,000 total estimates; 500,000 exposed). (Figures vary widely across sources; see AUDIT.) — Retrieved from: ____________________
3. New York Times investigation (1985) — "operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety" (quoted). — Retrieved from: ____________________
4. N. Meshkati, "Human Factors in Large-Scale Technological Systems' Accidents," Industrial Crisis Quarterly (1991); U.S. CSB training-cause pattern. — Retrieved from: ____________________
5. P. Shrivastava, Bhopal: Anatomy of a Crisis (1992); C. Perrow, Normal Accidents (1984). — Retrieved from: ____________________
6. The creation of the U.S. Chemical Safety Board and the post-Bhopal process-safety regime. — Retrieved from: ____________________

CASE 160 Davis-Besse Reactor Head Corrosion 2002

1. U.S. NRC Office of Inspector General, NRC's Regulation of Davis-Besse Regarding Damage to the Reactor Vessel Head (Case No. 02-03S, December 2002) — the post-TMI regulatory regime and oversight failures. — Retrieved from: ____________________
2. NRC event records and OIG (2002) — the 6-inch corrosion cavity, the remaining cladding, and the 2,200-psi coolant margin. — Retrieved from: ____________________
3. NRC Bulletin 2001-01, "Circumferential Cracking of Reactor Pressure Vessel Head Penetration Nozzles" (3 August 2001), and the compromise that let Davis-Besse run to a 16 February 2002 outage. — Retrieved from: ____________________
4. NRC OIG (2002), as quoted in U.S. GAO, GAO-04-415 (2004) — the finding that NRC's decision "was driven in large part by a desire to lessen the financial impact on FirstEnergy," and NRC's dispute of it. — Retrieved from: ____________________
5. D. Lochbaum / Union of Concerned Scientists analysis (2002); U.S. GAO, GAO-04-415 (2004). — Retrieved from: ____________________
6. J. V. Rees, Hostages of Each Other (1994) — INPO and the fragility of institutional safety capability. — Retrieved from: ____________________

CASE 161 Texas City BP Refinery Explosion 2005

1. U.S. Chemical Safety and Hazard Investigation Board, Refinery Explosion and Fire, Investigation Report 2005-04-I-TX (2007) — the startup, malfunctioning instruments, and 15 killed / 180 injured. — Retrieved from: ____________________
2. CSB (2007) — accumulated safety-culture decay, deferred maintenance, and the siting of occupied trailers beside a hazardous unit. — Retrieved from: ____________________
3. CSB (2007) — "indicators of personal safety are not indicators of process safety" (quoted). — Retrieved from: ____________________
4. Report of the BP U.S. Refineries Independent Safety Review Panel (the Baker Panel, 2007). — Retrieved from: ____________________

5. A. Hopkins, Failure to Learn: The BP Texas City Refinery Disaster (CCH, 2008). — Retrieved from: ____________________

6. OSHA Process Safety Management standard (29 CFR 1910.119, 1992) and the CCPS process-safety literature — the personal-vs-process-safety distinction. — Retrieved from: ____________________

CASE 162 Upper Big Branch Mine Explosion 2010

1. U.S. MSHA, Internal Review of MSHA's Actions at Upper Big Branch and the accident investigation (2011–2012) — the dual records and the 29 deaths. — Retrieved from: ____________________

2. Governor's Independent Investigation Panel (J. Davitt McAteer, 2011) — mine conditions: methane, ventilation, and coal-dust inerting. — Retrieved from: ____________________

3. MSHA and U.S. DOJ findings — suppressed methane readings, disabled monitors, and falsified pre-shift examinations. — Retrieved from: ____________________

4. United States v. Blankenship (S.D.W. Va., 2015–2016) — the misdemeanor conviction and felony acquittals. — Retrieved from: ____________________

5. H. Berkes / NPR investigative reporting on Massey Energy. — Retrieved from: ____________________

6. A. Hopkins, Disastrous Decisions: The Human and Organisational Causes of the Gulf of Mexico Blowout (2012) — corporate decision-making and safety (comparative). — Retrieved from: ____________________

CASE 163 Deepwater Horizon 2010

1. National Commission on the BP Deepwater Horizon Oil Spill, Deep Water: The Gulf Oil Disaster and the Future of Offshore Drilling (Report to the President, 2011) — systematic management failures as root cause; the misread negative-pressure test. — Retrieved from: ____________________

2. National Commission (2011) — the well-control sequence and mud-displacement decision. — Retrieved from: ____________________

3. National Commission (2011) — "systematic failures in risk management… place in doubt the safety culture of the entire industry" (quoted); U.S. Chemical Safety Board, Drilling Rig Explosion and Fire at the Macondo Well (final volumes, 2014–2016) — process-safety vs. personal-safety distinction and the improving personal-injury rates BP and Transocean reported as misleading indicators. — Retrieved from: ____________________

4. BOEMRE / U.S. Coast Guard Joint Investigation (2011) — blowout-preventer maintenance backlog and chain-of-command findings; National Academies, Macondo Well Deepwater Horizon Blowout: Lessons for Improving Offshore Drilling Safety (2012) — training gaps and the interlinked cascade of failed defenses. — Retrieved from: ____________________

5. Deepwater Horizon Study Group (UC Berkeley, 2011) final report; N. Leveson, systems-theoretic analysis of Deepwater Horizon — the multi-layer drift and the limits of single-cause framings. — Retrieved from: ____________________

6. Spill-volume estimates (4.9 million barrels) and BP cost disclosures (>$65 billion); the reorganization of the Minerals Management Service into BSEE/BOEM (Secretarial Order 3299, 2010); A. Lustgarten, Run to Failure: BP and the Making of the Deepwater Horizon Disaster (W.W. Norton, 2012) — long-arc account of accumulated procedural debt. — Retrieved from: ____________________

CASE 164 Grenfell Tower 2017

1. Grenfell Tower Inquiry, Phase 1 Report (2019) — the fire's spread up the cladding and the failure of "stay put." — Retrieved from: ____________________

2. Grenfell Tower Inquiry, Phase 2 Report (2024) — decades of failure and the combustible-cladding decision. — Retrieved from: ____________________

3. Phase 2 Report (2024) — cladding firms' "systematic dishonesty" and the RBKC building control failures (ch. 62). — Retrieved from: ____________________

4. Phase 1 Report (2019) — London Fire Brigade unpreparedness; preparation and planning "was gravely inadequate" (quoted). — Retrieved from: ____________________

5. UK Government response to the Grenfell Phase 2 report (2025) — building-safety and fire-service reform. — Retrieved from: ____________________

6. B. Hutter & M. Power (eds.), Organizational Encounters with Risk (2005) — distributed risk ownership. — Retrieved from: ____________________

CASE 165 Camp Fire / PG&E 2018

1. CalFire, Camp Fire Investigation Report (2019) — the worn transmission-line hardware as ignition source. — Retrieved from: ____________________

2. The Camp Fire record — 85 killed; PG&E's guilty plea to 84 counts of involuntary manslaughter (2020). — Retrieved from: ____________________

3. Butte County District Attorney, The Camp Fire Public Report (2020) — PG&E's knowledge and deferred maintenance (quoted). — Retrieved from: ____________________

4. California Public Utilities Commission, Order Instituting Investigation into PG&E (2019) — the regulatory dimension. — Retrieved from: ____________________

5. PG&E bankruptcy and California wildfire-mitigation reforms (2019–). — Retrieved from: ____________________

6. Cf. the Northeast Blackout (Case 146); climate-and-infrastructure literature. — Retrieved from: ____________________

CASE 166 Three Mile Island 1979

1. Kemeny Commission, Report of the President's Commission on the Accident at Three Mile Island (Oct. 1979) — operator training oriented to large design-basis accidents. — Retrieved from: ____________________

2. M. Rogovin & G. T. Frampton, Three Mile Island: A Report to the Commissioners and to the Public, NUREG/CR-1250 (U.S. NRC, 1980) — the September 1977 Davis-Besse stuck-PORV precursor and the failure to disseminate its lessons. — Retrieved from: ____________________

3. U.S. Nuclear Regulatory Commission, Backgrounder on the Three Mile Island Accident — accident sequence, misleading PORV indication, throttled high-pressure injection, and the off-site dose — about 1 millirem average, under 100 millirem at the site boundary. — Retrieved from: ____________________

4. Kemeny Commission (1979) — central conclusion that the fundamental problems were people-related, not equipment-related (quoted). — Retrieved from: ____________________

5. C. Perrow, Normal Accidents (1984); J. Reason, Human Error (1990) — why ambiguous cascades defeat design-basis training and command-not-state interfaces. — Retrieved from: ____________________

6. Institute of Nuclear Power Operations, "Our History" — INPO established December 1979 to set standards and force sharing of operating experience; see also NEI, "Lessons from the 1979 Accident at Three Mile Island." — Retrieved from: ____________________

7. J. V. Rees, Hostages of Each Other: The Transformation of Nuclear Safety since Three Mile Island (Univ. of Chicago Press, 1994) — TMI as the origin of systematic capability reform. — Retrieved from: ____________________

8. U.S. Nuclear Regulatory Commission, Regulatory Guide 1.97, Rev. 2, Instrumentation for Light-Water-Cooled Nuclear Power Plants to Assess Plant and Environs Conditions During and Following an Accident (Dec. 1980), and NUREG-0700, Human-System Interface Design Review Guidelines — the control-room and post-accident-instrumentation standards that grew out of TMI; see also J. S. Walker, Three Mile Island: A Nuclear Crisis in Historical Perspective (Univ. of California Press, 2004) on the operator-training-and-licensing overhaul. — Retrieved from: ____________________

CASE 167 Hurricane Katrina — The FEMA/DHS Response Failure 2005

1. Select Bipartisan Committee to Investigate the Preparation for and Response to Hurricane Katrina (2006), A Failure of Initiative, H. Rept. 109-377, U.S. House of Representatives, February 15, 2006 — including the dedicated Hurricane Pam findings chapter. — Retrieved from: ____________________

2. U.S. Senate Committee on Homeland Security and Governmental Affairs (2006), Hurricane Katrina: A Nation Still Unprepared, S. Rept. 109-322, May 2006 — 325+ interviews, 838,000+ pages of documents reviewed. — Retrieved from: ____________________

3. The White House (2006), The Federal Response to Hurricane Katrina: Lessons Learned (the Townsend report), February 23, 2006 — 17 lessons, 125 recommendations, including the situational-awareness finding. — Retrieved from: ____________________

4. U.S. Senate Committee on Homeland Security and Governmental Affairs (2006), Preparing for a Catastrophe: The Hurricane Pam Exercise, hearing, S. Hrg. 109-403, January 24, 2006 — testimony on the July 2004 exercise scope and unfinished follow-on work. — Retrieved from: ____________________

5. Post-Katrina Emergency Management Reform Act of 2006, Title VI of Public Law 109-295, signed October 4, 2006 — rebuilt FEMA authorities, accelerated pre-request federal assistance, national preparedness goal and system. — Retrieved from: ____________________

CASE 169 Fukushima Daiichi 2011

1. National Diet of Japan Fukushima Nuclear Accident Independent Investigation Commission (NAIIC; K. Kurokawa, chair), Report (2012) — the internal tsunami evidence and the regulatory-capture finding. — Retrieved from: ____________________

2. NAIIC (2012) — the accident sequence: seawall overtopping, generator inundation, and three core meltdowns. — Retrieved from: ____________________

3. NAIIC (2012) — the “made in Japan” cultural and regulatory-capture conclusion (quoted). — Retrieved from: ____________________

4. Investigation Committee on the Accident (Hatamura government commission, 2012); IAEA Director General, The Fukushima Daiichi Accident (2015) — external-hazard under-estimation. — Retrieved from: ____________________

5. D. Lochbaum, E. Lyman & S. Stranahan, Fukushima: The Story of a Nuclear Disaster (2014). — Retrieved from: ____________________

6. Y. Funabashi & K. Kitazawa, Fukushima in Review (2012); cf. INPO (Case 175) and Davis-Besse (Case 160). — Retrieved from: ____________________

CASE 170 West Africa Ebola — The Delayed International Response 2014 – 2016

1. Ebola Interim Assessment Panel (2015), Report of the Ebola Interim Assessment Panel (the “Stocking Report,” chair Dame Barbara Stocking), World Health Organization, 7 July 2015. — Retrieved from: ____________________

2. Moon, S., Sridhar, D., Pate, M. A., Jha, A. K., Clinton, C., et al. (2015), “Will Ebola change the game? Ten essential reforms before the next pandemic. The report of the Harvard-LSHTM Independent Panel on the Global Response to Ebola,” The Lancet 386(10009):2204–2221, 28 November 2015, doi:10.1016/S0140-6736(15)00946-0. — Retrieved from: ____________________

3. High-level Panel on the Global Response to Health Crises (2016), Protecting Humanity from Future Health Crises (chair Jakaya M. Kikwete), UN doc. A/70/723, 9 February 2016. — Retrieved from: ____________________

4. World Health Organization (2016), Ebola Situation Reports — West Africa: approximately 28,600 cases and 11,300 deaths reported to 10 June 2016 (reported figures; undercount acknowledged). — Retrieved from: ____________________

5. Médecins Sans Frontières (2015), Ebola: Pushed to the Limit and Beyond — MSF account of the international outbreak response, March 2015. — Retrieved from: ____________________

CASE 171 Hurricane Maria — Puerto Rico Response and Logistics Failure 2017 – 2018

1. Federal Emergency Management Agency (2018), 2017 Hurricane Season FEMA After-Action Report, July 12, 2018 — the agency's own self-assessment: force strength below target, Caribbean Distribution Center depletion, commodity-tracking shortfalls. — Retrieved from: ____________________

2. U.S. Government Accountability Office (2018), 2017 Hurricanes and Wildfires: Initial Observations on the Federal Response and Key Recovery Challenges, GAO-18-472, September 4, 2018 — 54% of FEMA staff serving in a capacity in which they did not hold the "Qualified" title at the October 2017 peak. — Retrieved from: ____________________

3. Milken Institute School of Public Health, George Washington University (2018), Ascertainment of the Estimated Excess Mortality from Hurricane María in Puerto Rico, August 2018 — commissioned by the Government of Puerto Rico; 2,975 excess deaths, September 2017 – February 2018; basis for the official toll revision from 64. — Retrieved from: ____________________

4. Kishore, N., Marqués, D., Mahmud, A., et al. (2018), "Mortality in Puerto Rico after Hurricane Maria," New England Journal of Medicine 379(2):162 – 170, doi:10.1056/NEJMsa1803972 — household-survey estimate of 4,645 excess deaths (95% CI 793 – 8,498), September 20 – December 31, 2017. — Retrieved from: ____________________

5. Houser, T., & Marsters, P. (2018), "The World's Second Largest Blackout," Rhodium Group, April 12, 2018 — approximately 3.4 billion lost customer-hours; the largest blackout in U.S. history by that metric. — Retrieved from: ____________________

CASE 172 CrowdStrike Falcon Outage 2024

1. CrowdStrike, Falcon Content Update Preliminary Post Incident Report (24 July 2024) — the Rapid-Response-Content-versus-Sensor-Content testing and staged-rollout gap. — Retrieved from: ____________________

2. CrowdStrike PIR (2024) — the configuration-file fault and the kernel crash; the 8.5 million affected Windows machines figure is Microsoft's (D. Weston, 20 July 2024). — Retrieved from: ____________________

3. Microsoft resilient-engineering analyses and Windows kernel-access review (2024). — Retrieved from: ____________________

4. U.S. Government Accountability Office, Cyber Resiliency: CrowdStrike Outage Highlights Challenges, GAO-24-107733 (23 September 2024); House Committee on Homeland Security, Subcommittee on Cybersecurity and Infrastructure Protection hearing, 24 September 2024 — supply-chain and concentration risk in endpoint security. — Retrieved from: ____________________

5. B. Beyer et al. (eds.), Site Reliability Engineering (2016) — staged rollout and canarying; cf. Knight Capital (Case 143). — Retrieved from: ____________________

CASE 173 Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable 1963 – present

1. Rear Admiral Paul E. Sullivan, statement to House Science Committee (2003), NASA/Columbia archive — primary congressional record on SUBSAFE. — Retrieved from: ____________________

2. MIT Press, "SUBSAFE: An Example of a Successful Safety Program" — book chapter (open access). — Retrieved from: ____________________

3. NASA SMA (2006), "USS Thresher Lessons Learned" briefing — safety-message archive. — Retrieved from: ____________________

4. Columbia Accident Investigation Board (2003), final report — Volume I, endorsement of SUBSAFE as a model for NASA. — Retrieved from: ____________________

5. U.S. Navy NAVSEA, SUBSAFE program documentation — operating program publications. — Retrieved from: ____________________

CASE 174 Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed 1996 – 2000

1. General Accounting Office, Year 2000 Computing Challenge report series (1997–2000), particularly GAO/AIMD-00-290, Lessons Learned Can Be Applied to Other Management Challenges (September 2000) — the federal-program-management record and GAO's own retrospective. — Retrieved from: ____________________
2. Office of Management and Budget, quarterly reports on federal Y2K remediation status (1997–1999) — program-self-report tier. — Retrieved from: ____________________
3. President's Council on Year 2000 Conversion, The Journey to Y2K final report (2000) — institutional retrospective of the federal coordination effort. — Retrieved from: ____________________
4. Manion & Evan (2000), "The Y2K Problem and Professional Responsibility: A Retrospective Analysis," Technology in Society 22(3), 361–387 — retrospective literature on the counterfactual debate. — Retrieved from: ____________________
5. U.S. Department of Commerce, Economics and Statistics Administration, The Economics of Y2K and the Impact on the United States (November 1999) — total U.S. Y2K spending of about $100 billion covering repairs started 1995 through 2001, of which the federal share was about $8 billion. — Retrieved from: ____________________

CASE 175 INPO and the Nuclear Academy 1979 – present

1. Rees, J. (1994), Hostages of Each Other: The Transformation of Nuclear Safety since Three Mile Island — INPO's design and the "hostages" premise (paraphrased). — Retrieved from: ____________________
2. Report of the President's Commission on the Accident at Three Mile Island (Kemeny Commission, 1979) — the pre-TMI culture. — Retrieved from: ____________________
3. Nuclear Energy Institute, "Lessons from the 1979 Accident at Three Mile Island"; National Academy for Nuclear Training — accreditation and peer evaluation. — Retrieved from: ____________________
4. World Nuclear Association — Three Mile Island Accident; INPO/WANO performance indicators. — Retrieved from: ____________________
5. Willard, R. F. (2019), The Role of the Institute of Nuclear Power Operations in Supporting the United States Commercial Nuclear Power Industry's Focus on Nuclear Safety, testimony for the record, U.S. Senate Committee on Environment and Public Works — INPO's design, funding, peer evaluations, and the National Academy. — Retrieved from: ____________________

CASE 176 Tylenol Recall 1982

1. Kaplan, T. (2014), The Tylenol Crisis — the recall and corporate response. — Retrieved from: ____________________
2. James Burke (J&J CEO), in Lasting Leadership (Wharton) — the Credo quote. — Retrieved from: ____________________
3. Greyser, S., Johnson & Johnson: The Tylenol Tragedy (HBS case, 1992) — market recovery and crisis management. — Retrieved from: ____________________
4. FDA Final Rule on Tamper-Resistant Packaging (1982) — the packaging reform. — Retrieved from: ____________________
5. Edmondson, A. (2018), The Fearless Organization — pre-committed values under stress. — Retrieved from: ____________________

CASE 177 CIRAS — Confidential Incident Reporting for UK Rail 1996 – present

1. Davies, Wright, Courtney, & Reid (2000), "Confidential Incident Reporting on the UK Railways: The 'CIRAS' System," Cognition, Technology & Work 2, 117–125, doi:10.1007/PL00011494. — Retrieved from: ____________________

2. Rail Safety and Standards Board (RSSB), CIRAS program documentation 2008 – present — operating-program publications. — Retrieved from: ____________________

3. University of Strathclyde, CIRAS impact case study — the operating-program-self-report on outcomes between 2008 and 2012. — Retrieved from: ____________________

4. Ladbroke Grove Rail Inquiry (Cullen, 2001), final report — the public inquiry into the crash that preceded the national mandate. — Retrieved from: ____________________

CASE 178 The UN Cluster Approach — Engineering Humanitarian Coordination After the Tsunami 2005 – 2020

1. Adinolfi, Bassiouni, Lauritzsen, & Williams (2005), Humanitarian Response Review, commissioned by the UN Emergency Relief Coordinator, New York/Geneva: OCHA. — Retrieved from: ____________________

2. Stoddard, Harmer, Haver, Salomons, & Wheeler (2007), Cluster Approach Evaluation, Final Report, OCHA Evaluation and Studies Section, with NYU Center on International Cooperation, ODI Humanitarian Policy Group, and the Praxis Group. — Retrieved from: ____________________

3. Steets, Grünewald, Binder, de Geoffroy, Kauffmann, Krüger, Meier, & Sokpoh (2010), Cluster Approach Evaluation 2: Synthesis Report, GPPi / Groupe URD, commissioned by the IASC, Geneva: OCHA. — Retrieved from: ____________________

4. Inter-Agency Standing Committee (2011 – 2012), IASC Transformative Agenda — Principals' decisions of December 2011 and associated protocols. — Retrieved from: ____________________

5. Grünewald & Binder (2010), Inter-agency Real-Time Evaluation in Haiti: 3 Months After the Earthquake, Groupe URD / GPPi. — Retrieved from: ____________________

CASE 179 The Johns Hopkins COVID-19 Dashboard — Evidence Infrastructure at Decision Speed 2020 – 2023

1. Dong, Du, & Gardner (2020), "An interactive web-based dashboard to track COVID-19 in real time," The Lancet Infectious Diseases 20(5), 533 – 534, doi:10.1016/S1473-3099(20)30120-1. — Retrieved from: ____________________

2. Dong et al. (2022), "The Johns Hopkins University Center for Systems Science and Engineering COVID-19 Dashboard: data collection process, challenges faced, and lessons learned," The Lancet Infectious Diseases 22(12), e370 – e376, doi:10.1016/S1473-3099(22)00434-0. — Retrieved from: ____________________

3. Johns Hopkins University Hub (10 March 2023), "Johns Hopkins COVID-19 data hub ends after three years" — Coronavirus Resource Center closure announcement. — Retrieved from: ____________________

4. Lasker Foundation (2022), Lasker-Bloomberg Public Service Award citation — "The COVID-19 Dashboard" (Lauren Gardner); companion award essay in JAMA. — Retrieved from: ____________________

5. NPR (10 February 2023), "As the pandemic ebbs, an influential COVID tracker shuts down." — Retrieved from: ____________________

CASE 180 Healthcare.gov Launch 2013

1. U.S. GAO, Healthcare.gov: CMS Has Taken Steps to Address Problems, but Needs to Further Implement Systems Development Best Practices, GAO-15-238 (2015) — "end-to-end testing of Healthcare.gov and its supporting systems did not occur prior to system launch as required." — Retrieved from: ____________________

2. HHS Office of Inspector General, Case Study of CMS Management of the Federal Marketplace, OEI-06-14-00350 (2016) — unclear ownership and the CMS/CGI integration confusion. — Retrieved from: ____________________

3. HHS OIG, OEI-06-14-00350 (2016), p. iii — "HealthCare.gov lacked clear project leadership to give direction and unity of purpose, responsiveness in execution, and a comprehensive view of progress." — Retrieved from: ____________________

4. J. Pahlka, Recoding America (2023) — the founding of the U.S. Digital Service out of the rescue. — Retrieved from: ____________________

5. L. A. Schlesinger & P. D. Bhayani, "HealthCare.gov: The Crash and the Fix (A)," Harvard Business School case 315129 (2015); Mergel et al. (2018), digital-government literature. — Retrieved from: ____________________

CASE 181 EU Human Brain Project — Top-Down Vision That Unraveled 2013 – 2023

1. "How big science failed to unlock the mysteries of the human brain," MIT Technology Review, 25 August 2021 — the principal critical assessment of HBP alongside BRAIN. — Retrieved from: ____________________

2. In Silico (2020), documentary by Noah Hutton — long-form follow of Markram and the HBP through the contestation period. — Retrieved from: ____________________

3. Science and Nature contemporaneous coverage of the open letters and the mediation/restructuring (2014–2016). — Retrieved from: ____________________

4. Human Brain Project final reports (2023) — the project's own restructuring and concluding documentation. — Retrieved from: ____________________

5. Alivisatos et al. (2012), Neuron — the BRAIN position paper that is the paired contrast (Case 198). — Retrieved from: ____________________

CASE 182 Amazon Hiring AI — Trained Bias, Deprecated 2017 2014 – 2018

1. Dastin, J. (2018), "Amazon scraps secret AI recruiting tool that showed bias against women," Reuters, October 10, 2018 — the primary investigation. — Retrieved from: ____________________

2. Subsequent commentary: Kim, P. T. (2017), "Data-Driven Discrimination at Work," William & Mary Law Review 58(3):857–936 — academic frame for the structural pattern the Amazon case instantiates. — Retrieved from: ____________________

3. Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020), "Mitigating Bias in Algorithmic Hiring: Evaluating Claims and Practices," Proceedings of FAT* 2020, pp. 469–481 — survey of the algorithmic-hiring landscape into which the Amazon case is positioned. — Retrieved from: ____________________

4. Bogen, M., & Rieke, A. (2018), Help Wanted: An Examination of Hiring Algorithms, Equity, and Bias, Upturn report — contemporary practice survey of algorithmic hiring at the time of the Amazon deprecation. — Retrieved from: ____________________

CASE 183 Uber ATG / Tempe Fatality 2018

1. NTSB, Collision Between Vehicle Controlled by Developmental Automated Driving System and Pedestrian, Highway Accident Report HAR-19/03 (2019) — the Tempe collision and probable cause. — Retrieved from: ____________________

2. NTSB HAR-19/03 (2019) — the safety operator's distraction (watching a video) before impact. — Retrieved from: ____________________

3. NTSB HAR-19/03 (2019) — Uber "did not adequately recognize the risk of automation complacency"; training and policy failures (quoted in part). — Retrieved from: ____________________

4. NTSB HAR-19/03 (2019) — the suppressed emergency braking and the absent functionality to anticipate a pedestrian crossing midblock. — Retrieved from: ____________________

5. NTSB HAR-19/03 (2019), Board Member Statement, p. 62 — Vice Chairman Bruce Landsberg: "Automation performs remarkably well most of the time and therein lies the problem." — Retrieved from: ____________________

6. R. Parasuraman & D. Manzey (2010), complacency in automation; L. Bainbridge (1983), "Ironies of Automation." — Retrieved from: ____________________

CASE 184 UK Post Office Horizon Scandal 1999 – 2015

1. Post Office Horizon IT Inquiry hearings and exhibits (2020–) — the system, the prosecutions, and the human toll. — Retrieved from: ____________________

2. Hamilton & Others v. Post Office Limited (Court of Appeal, 2021) — quashed convictions. — Retrieved from: ____________________

3. Internal Fujitsu and Post Office documents released through litigation — engineers' knowledge of Horizon bugs. — Retrieved from: ____________________

4. Sir Wyn Williams, Post Office Horizon IT Inquiry, Volume 1 (July 2025) — senior employees "knew… that Legacy Horizon was capable of error" (quoted). — Retrieved from: ____________________

5. N. Wallis, The Great Post Office Scandal (2021). — Retrieved from: ____________________

6. The Post Office (Horizon System) Offences Act 2024 (mass exoneration) and the compensation schemes. — Retrieved from: ____________________

CASE 185 Air Canada Chatbot Liability — Delegation Without Revocation 2022 – 2024

1. Moffatt v. Air Canada, 2024 BCCRT 149 (British Columbia Civil Resolution Tribunal, February 14, 2024), Tribunal Member Christopher Rivers presiding. — Retrieved from: ____________________

2. Cecco, L. (2024), "Air Canada ordered to pay customer who was misled by airline's chatbot," The Guardian, February 16, 2024 — contemporaneous press coverage of the ruling. — Retrieved from: ____________________

3. Air Canada bereavement-fare policy text (as in effect November 2022 and through the period covered by the ruling) — referenced in the tribunal decision as the divergence the chatbot's representation produced. — Retrieved from: ____________________

4. Sookman, B. (McCarthy Tétrault LLP, 2024), "Moffatt v. Air Canada: A Misrepresentation by an AI Chatbot," McCarthy Tétrault TechLex Blog, February 19, 2024 — practitioner-tier analysis of the tribunal's negligent-misrepresentation holding, the rejection of the "separate legal entity" defense, and the duty-of-care framing for AI-mediated consumer interactions. mccarthy.ca. — Retrieved from: ____________________

CASE 186 Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity 2009 – 2022

1. Bartlett, Morse, Stanton, & Wallace (2022), "Consumer-lending discrimination in the FinTech era," Journal of Financial Economics 143(1):30–56, doi:10.1016/j.jfineco.2021.05.047. — Retrieved from: ____________________

2. Consumer Financial Protection Bureau, Mortgage Market Activity and Trends (annual HMDA reports), supporting the population-level disparities backdrop. — Retrieved from: ____________________

3. Dwork et al. (2012), "Fairness Through Awareness," ITCS 2012 — the foundational technical statement of the limits of unawareness. — Retrieved from: ____________________

4. Mitchell, Potash, Barocas, D'Amour, & Lum (2021), "Algorithmic Fairness: Choices, Assumptions, and Definitions," Annual Review of Statistics and Its Application 8:141–163 — the competing-definitions framing. — Retrieved from: ____________________

CASE 187 COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate 2014 – 2018

1. Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016), "Machine Bias," ProPublica, May 23, 2016 — the audit investigation of COMPAS scores in Broward County, Florida. — Retrieved from: ____________________

2. Dieterich, W., Mendoza, C., & Brennan, T. (2016), COMPAS Risk Scales: Demonstrating Accuracy Equity and Predictive Parity, Northpointe Inc. response document. — Retrieved from: ____________________

3. Chouldechova, A. (2017), "Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments," Big Data 5(2):153–163, doi:10.1089/big.2016.0047. — Retrieved from: ____________________

4. Kleinberg, J., Mullainathan, S., & Raghavan, M. (2017), "Inherent Trade-Offs in the Fair Determination of Risk Scores," Proceedings of ITCS 2017, doi:10.4230/LIPIcs.ITCS.2017.43. — Retrieved from: ____________________

5. Dressel, J., & Farid, H. (2018), "The accuracy, fairness, and limits of predicting recidivism," Science Advances 4(1):eaao5580 — COMPAS no more accurate than untrained humans or a two-variable model. — Retrieved from: ____________________

6. Rudin, C., Wang, C., & Coker, B. (2020), "The age of secrecy and unfairness in recidivism prediction," Harvard Data Science Review 2(1) — the age-nonlinearity reanalysis of ProPublica's disparity claim. — Retrieved from: ____________________

7. Bao, M., et al. (2021), "It's COMPASlicated: The Messy Relationship between RAI Datasets and Algorithmic Fairness Benchmarks," NeurIPS Datasets & Benchmarks — the case against COMPAS as a fairness benchmark. — Retrieved from: ____________________

CASE 189 Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds 2014 – 2020

1. District Court of The Hague (2020), judgment of 5 February 2020 (NJCM et al. v. State of the Netherlands), ECLI:NL:RBDHA:2020:865. — Retrieved from: ____________________

2. Rachovitsa & Johann (2022), "The Human Rights Implications of the Use of AI in the Digital Welfare State: Lessons Learned from the Dutch SyRI Case," Human Rights Law Review 22(2), doi:10.1093/hrlr/ngac010. — Retrieved from: ____________________

3. Library of Congress Global Legal Monitor (2020), report on the SyRI ruling. — Retrieved from: ____________________

4. Algorithm Watch (2020), case study of the SyRI ruling and its implications for public-sector AI in Europe. — Retrieved from: ____________________

CASE 190 Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment 2023 – 2024

1. California DMV (24 October 2023), Order of Suspension, Cruise LLC — driverless deployment and testing permits. — Retrieved from: ____________________

2. NBC News (October 2023), reporting on the Cruise pedestrian incident and DMV suspension. — Retrieved from: ____________________

3. TechCrunch (October 2023), incident-disclosure reconstruction. — Retrieved from: ____________________

4. SF Standard (October 2023), San Francisco AV regulatory reporting. — Retrieved from: ____________________

5. Mission Local (October 2023), San Francisco-specific incident reporting. — Retrieved from: ____________________

6. Cruise public statements, October–November 2023, and the company's blog post accepting the Quinn Emanuel findings (January 2024). — Retrieved from: ____________________

CASE 191 Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict 2016 – 2023

1. Royal Commission into the Robodebt Scheme, Final Report, Commissioner Catherine Holmes AC SC, July 7, 2023 (Commonwealth of Australia). — Retrieved from: ____________________

2. Prygodicz v Commonwealth of Australia (No 2) [2021] FCA 634 — Federal Court class-action settlement following the 2019 unlawfulness finding. — Retrieved from: ____________________

3. Australian National Audit Office, Centrelink's Compliance Activities — Income Compliance Program, performance audit reports across 2017–2020. — Retrieved from: ____________________

4. Whiteford, P. (2021), "Debt by Design: The Anatomy of a Social Policy Fiasco," Australian Journal of Public Administration 80(2):340–360 — academic synthesis of the policy and administrative history. — Retrieved from: ____________________

CASE 192 Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model 2019 – 2021

1. New York State Department of Financial Services, Report on Apple Card Investigation, March 23, 2021. — Retrieved from: ____________________

2. Heinemeier Hansson, D. (2019), Twitter thread of November 7, 2019, archived in DFS investigation record and contemporaneous press coverage (Bloomberg, New York Times, Wall Street Journal, November 2019). — Retrieved from: ____________________

3. Vigdor, N. (2019), "Apple Card Investigated After Gender Discrimination Complaints," The New York Times, November 10, 2019. — Retrieved from: ____________________

4. Goldman Sachs Bank USA, public response and credit-line-review process documentation submitted to DFS during the investigation (2019 – 2021). — Retrieved from: ____________________

CASE 193 Bernard Madoff / SEC Failures 1992 – 2008

1. SEC Office of Inspector General, Investigation of Failure of the SEC to Uncover Bernard Madoff's Ponzi Scheme, Report of Investigation Case No. OIG-509 (31 August 2009) — the quoted finding. — Retrieved from: ____________________

2. Markopolos, H. (2010), No One Would Listen — the 2005 memo and its dismissal. — Retrieved from: ____________________

3. United States v. Madoff (2009) — guilty plea and the $65B figure. — Retrieved from: ____________________

4. SEC OIG (2009) — staff expertise gap and deference to Madoff's stature. — Retrieved from: ____________________

5. SEC post-Madoff reforms (2009–2010), including the Office of Market Intelligence for tip and referral triage. — Retrieved from: ____________________

CASE 194 Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox) 2001 – present

1. Kattel, R., & Mergel, I. (2019), "Estonia's Digital Transformation: Mission Mystique and the Hiding Hand," in Compton, M., & 't Hart, P. (eds.), Great Policy Successes (Oxford University Press, 2019) — peer-reviewed analytical chapter. — Retrieved from: ____________________

2. Jackson, Dreyling, & Pappel (2021), "A Historical Analysis on Interoperability in Estonian Data Exchange Architecture," ICEGOV 2021 proceedings, doi:10.1145/3494193.3494209. — Retrieved from: ____________________

3. X-Road Global / Nordic Institute for Interoperability Solutions (NIIS) — program documentation and deployment-partner case studies. — Retrieved from: ____________________

4. Republic of Estonia, e-Estonia briefing materials and Year-in-Review documentation (2024) — program-report sourcing. — Retrieved from: ____________________

CASE 195 Tesla Autopilot — Recurring Fatalities 2016 – present

1. NTSB, Highway Accident Report HAR-17/02 (Williston, FL, 2016) — the quoted disengagement finding. — Retrieved from: ____________________

2. NTSB, Highway Accident Report HAR-20/01 (Mountain View, CA, 2018) — Autopilot crash analysis. — Retrieved from: ____________________

3. NHTSA Standing General Order 2021-01 reports — documented Autopilot fatal crashes. — Retrieved from: ____________________

4. NHTSA Office of Defects Investigation EA22-002 (opened 2022; closed April 2024, superseded by recall 23V-838) — driver-engagement adequacy. — Retrieved from: ____________________

5. NHTSA recall-query RQ24009 (April 2024) into the adequacy of the December 2023 Autopilot remedy, and investigation PE24031 (October 2024) into Full Self-Driving; Tesla's June 2025 Austin robotaxi launch and the ensuing regulator scrutiny. — Retrieved from: ____________________

6. Parasuraman, R. & Manzey, D. (2010) — automation complacency and monitoring. — Retrieved from: ____________________

CASE 196 Fintech Lending Fairness Audit — When Including Race Reduces Disparity 2025

1. Coots, Bartlett, Nyarko, & Goel (2025), "Algorithmic Bias in Lending: Evidence from a Fintech Audit," arXiv:2512.20753 — audit of 80,000 personal loans showing model miscalibration disparities and that controlled inclusion of protected attributes could correct them; the evidence-tier flag is binding until peer review completes. — Retrieved from: ____________________

2. Bartlett, Morse, Stanton, & Wallace (2022), "Consumer-lending discrimination in the FinTech era," Journal of Financial Economics 143(1):30–56 — the paired big-tier case (186). — Retrieved from: ____________________

3. Chouldechova (2017), "Fair Prediction with Disparate Impact," Big Data 5(2):153–163 — the impossibility result for calibration and error-rate parity. — Retrieved from: ____________________

4. Kleinberg, Mullainathan, & Raghavan (2017), "Inherent Trade-Offs in the Fair Determination of Risk Scores," ITCS — competing fairness definitions are not jointly achievable. — Retrieved from: ____________________

5. Mitchell, Potash, Barocas, D'Amour, & Lum (2021), "Algorithmic Fairness: Choices, Assumptions, and Definitions," Annual Review of Statistics and Its Application 8:141–163 — practitioner-facing survey. — Retrieved from: ____________________

CASE 197 Predictive Policing — PredPol 2011 – present

1. Lum, K. & Isaac, W. (2016), "To Predict and Serve?," Significance — the enforcement-vs-crime feedback loop (paraphrased). — Retrieved from: ____________________

2. Richardson, Schultz & Crawford (2019), "Dirty Data, Bad Predictions" — biased records feeding predictive systems. — Retrieved from: ____________________

3. Brantingham et al. (2018) — predictive-policing field experiments. — Retrieved from: ____________________

4. Brayne, S. (2017), "Big Data Surveillance: The Case of Policing." — Retrieved from: ____________________

5. Municipal records on suspension and abandonment of predictive policing (Santa Cruz, New Orleans, Los Angeles). — Retrieved from: ____________________

6. Sankin, A. & Mattu, S. (2023), "Predictive Policing Software Terrible at Predicting Crimes," The Markup, 2 October 2023 (copublished with Wired) — the Plainfield, NJ accuracy analysis; and reporting on PredPol/Geolitica's 2021 rebrand and 2023 wind-down into SoundThinking. — Retrieved from: ____________________

CASE 198 Launching the BRAIN Initiative — Governance of a Grand Challenge 2011 – 2015 – present

1. Alivisatos, Chun, Church, Greenspan, Roukes, & Yuste (2012), "The Brain Activity Map Project and the Challenge of Functional Connectomics," Neuron 74(6):970–974, doi:10.1016/j.neuron.2012.06.006. — Retrieved from: ____________________

2. Jorgenson et al. (2015), "The BRAIN Initiative: developing technology to catalyse neuroscience discovery," Phil. Trans. R. Soc. B 370(1668):20140164, doi:10.1098/rstb.2014.0164 — the BRAIN 2025 plan. — Retrieved from: ____________________

3. Yuste & Bargmann (2017), "Toward a Global BRAIN Initiative," Cell 168(6):956–959, doi:10.1016/j.cell.2017.02.023. — Retrieved from: ____________________

4. Underwood (2013), "As White House Embraces BRAIN Initiative, Questions Linger," Science / ScienceInsider (April 3, 2013) — source of the Yuste and Bargmann public-record contestation quotes. — Retrieved from: ____________________

5. Mullin, E. (2021), "How big science failed to unlock the mysteries of the human brain," MIT Technology Review (25 August 2021) — the critical ten-year assessment. — Retrieved from: ____________________

CASE 199 Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact 2023 – 2026

1. Waymo (2023), "A Blueprint for AV Safety: Waymo's Toolkit For Building a Credible Safety Case," whitepaper. — Retrieved from: ____________________

2. Waymo (November 2025), "Independent Audits of Waymo's Safety Case and Remote Assistance Programs," summary release. — Retrieved from: ____________________

3. NHTSA recall and investigation record (December 2025 school-bus recall; January 2026 NHTSA/NTSB investigation of a Santa Monica child-strike; June 2026 recall of 3,900 vehicles over freeway construction zones) — the post-deployment failure record under regulator action. — Retrieved from: ____________________

4. Montreal AI Ethics Institute (2023), summary and analysis of the Waymo safety case framework. — Retrieved from: ____________________

5. California Public Utilities Commission, AV passenger-service permit framework documents — paired Case 200 for the regulator-side artifact. — Retrieved from: ____________________

6. Cruise / California DMV Order of Suspension (October 2023) — paired Case 190 as the foil. — Retrieved from: ____________________

CASE 200 CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver 2020 – 2026

1. California Public Utilities Commission, "Autonomous Vehicle Passenger Service Programs" — program page for the Drivered and Driverless pilot and deployment programs, including the Passenger Safety Plan requirement. — Retrieved from: ____________________

2. CPUC permit decisions for Cruise and Waymo, 2020–2024. — Retrieved from: ____________________

3. California Department of Motor Vehicles, "New Autonomous Vehicle Regulations Strengthen Oversight and Enforcement, Authorize Trucks and Transit" (28 April 2026) — adopted authority to impose targeted operational restrictions on fleet size, location, speed and weather; draft language circulated for comment 30 August 2024. — Retrieved from: ____________________

4. California Public Utilities Commission, Resolution TL-19144 (Waymo), issued 11 August 2023 — approves driverless deployment in all of San Francisco 24 hours a day per the DMV-approved ODD, and declines to set limits on operating hours, geography or fleet size. — Retrieved from: ____________________

CASE 201 Aadhaar Exclusion — Biometric Welfare Delegation and Its Judicial Reckoning in India 2018 – 2025

1. Supreme Court of India (2018), Justice K. S. Puttaswamy (Retd.) v. Union of India, (2019) 1 SCC 1 — the Aadhaar judgment; 4–1 majority upholding Section 7, with Justice D. Y. Chandrachud's dissent on technological exclusion and dignity. — Retrieved from: ____________________

2. Drèze, J., Khalid, N., Khera, R., & Somanchi, A. (2017), "Aadhaar and Food Security in Jharkhand: Pain without Gain?," Economic & Political Weekly 52(50) — field study of biometric-authentication exclusion in the PDS. — Retrieved from: ____________________

3. Right to Food Campaign compilation and Indian journalism (The Hindu, The Wire, Scroll.in, 2017–2019) on the Jharkhand starvation deaths (Santoshi Kumari, September 2017) — lived-exclusion sourcing with journalism-tier flag. — Retrieved from: ____________________

4. Supreme Court of India (2025), Pragya Prasun & Ors. v. Union of India, 2025 INSC 599 — the distinct, later ruling reading a right to accessible digital access into Article 21 in the e-KYC context for persons with disabilities. — Retrieved from: ____________________

5. Dixon, P. (2017), "A Failure to 'Do No Harm' — India's Aadhaar biometric ID program and its inability to protect privacy in relation to measures in Europe and the U.S.," Health and Technology 7(4):539–567 — comparative privacy and data-protection analysis, reporting biometric-authentication failure rates alongside it. — Retrieved from: ____________________

CASE 203 NYC Local Law 144 — Bias Audits as Governance Artifact

2023 – present

1. New York City Department of Consumer and Worker Protection, Rules Implementing Local Law 144 of 2021 (Automated Employment Decision Tools), effective July 5, 2023. — Retrieved from: ____________________

2. Wright, L., Muenster, R. M., Vecchione, B., Qu, T., Cai, S., Smith, A., Metcalf, J., & Matias, J. N. (2024), "Null Compliance: NYC Local Law 144 and the Challenges of Algorithm Accountability," in Proceedings of FAccT 2024, doi:10.1145/3630106.3658998. — Retrieved from: ____________________

3. Engler, A. (2023), "The EU and U.S. diverge on AI regulation: A transatlantic comparison and steps to alignment," Brookings Institution commentary — regulatory-comparative frame for the municipal intervention. — Retrieved from: ____________________

4. Office of the New York State Comptroller (2025), audit of the Department of Consumer and Worker Protection's enforcement of Local Law 144 — finding enforcement "ineffective" (December 2025). — Retrieved from: ____________________

CASE 205 The Discipline We Build Next

ongoing

1. LDT / LENS program statement, Johns Hopkins School of Education — the program and its commitments. — Retrieved from: ____________________

2. Goodell & Kolodner (2022), Learning Engineering Toolkit — the discipline's methods. — Retrieved from: ____________________

3. The preceding cases of this volume — the cumulative record of cost and of possibility. — Retrieved from: ____________________

4. INPO (Case 175) and CRM/CAST (Case 117) — institution-building precedents (1979, 1981). — Retrieved from: ____________________

5. IEEE ICICLE Learning Engineering Body of Knowledge (LEBoK); the introduction to this volume. — Retrieved from: ____________________

AUDIT SURFACE FOR

Capability Matters: A Casebook

Edited by William Gray-Roncal, PhD and James Diamond, PhD

Learning Engineering for Next-Generation Systems · v2.1 · Edition · 15 June 2026

LDT · School of Education · Johns Hopkins University